Post-Capitalist Labs Imprint · Working Draft Monograph

DISTRIBUTED PARTICIPATORY ECONOMICS PROTOCOL

A Rigorous, Source-Grounded Feasibility Study and Post-Quantum Technical Architecture for Democratic Planning, Self-Management, and the Productive Commons

Titan Cassini



DESIGNED FOR THREE INTERDISCIPLINARY AUDIENCES:

- Advocates & Cooperative Practitioners: Council democracy, balanced job complexes, and historical lessons from Catalonia (1936), the Paris Commune (1871), and Classical Athens.
- Economists & Mechanism Designers: Formal welfare theorems, Sandström's double-entry accounting, pollution demand revelation (CAPs), and candidate ADMM regularization under convex restrictions.
- Systems Programmers & Cryptographers: Local-first CRDTs, Two-Root BLAKE3 aggregation, deterministic WASM IFB engines, and NIST FIPS 203/204/205 post-quantum primitives.

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Leveraging technology to amplify social movements · Decentralized consensus, mechanism design & democratic coordination

Post-Capitalist Labs Imprint · Open-Source Research Monograph Series

Distributed Participatory Economics Protocol (DPEP)

A Rigorous, Source-Grounded Feasibility Study and Post-Quantum Technical Architecture for Democratic Planning, Self-Management, and the Productive Commons

Titan Cassini

About Post-Capitalist Labs (PCL)

At Post-Capitalist Labs, we harness the transformative power of technology to bolster social movements and build directly democratic alternatives to capitalism. Our work spans decentralized consensus, applied cryptography, operations research, and democratic mechanism design.

Our Vision: We envisage a world where solidarity, care, classlessness, diversity, self-reflection, and direct democracy are not just ideals, but the foundations of everyday life. Our goal is to make this vision a reality through the thoughtful application of open-source technology, where people remain in control.

Our Approach:

- Learning from History: Analyzing past and present democratic movements and self-managed collectives to guide institutional design.
- Emancipatory Modeling: Developing rigorous mathematical and computational models for equitable, non-market resource coordination.
- Open-Source Engineering: Turning theoretical models into verifiable, accessible, community-driven proofs of concept and protocols.

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Interdisciplinary Navigation Guide

How to Read This Book Across Disciplines

This monograph bridges libertarian-socialist institutional theory, formal welfare and planning economics, and modern distributed systems cryptography. Depending on your primary background, we recommend the following reading paths:

FOR ADVOCATES, COOPERATORS & ORGANIZERS — Institutional Self-Management Without Hierarchy

Start with Chapter 1 (Executive Summary) and Chapter 2 (Historical & Institutional Inspirations) to see how lessons from the 1936–1939 Spanish Revolution, the 1871 Paris Commune, Classical Athens, Occupy Wall Street, Mondragón, Porto Alegre Participatory Budgeting, Rojava, Elinor Ostrom's commons principles, the Russian factory committees (read through Maurice Brinton), the Hungarian workers' councils of 1956 (read with Cornelius Castoriadis), Yugoslav self-management, Chile 1970–1973, Argentina 2001, the Zapatistas, and the uprisings of 2010–2011 and Sudan's resistance committees are translated into concrete protocol safeguards against bureaucratic capture, meeting fatigue, and inter-regional inequality. Then read Chapter 3 for the complete traceability matrix linking Michael Albert and Robin Hahnel's 1991 and 2021 books to our design.

FOR ECONOMISTS & OPERATIONS RESEARCHERS — Formal Models, Accounting & Non-Convex Convergence

Focus on Chapter 4 (Formal Economic Model), which unites Albert and Hahnel's Formal Model of a Participatory Economy (FMPE) with Anders Sandström's double-entry Society Account and Robin Hahnel's Pollution Demand-Revealing Mechanism (PDRM). See Section 4.4 for our source-code audit of pequod-cljs and the four research-candidate solvers (Algorithms 3a–3d), alongside Chapter 8 and Chapter 10 for engagement with John O'Neill (2024a) and the non-convex experiments still required.

FOR SYSTEMS PROGRAMMERS & CRYPTOGRAPHERS — Local-First P2P, Two-Root Aggregation & Post-Quantum Security

Jump directly to Chapter 4 (Section 4.5 State Machine & Canonical CBOR Schemas), Chapter 5 (Four-Layer Architecture D & 6-Role Mobile Taxonomy), Chapter 6 (Weighted Stack Comparison), Chapter 7 (NIST FIPS 203/204/205, Sybil Resistance & Prio3 VDAF Aggregation), and Chapter 9 (Experimental Program, Kill Gates & Multi-Scale Estimates).

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Key Abbreviations

The thirty abbreviations used most often in this book. Every abbreviation is defined in Appendix B, and the core mathematical notation is summarized at the start of Chapter 4.

Abbreviation	Full Term	Definition & Role in Monograph
ADMM	Alternating Direction Method of Multipliers	Augmented-Lagrangian primal-dual method; Algorithm 3b is the proximal Jacobian variant of Deng et al. (2017), a research candidate.
BFT	Byzantine Fault Tolerance	Distributed consensus property guaranteeing safety even when up to $f < n/3$ validator nodes act maliciously.
BJC	Balanced Job Complex	Bundle of workplace tasks balanced for empowerment and desirability within and across workplaces (PEPE Ch. 2).
BLAKE3	BLAKE3 Cryptographic Hash	Hash function with 256-bit output and a 128-bit security target; used for the Two-Root aggregation trees and content addressing.
BLS	Boneh–Lynn–Shacham signatures	Pairing-based aggregatable signatures over BLS12-381; used in threshold randomness beacons and proposed for aggregated round checkpoints; not post-quantum.
C2SP	Community Cryptography Specification Project	Publishes the tlog-checkpoint, tlog-cosignature and tlog-witness specifications for witnessed transparency logs (Section 5.3.2).
CAP	Community of Affected Parties	Ecological/geographic collective of residents impacted by a specific pollutant in an airshed or watershed (DEP Ch. 7).
CBOR (CDE)	Concise Binary Object Representation	RFC 8949 Canonical Deterministic Encoding used for all signed DPEP messages with fixed 10^{-6} micro-unit scale.
CC (CC_h)	Consumers' Council	Democratic neighborhood consumer council or nested ward/city/regional consumer federation.
CRQC	Cryptographically Relevant Quantum Computer	Future fault-tolerant quantum computer capable of breaking classical elliptic-curve and RSA cryptography via Shor's algorithm.
DAP	Distributed Aggregation Protocol	IETF protocol for running VDAFs between clients and aggregators.
DEP	Democratic Economic Planning	Primary reference volume by Robin Hahnel et al. (Routledge, 2021); cited as Hahnel 2021.
DPEP	Distributed Participatory Economics Protocol	The open-source decentralized coordination protocol specified in this monograph.
FIPS	Federal Information Processing Standards	NIST post-quantum cryptographic standards finalized in August 2024: FIPS 203, FIPS 204, and FIPS 205.

Continued from previous page

Abbreviation	Full Term	Definition & Role in Monograph
IFB	Iteration Facilitation Board	Informational board/engine that aggregates supply and demand and updates indicative prices via public rules.
LF	Looking Forward	Primary reference volume by Michael Albert and Robin Hahnel (South End Press, 1991); cited as Albert and Hahnel 1991b.
LP / QP / MILP	Linear / Quadratic / Mixed-Integer Linear Programming	Mathematical optimization paradigms solved locally via Clarabel.rs and HiGHS.
ML-DSA-65	Module-Lattice Digital Signature Algorithm	NIST FIPS 204 (Dilithium3) post-quantum digital signature scheme combined in hybrid mode with Ed25519.
ML-KEM-768	Module-Lattice Key-Encapsulation Mechanism	NIST FIPS 203 (Kyber-768) post-quantum key-establishment scheme combined in hybrid mode with X25519.
NIST	National Institute of Standards and Technology	US standards body that published FIPS 203–205 (post-quantum cryptography).
Parecon / PE	Participatory Economics	Post-capitalist economic model developed by Michael Albert and Robin Hahnel.
PEPE	The Political Economy of Participatory Economics	Primary technical reference volume by Michael Albert and Robin Hahnel (Princeton University Press, 1991); cited as Albert and Hahnel 1991a.
Prio3 / VDAF	Verifiable Distributed Aggregation Function	IETF CFRG draft-irtf-cfrg-vdaf-22 (Informational). Privacy needs one honest aggregator; correct totals need all of them.
Role-LP / CC / SR / PCW / BFT / GW	DPEP Node Roles 1–6	Lightweight Participant (LP), Council Coordinator (CC), Storage Replica (SR), Planning Compute Worker (PCW), Epoch Validator (BFT), and Delay-Tolerant Gateway (GW).
SB / SC	Social Benefit / Social Cost	Ratio $SB_j(P)/SC_j(P)$ measuring the social value of a worker council's outputs relative to the opportunity cost of its inputs and endowments (≥ 1 constraint).
SLH-DSA	Stateless Hash-Based Digital Signature Algorithm	NIST FIPS 205 (SPHINCS+) signature standard relying solely on hash functions, used for long-term root anchors.
Two-Root	Two-Root BLAKE3 aggregation	Proposal Merkle root plus category-totals root (Section 5.3.1). Replaces internal-vector Sparse Merkle-Sum Trees.
WASM	WebAssembly	Deterministic, sandboxed binary instruction format (Wasmtime) executing IFB price-update rules.
WC (WC_j)	Workers' Council	Democratic workplace collective where each worker has one vote and holds a Balanced Job Complex.
X25519 / Ed25519	Curve25519 key exchange / signatures	Classical elliptic-curve halves of DPEP's hybrid schemes.

CHAPTER 1

Executive Summary: Feasibility, Architecture, and Strategic Roadmap

> No central agency composes the final plan. No market competition generates the final plan. A decentralized, social, iterative, communication process allows all actors to democratically formulate the final plan.
>
> — Michael Albert and Robin Hahnel, Looking Forward: Participatory Economics for the Twenty First Century (1991), Ch. 6[1]

1.1 What Is Being Proposed

This monograph sets out a source-grounded feasibility assessment, a gated research program, and a technical architecture proposal for the Distributed Participatory Economics Protocol (DPEP): an open-source, local-first, cryptographically verifiable coordination protocol intended to enable democratic worker councils, neighborhood consumer councils, and nested federations to formulate, iterate, ratify, and execute comprehensive economic plans without capitalist markets or authoritarian central planning.

The intellectual foundation of this protocol is Participatory Economics (Parecon), originally developed by Michael Albert and Robin Hahnel in The Political Economy of Participatory Economics (1991a) and Looking Forward (1991b), elaborated across daily-life institutions and critic rebuttals in Michael Albert's Parecon: Life After Capitalism (2003),[2] and mathematically extended in Robin Hahnel's Democratic Economic Planning (2021), Anders Sandström's Anarchist Accounting (2020; see also Sandström 2023), and Mitchell Szczepanczyk's computer simulation work (pequod-cljs, 2023).[3]

DPEP provides the computational substrate for four core institutional pillars:

1. Self-Managed Work and Consumption: Democratic worker councils (WC_j) and consumer councils (CC_h) in which participants exercise decision-making input proportional to the degree they are affected by an economic choice (Albert and Hahnel 1991a, chap. 1, 8, 16, and chap. 2, 24; Hahnel 2021, chap. 1, 14–16).

[1] Albert and Hahnel 1991b, chap. 6 ("Participatory Allocation"), accessed via the authorized ZNetwork online edition (https://znetwork.org/wp-content/uploads/zbooks/htdocs/ParEcon/lookingforward/lf6_1.htm).

[2] Albert 2003; full text archived at ZCommunications/ZNetwork (http://books.zcommunications.org/books/pareconv/).

[3] Szczepanczyk 2023; open-source repository at https://github.com/msszczep/pequod-cljs.

2. Equitable Remuneration and Balanced Job Complexes (BJCs): Non-transferable consumption allowances calibrated to the duration, intensity, and onerousness (effort and sacrifice) of socially valued labor, supplemented by universal free public goods and council-approved special-need allowances, paired with job complexes balanced for empowerment and desirability within and across workplaces (Albert and Hahnel 1991a, chap. 2, 25–30, and chap. 3, 44–55).

3. Decentralized Iterative Planning: A multi-round coordination protocol in which councils submit production and consumption proposals, federated aggregators compute excess demands across seven economic categories (private goods, public goods, intermediate inputs, natural resources, labor categories, capital stocks, and pollutant emissions), and a deterministic Iteration Facilitation Board (IFB) engine updates indicative social opportunity-cost prices until a feasible, ecologically bounded plan is ratified (Albert and Hahnel 1991a, chap. 4, 57–72, and chap. 5, 73–106; Hahnel 2021, chap. 7, 130–162).

4. Intra-Year Operational Clearing and Scoped Replanning: Continuous tracking of deliveries and inventory drawdowns against the ratified plan, absorbing routine consumer preference shifts locally within neighborhood and city slack buffers while mediating larger supply shocks across affected industry federations via Updating Facilitation Boards (UFBs).[4]

1.1.1 What a Demonstration Would Look Like

What follows describes a planned demonstration, not software that exists. A visitor opens an application representing a small economy of 20 workplaces and 20 consumer councils. Worker councils propose what they can produce and what they need; consumer councils propose what they want, including collective goods. Everyone sees where proposals exceed what is available, and councils revise, round by round, while the gap closes. A facilitation board computes new prices but cannot write or change any council's proposal. One council is then disconnected: it keeps working locally and, under rules it knew in advance, its proposal either counts or is carried forward. An operator who alters a council's signed proposal, or quietly leaves it out, is exposed by the council's receipt and the published totals. Finally, households submit consumption requests split between two independent operators: the neighborhood total is correct, and neither operator sees an individual request.

Each scene has a pre-registered pass and fail condition, and two of the claims behind them are still open: whether rounds converge with indivisible plants and price-sensitive incomes, and how to prove privately that each household request stays within budget. Section 9.2 sets out what each scene tests, in which phase it could be shown, and how it could fail.

[4] Albert and Hahnel 1991a, chap. 4, 66–67; Albert and Hahnel 1991b, chap. 6 (lf6_1.htm); Hahnel 2021, chap. 8, 167–170.

1.2 Why It Matters: Breaking the "No Third Way" Dogma

For over a century, mainstream and socialist economists alike have been haunted by Alec Nove's famous assertion in The Economics of Feasible Socialism (1983):

> In a complex industrial economy the interrelation between its parts can be based in principle either on freely chosen negotiated contracts [i.e., markets], or on a system of binding instructions from planning offices [i.e., command planning]. There is no third way.[5]

Yet historical experiments in democratic self-management—most notably the industrial and agrarian collectives of the Spanish Revolution (1936–1939), the Paris Commune (1871), federated cooperatives like Mondragón, and horizontal movements like Occupy Wall Street (2011)—repeatedly encountered two structural barriers when attempting to coordinate beyond single workplaces or rural villages:

- The Accounting and Valuation Barrier: Without a unified social accounting and indicative-price system, wealthier collectives hoarded superior inherited machinery (what Gaston Leval and Frank Mintz termed "collective egoism" or neocapitalismo colectivo in 1936 Barcelona; Leval 1975; Mintz 2013; Dolgoff 1974, chap. 6), while inter-industry coordination of complex intermediate inputs broke down into cumbersome bilateral barter.
- The Deliberative Scaling Barrier: Without structured, asynchronous computational protocols, horizontal deliberation tended toward exhausting multi-hour plenary meetings (speaking time grows at least linearly with attendance, and possible pairwise exchanges quadratically; this is arithmetic, not an empirical threshold, Section 2.10) or unaccountable informal cliques (Jo Freeman's "tyranny of structurelessness"; Freeman 1972; Graeber 2013).

Contemporary distributed computing, post-quantum cryptography, and mathematical optimization make it conceivable to attack both barriers at once. Whether millions of participants can coordinate industrial supply chains and ecological boundaries democratically, and at what wall-clock cost per round, is an experimental claim: Category B estimates in Table 1.1 and Table 9.4 are schema-derived, not measured at that scale.

1.3 Technical Feasibility: Demonstrated Results, Estimates, and Open Targets

A core mandate of this monograph is to separate what has been shown, what has been estimated, and what is unknown.

[5] Nove 1983, ix–x, 44; quoted directly at the opening of chapter 4 in Albert and Hahnel 1991a, 57.

What is established, estimated, and unknown (plain statement). Established: the 1991 model and its later formalizations exist and are internally coherent at the level of accounting identities; the pequod-cljs Cobb–Douglas simulations reach a 5% imbalance in about 12 rounds (fixed incomes, smooth convex technology); the unmodified price rule fails on Scarf's cycle, where incomes move with prices, and on a single indivisible plant, while with fixed budgets it clears Scarf's Leontief consumers; the NIST post-quantum primitives are standardized, and every function in the stack except threshold signatures, anonymous credentials, the sortition beacon, capability-certificate formats, and post-quantum HPKE has a standardized, deployed primitive (Table 6.3), although almost every way DPEP combines them is original; Prio3SumVec report sizes and desktop timings have been measured with libprio; and signature verification has been measured on a desktop, where even 200,000 hybrid verifications take about 10 s on one core. Estimated (not measured at scale): round latency, archive sizes, mobile bandwidth, hosting cost, and aggregator CPU at 10^6 participants. Unknown: whether any candidate solver clears economies with Leontief inputs, lumpy plants, endogenous incomes, and strategic reporting; whether ordinary people can and will take part in multi-round planning without a technical elite forming around the software; how household budgets can be checked privately when prices change between rounds; whether a credential scheme can enforce one report per person without a population register; and how consumption at the point of use (intra-year fulfillment, state S7) can be kept private and protected against double spending while offline. The household privacy claim of Chapter 7 therefore covers annual requests, not what a person collects at a depot (Section 7.5). Phase 1 and Phase 1H (Chapter 9) are designed to settle the first two unknowns, or to stop the program if they cannot be settled.

1.4 The Most Promising Architecture: Why We Reject a Global Blockchain

We explicitly reject building a per-transaction global blockchain (Architecture C), a centralized cloud database (Architecture A), or an unstructured gossip network without finality (Architecture B).

Participatory planning is not a stream of high-frequency financial trades; it is an epoch-synchronized, hierarchical bulk-synchronous coordination process followed by localized intra-year fulfillment. Furthermore, putting personal consumption requests on a public ledger violates personal privacy, and broadcasting 3.3 KB post-quantum signatures for every household action across all global nodes wastes bandwidth and energy.

Instead, our weighted evaluation (Chapter 6: 4.07 / 5.00 for D with BFT checkpoints and 4.10 with a witnessed log, against 3.10 for a federated SQL service

[6] Rows 1 and 2 of Category A are computed by research/analysis/verify_r2.py (Scarf's cycle by research/analysis/verify_economics.py) in the project repository.

[7] Hurwicz 1972, 1973.

Table 1.1: Three-tier evidence classification: demonstrated results, analytical estimates, and research targets

Category A: Empirically Demonstrated (Code / Data Audited)	Category B: Analytical Estimates (Schema / Complexity Models)	Category C: Open Research Targets (Require Phase 1–3 Validation)
1. Smooth Cobb-Douglas Convergence (pequod-cljs CSVs): Across the 40 large-run logs the book cites (dep1ex61–100.csv),[6] Year 1 reaches $\rho_i \leq 5\%$ in 11.85 rounds and $\rho_i \leq 3\%$ in 19.20 rounds; Year 2 reaches 5% in 6.58 rounds. The 50 small-run files (dep1ex01–50.csv) average 11.18 Year-1 5% rounds. All 70 dep1ex*.clj generators (20 stored via Git LFS) were audited.	1. Distributed Round Latency ($N = 10^6$): A round moves a 0.10–1.45 GB raw archive of 40k council proposals (Table 5.2; the lower figures use classical council signatures) plus a Layer-1 checkpoint. Verifying 10^6 household reports costs each aggregator ≈ 4 (coarse) to ≈ 61 (dense) desktop core-minutes; verifying 40k–200k council signatures takes 2–10 s on one desktop core even when every signature is hybrid (measured on an Apple M4 Pro; Raspberry Pi and community-server timings are a Kill Gate 2 measurement). No round-latency figure is claimed; Kill Gate 2 measures it against a pre-registered threshold.	1. Non-Convex & Leontief Convergence (Kill Gate 1): Whether any of Algorithms 3a–3d clears (not merely satisfies one-sided feasibility; Definition 4.2) Leontief + lumpy setup-cost economies in ≤ 25 rounds on pre-registered instances, including Scarf cycles, infeasible plans, and endogenous income.
2. Limits of Unregularized Tâtonnement (Scarf & Single-Plant Jumps): Fails Scarf's 3-good cycle with endowment incomes ($\min \rho = 10.5\%$ over 3,000 rounds; with fixed budgets the same consumers clear to $< 0.02\%$) and single indivisible plants ($\min \rho = 50.4\%$).	2. One-Good Shapley–Folkman Stress Test ($K = 20$): The minimum imbalance across 500 rounds is 0.40%, but the maximum in the last 100 rounds is 8.4% at $a = 70$; a sweep $a \in [60, 80]$ has median last-100-round max 6.0% and worst 9.5%. This says nothing about interdependent Leontief markets.	2. Human Override & Agency (Kill Gate 1H): Whether 30–60 self-selected volunteers modify solver drafts, and what that can and cannot show about democratic agency.
3. NIST PQC Standards and measured Prio3 sizes: FIPS 203/204/205 are standardized. draft-irtf-cfrg-vdaf-22 is an Informational IRTF draft with no post-quantum claim. Prio3SumVec sizes and desktop timings were measured with libprio 0.18.1 (Table 7.6): a dense 3,000-category, 20-bit report is 975,760 B to the leader and costs ≈ 3.5 ms to verify per aggregator on one desktop core.	3. Mobile Round Bandwidth (Role-LP): Checking a checkpoint's 86 quorum signatures costs ≈ 290 kB if they are hybrid ($86 \times 3{,}373$ B), 5.5 kB if they are Ed25519, and about 0.1 kB as one aggregated BLS signature with an engine or log that aggregates signatures (CometBFT as configured does not, and gives 5.5 kB with Ed25519). Under the proposed tiered signature policy every phone can check every checkpoint within ≈ 46 kB per automated round and ≈ 101 kB per milestone round, including the coarse 55 kB household report (Table 7.10).	3. Incentive & Soft-Budget Resistance: Mitigating capacity understatement (Weitzman 1980), soft-budget appeals (Kornai 1980), and carousel fraud (Hurwicz 1972) remains an open mechanism-design problem.[7]

and 2.54 for a blockchain; expert judgements, not measurements, and A overtakes D with BFT checkpoints when simplicity, throughput, and cost carry 63% of the weight) selects Architecture D—A Four-Layer Hybrid Local-First, Federated P2P, and Selective Epoch-BFT Architecture:

1. Layer 4 (Human Deliberation & Local Solver Edge): Native Android and iOS clients (Kotlin and Swift interfaces over the shared Rust core through UniFFI) as the primary participant client, with a web client for read-only exploration, accessibility, and the Phase 1H probe (proposed; Section 5.5), running an embedded interior-point conic/quadratic solver (Clarabel.rs / HiGHS) for local proposal drafting and "what-if" collective-good exploration.

2. Layer 3 (Local-First Council & Household State): Encrypted local SQLite (SQLCipher) for structured vectors and accounting ledgers + Loro/Automerge CRDTs for offline-capable workplace and neighborhood council deliberation, authorized via revocable capability certificates (CapCert; format to be chosen, signed under the active CryptoSuiteId).

3. Layer 2 (Federated Aggregation & Content-Addressed Data Plane): Peer-to-peer QUIC (quinn / iroh) transport organizing councils into nested Ward → City → Region and Industry Federation Two-Root BLAKE3 authenticated aggregation (proposal Merkle root + category-totals root; Section 5.3.1), paired with content-addressed storage for qualitative council narratives (Lists).

4. Layer 1 (Selective Checkpoint & Deterministic IFB Engine): Either a witnessed transparency log, in which sortitioned witnesses cosign each round's checkpoint after recomputing it, or a rotating BFT committee; Kill Gate 2 tests both (Section 5.3.2). The BFT form is a committee of $n = 128$ validators (drawn by sortition; an interim classical threshold beacon is adopted and a post-quantum beacon is open, Sections 7.2.1 and 7.7) that commits one checkpoint per planning round, plus daily operational rollups and scoped replans—425 blocks/year at a 10-round plan $(10 + 365 + 50)$, or 435 at 20 rounds—anchoring the two aggregation roots, executing the deterministic WASM price-update rule, and settling Sandström's macro Society Account. "10–20 checkpoint blocks" refers only to the annual planning rounds, not the Layer-1 archive.

The architecture is decentralized, trust-minimized, censorship-resistant, and open source because participatory planning must work among people as they are, protect councils and households from surveillance and from capture within the federation, and leave no single operator able to seize or quietly edit the plan. Section 5.1 gives the argument and the limits of each property.

1.5 Where AI Adds Value—and Where It Must Be Excluded

In strict adherence to Parecon's anti-coordinator principles,[8] AI must never become an unaccountable planning authority:

- Deterministic Core (by design, not yet implemented): Indicative price adjustments, SB/SC ratios, and Society Account clearing are specified as deterministic fixed-point arithmetic and convex operations-research solvers with no neural-network weights. Local IEEE-754 drafts are advisory only (Section 4.5).
- Accountable Edge Support: Classical statistical models (hierarchical ETS/ARIMA) forecast SKU size/color splits; local conic QP solvers (Clarabel.rs) automatically draft price-adjusted council revisions; and optional small local open-weight LLMs are restricted to multi-lingual translation, screen-reader accessibility, and summarizing qualitative Lists with mandatory links to raw council source text.

1.6 Estimated Effort, Budget Scenarios, and the Five Key Decisions

The Five Most Important Decisions for Project Leadership

1. Treat Architecture D as the leading long-term hypothesis (local-first + Two-Root aggregation + epoch checkpoints) and reject a per-transaction blockchain. Use Architecture A (a disclosed trusted server) for Phase 1H and the earliest pilots: it is simpler to operate and explain, and it wins whenever simplicity, throughput, and cost dominate the weighting (Section 6.1.1).
2. Replace unconstrained Cobb-Douglas with constrained local solvers, and treat Algorithms 3a–3c as research candidates to be benchmarked in Phase 1 rather than as a committed production solver.
3. Enforce Hard Biophysical Caps ($r^*, E_z^{\max}$) Lexicographically Above PDRM Pollution Pricing.
4. Design for a "3-Milestone Hybrid Deliberation Schedule" separating automated intermediate input-clearing sub-iterations from human value-deliberation milestones.
5. Target Dual Utility: Support both full-scale Parecon simulation research and immediate deployment for federated worker cooperatives and participatory budgeting communities ("Experiment 2: Developing a Parallel Economy"; Albert and Hahnel 1991a, chap. 6, 129–130).

Final Recommendation: PROCEED WITH PHASE 1 AND PHASE 1H AS KILL GATES—the Python-first Economic Coordination & Non-Convexity Simulator (Phase 1) and the Low-Technology Human Participation Probe (Phase 1H)

[8] Albert and Hahnel 1991a, chap. 1, 15–17; Albert and Hahnel 1991b, chap. 9 (lf9_2.htm) and chap. 10 (lf10_1.htm).

Table 1.2: Proposed budget scenarios at one assumed loaded rate (for author ratification)

Phase	Staff & Time	Non-labor	Estimate & Gate
Phase 1 Economic simulator	1.0 FTE 4–6 months	Compute $1k–$3k	$41k–$93k. Pre-registration in month 1. Kill Gate 1.
Phase 1H Low-tech human probe	1.0 FTE 3–4 months (in parallel)	Ethics review, consent, compensation, data protection: $8k–$25k	$38k–$85k. Kill Gate 1H.
Phase 2 Distributed coordination	2.5 FTE 6–8 months	WAN cluster $2k–$5k	$152k–$305k, including a model-checked formal specification written before Rust work ($20k–$45k). Kill Gate 2.
Phase 3 PQC, Prio3, native clients, co-op pilot	4.5 FTE 9–12 months	Audit $60k–$180k (including fulfillment and clients); hosting $10k–$40k; pilot ethics and data protection $20k–$40k	$495k–$1.07M for Phase 3 alone, with two aggregators and an existing DAP server. A three-aggregator protocol of DPEP's own adds $40k–$95k. Kill Gate 3.
Program			$0.73M–$1.55M if every gate passes.

Basis: an assumed fully loaded cost of $120k–$180k per FTE-year (low end of each range uses the shorter duration and lower rate). This is a planning assumption, not a sourced market rate. Figures from research/analysis/verify_r3.py. Two to three months is not realistic for building a simulator, pre-registering instances, and implementing an endogenous-income solver, so Phase 1 is planned at 4–6 months.

first, each with pre-registered pass and fail conditions (Table 9.2), holding Phase 2 (distributed Rust nodes) and Phase 3 (post-quantum and Prio3 field pilots, with a mandatory $60k–$180k external audit) contingent on passing Kill Gates 1 and 1H. A failed gate stops or narrows the program; it is not reported as a partial success.

CHAPTER 2

Historical and Institutional Inspirations: From Council Practice to Protocol Design

> In almost all the industries, factories, mills, workshops, transportation services, public services, and utilities, the rank and file workers, their revolutionary committees, and their syndicates reorganized and administered production, distribution, and public services without capitalists, high salaried managers, or the authority of the state… They instituted not bourgeois formal democracy but genuine grass roots functional libertarian democracy, where each individual participated directly in the revolutionary reorganization of social life.
>
> — Gaston Leval, Espagne Libertaire (1971), p. 11, translated in Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (1974), Ch. 1, p. 6[1]

2.1 Epistemological Premise: Institutions Before Cryptography

A recurring failure mode of contemporary "decentralized governance" (DAO) architectures is the conflation of cryptographic trustlessness with democratic self-management. Recording a tally on an immutable ledger does not render an economy democratic if:

1. Proposal drafting and agenda-setting are monopolized by an unaccountable technical or managerial elite—what Albert and Hahnel diagnose as the coordinator class.[2]
2. Decision-making influence is proportional to token wealth or meeting endurance rather than proportional to the degree participants are affected.
3. Wealthier production collectives hoard superior inherited productive endowments and refuse to equalize resources across regions.

This chapter reads historical and contemporary experiences from one standpoint: what people built for themselves when they took decisions into their own hands, what destroyed or captured it, and what must be carried forward so that the impulse to freedom, self-management, and solidarity ends in direct democracy rather than in better government by others. Reforms granted by rulers are examined only for what they teach about that impulse. The cases fall into two groups, each in date order, so that each experience can be read against those it

[1] Leval 1971, 11; verbatim English translation in Dolgoff 1974, chap. 1, 6.

[2] Albert and Hahnel 1991a, chap. 1, 14–17, 20–22; Albert and Hahnel 1991b, prologue ("Coordinatorism" and "The Origins of Coordinatorism").

learned from. The first is self-management built from below: classical Athens as the ancient antecedent, with its exclusions; the Paris Commune; the Russian factory committees and soviets read through Maurice Brinton; the Spanish Revolution; the Hungarian councils of 1956 read with Cornelius Castoriadis's design for a self-managed economy; Chile 1970–1973; the Zapatistas; Argentina 2001–2003; the uprisings of 2010–2011, Occupy, and their successors in Sudan; and Rojava (Sections 2.2–2.11). The second is self-management granted from above, together with institutions that have lasted inside market and state economies: Yugoslavia, Porto Alegre, Kerala, Ireland's Citizens' Assembly, Decidim, Mondragón and Ostrom's commons (Section 2.12). The cases are not ordered by lesson, because most teach several; the synthesis does that, mapping mechanisms to protocol capabilities (Table 2.2) and lessons to requirements (Table 2.3). Section 2.13.2 draws the enduring lessons and the binding architectural rules that follow from them. Section 5.1 states why DPEP nonetheless adopts decentralization, trust minimization, censorship resistance, open source, tamper-evident history, and disintermediation, and where each of them stops.

2.2 Classical Athenian Democracy (∼508/7–322 BCE)

Over nearly two centuries from the reforms of Kleisthenes (508/7 BCE) to the Macedonian conquest (322 BCE), classical Athens developed the most sophisticated institutional machinery for direct citizen self-governance in antiquity, analyzed definitively by Mogens Herman Hansen (1991).

2.2.1 Core Institutional Mechanisms Adaptable to DPEP

1. Direct Assembly (Ekklesia) vs. Norm-Setting (Nomothetai): Open to all adult male citizens (∼30,000–40,000 eligible in the 4th century BCE, with ∼6,000 attending the 40 annual meetings on the Pnyx), the Ekklesia voted on decrees (psephismata), while permanent constitutional laws (nomoi) were ratified by a separate panel of 1,000 nomothetai selected by lot.

2. Sortition (Kleroterion) and Strict Rotation: Roughly 600 of the ∼700 domestic magistrates (archai) and the 6,000 annual jurors (dikastai) were selected by lot using bronze allotment machines (kleroteria) rather than election—following Aristotle's principle (Politics 1294b) that elections are intrinsically oligarchic (favoring the wealthy, famous, and rhetorically polished) whereas sortition is democratic (Hansen 1991, chap. 9, 230–245). Almost all offices were restricted to a non-repeatable one-year term (or repeatable only once in a lifetime for the Boule), preventing a permanent coordinator class.

3. Probouleutic Agenda Facilitation (Boule of 500): Five hundred citizens (50 from each of the 10 Kleisthenic tribes, chosen by lot from local demes) served for one year on the Council (Boule), with each tribe serving as the standing executive committee (prytaneis) for 1/10th of the year and a new chair

(epistates) drawn by lot every 24 hours. Crucially, the Boule's role was probouleutic: it prepared structured proposals (probouleumata) for the Ekklesia to debate, amend, or reject, but could not impose major policies on its own (Hansen 1991, chap. 10, 246–265). This is the closest historical analogy (not a direct institutional ancestor) of Parecon's Facilitation Boards.

4. Public Scrutiny and Mandatory Audit (Dokimasia, Euthynai, Graphe Paranomon): Every magistrate underwent pre-office eligibility scrutiny (dokimasia) and a mandatory end-of-term financial and conduct audit (euthynai) before 10 logistai (accountants) and 10 euthynoi chosen by lot (Hansen 1991, chap. 9, 218–224).

5. Civic Remuneration (Misthophoria): Introduced by Pericles for jurors and councilors and expanded in the 4th century BCE to assembly attendance (misthos ekklesiastikos, 3 to 6 obols per day), ensuring working citizens (thetes) could afford to participate.

2.2.2 Explicit Examination of Structural Exclusions

Athenian democracy excluded women (who lacked political citizenship and independent property control), metics (resident non-citizens, ∼20,000–30,000 artisans and merchants), and enslaved people (∼80,000–150,000 forced laborers in the Laurion silver mines, agriculture, crafts, and households), leaving only ∼10–15% of the total population with full political rights (Hansen 1991, chap. 4, 86–124). Furthermore, as Moses Finley and G. E. M. de Ste. Croix demonstrate, Athens' fiscal capacity to fund misthophoria and public works depended heavily on the Laurion silver mines (worked by tens of thousands of enslaved miners) and 5th-century Delian League imperial tribute (phoros) (Finley 1981, chap. 5; Ste. Croix 1981). In a universal, non-extractive participatory economy, every hour spent in deliberation and euthynai audits carries a real opportunity cost in productive labor; governance time must therefore be budgeted inside paid Balanced Job Complexes and economized carefully, while sortition should draw from all citizens regardless of hardware ownership (the beacon design is open; Section 7.7).

2.3 The Paris Commune (March 18–May 28, 1871)

During its 72 days of existence before its bloody suppression during La Semaine Sanglante, the Paris Commune established foundational principles of council democracy celebrated in Karl Marx's The Civil War in France (1871; see also Merriman 2014). A rigorous institutional analysis must distinguish what the Commune actually implemented from unimplemented proposals and later theoretical extrapolations.

[3] Marx 1871, sec. III.
[4] Merriman 2014, chap. 5.

Table 2.1: Paris Commune (1871): institutions actually implemented vs. proposals and extrapolations

Dimension	Actually Implemented (March–May 1871)	Proposed on Paper or Later Extrapolation
Representation & Recall	Municipal elections across 20 arrondissements (March 26, 1871) by universal male suffrage; councilors declared revocable at any time; merged legislative and executive work across 10 Commissions.[3]	Formal recall was rarely exercised during the 72-day military siege; women were excluded from formal voting despite leading grass-roots defense and cooperative organizing via the Union des Femmes (Élisabeth Dmitrieff, Nathalie Lemel).
Remuneration Cap	Decree of April 2, 1871: Capped maximum salaries of all Commune officials and municipal administrators at 6,000 francs/year (equivalent to a skilled Parisian workman's wage) and banned multiple stipends.	Did not abolish wage hierarchies inside private Parisian enterprises or establish balanced job complexes inside workshops.
Workplace Self-Management	Decree of April 16, 1871 (drafted by Léo Frankel's Commission of Labor and Exchange): Ordered a census of workshops abandoned by fleeing owners and their transfer to worker cooperative associations; abolished night baking (April 20) and employer wage fines (April 27).[4]	Only a few dozen cooperative workshops (e.g., the Louvre armaments workshop electing its own council and foremen; women's uniform cooperatives) actually began operating before Versailles troops entered Paris on May 21.
National Federation & Credit	Operated strictly within besieged Paris; left the Banque de France intact (negotiating advances via Charles Beslay rather than socializing credit).	Declaration to the French People (April 19, 1871) (drafted by Proudhonian Pierre Denis) and Marx envisioned a national federation of autonomous communes coordinating via delegated national assemblies—never realized due to the crushing of provincial communes in Lyon and Marseille.
Crisis Centralization	On May 1, 1871, under bombardment, the Jacobin/Blanquist majority voted (45 to 23) to create a 5-member Committee of Public Safety with sweeping powers over the objection of the Federalist minority (Frankel, Varlin, Longuet, Courbet).	Demonstrates the acute vulnerability of council democracies during crisis to revert to emergency command authority unless crisis-mode bounds (TTLs and post-incident audits) are constitutionally hard-coded.

2.4 Russia 1917–1921: Factory Committees, Soviets, and the Question of Who Manages

What people built. In 1917 Russian workers created factory committees in the workplaces and soviets (councils of delegates) in the towns: organs that, in Maurice Brinton's words, "might have ensured the management of society by the workers themselves." The First All-Russian Conference of Factory Committees met on October 17–22, 1917.[5]

How it was captured. Brinton documents, step by step, how these organs were absorbed. The decree of November 14, 1917, confined the committees to "control" (supervision) rather than management. A decree of December 5, 1917, set up a Supreme Economic Council (Vesenkha), attached to the Council of People's Commissars, to work out "a plan for the organization of the economic life of the country" and to direct all economic authorities, including the workers' control bodies. In January 1918 the First Trade Union Congress resolved to turn the factory committees into union organs. By late 1920, of 2,051 important enterprises for which data were available, 1,783 were under "one-man management." The Kronstadt rising of March 1921, whose sailors declared that they fought "for the genuine power of the working people" rather than for the power of the Party, was crushed by the Red Army (Brinton 1970, chronology entries for November 14 and December 5, 1917, January 1918, 1920, and 1921).

Brinton's analysis. Brinton's central claim is that property relations and relations of production are different things. The revolution expropriated the owners but left the "order-giving, order-taking" relations of production intact, and the means of production can pass from private hands to a bureaucracy "without this revolutionizing the relations of production." He therefore put one question at the center of any serious discussion of socialism: "who manages production after the overthrow of the bourgeoisie?" (Brinton 1970, introduction). This is the historical record behind Albert and Hahnel's coordinator class (Section 2.1). Managers and technical experts, many from the old order, returned "by the back door" as managers of the labor process in the new state, and the party claimed to exercise workers' power on the workers' behalf.

Lesson carried forward. Nationalization, a workers' party in office, or a plan drawn up on workers' behalf does not change who manages. Self-management exists only where producers and consumers themselves draft, revise, and decide, and every body above them is their instrument. In DPEP, proposals originate only in councils. No federation, facilitation board, or validator committee can draft a council's proposal, alter it, or substitute its own, and the price-update rule is ratified, published, and recomputable by anyone (Sections 5.3.1 and 5.1).

[5] Brinton 1970, introduction and chronology for 1917; reprinted in Brinton 2004. The 1970 text was read in full; Brinton (2004) is cited at bibliographic level.

2.5 The Spanish Revolution (1936–1939): Industrial and Agrarian Collectives

Following the defeat of the Nationalist military coup in July 1936 across Catalonia, Aragon, Valencia, and parts of Castile and Andalusia, workers and peasants affiliated with the anarchist CNT (Confederación Nacional del Trabajo) and socialist UGT (Unión General de Trabajadores) collectivized a large share of Catalan industry (figures of 70–80% are commonly repeated in the secondary literature; this monograph has not traced them to a primary count) and several hundred agrarian collectives in Aragon. Estimates of participation vary widely and come mostly from participants and sympathizers: Frank Mintz gives a minimum of about 1.84 million collectivists (1.08 million industrial, 0.76 million agrarian), revised downward in 1977 from his earlier 2.44–3.2 million, while the eyewitness and CNT militant Gaston Leval gave figures of up to 3 million. Mintz's figures are cited here from the bibliographic record and were not checked against the volume.[6]

2.5.1 Internal Self-Management, Delegation, and Recall

Across both Barcelona factories (tramways, textiles, metalworks, water/gas/electric utilities) and Aragonese villages (Alcañiz, Calanda, Binéfar, Graus, Mas de las Matas), supreme authority resided in the weekly or biweekly General Assembly (Asamblea General) of all workers or villagers (Dolgoff 1974, chaps. 2 and 9; Bookchin 1974, xxvii–xxx). Assemblies elected specialized administrative committees (Comités de Administración) to manage daily coordination, input procurement, and statistical reporting. As documented by Bookchin, Peirats, and Leval, these delegates operated under strict anti-bureaucratic constraints:

> Where such popular assemblies were possible, their decisions were executed by a committee elected from the assembly. Apparently, the right to recall committee members was taken for granted and they certainly enjoyed no privileges, emoluments, or institutional power… It remained a cardinal principle of Spanish anarchists never to pay their delegates, even when the CNT numbered a million members. Normally, the responsibilities of elected delegates had to be discharged after working hours. (Bookchin 1974, xxviii)

2.5.2 Inter-Collective Coordination, Reorganization, and Accounting

- Regional Equalization in Aragon: The Regional Defense Council of Aragon was created at a plenary of unions and militia columns in Bujaraloz on October 6, 1936. The Aragon Federation of Collectives was constituted later, at the Caspe congress of February 14–15, 1937. Accounts of its size differ by a factor of five, and all come from sympathetic sources: Leval, who attended, lists 24 cantonal federations covering 275 villages and 141,430 member families;

[6] Mintz 2013; Leval 1975; Dolgoff 1974, available in full at https://theanarchistlibrary.org/library/sam-dolgoff-editor-the-anarchist-collectives.

Peirats, citing Souchy, gives 500 collectives with 433,000 members; a widely reproduced summary gives 500 delegates representing 80,000 collectivists. The federation pooled surpluses at cantonal depots, moved machinery and seed between villages, and traded directly with Catalan syndicates. The Council was dissolved by decree on August 10–11, 1937, after Republican troops under Líster entered Aragon.[7]

- Unified Export Coordination in Valencia (CLUEA): In the Levante, the CNT and UGT established the Consejo Levantino Unificado de Exportación Agrícola (CLUEA), federating ∼270 local orange and citrus committees to standardize grading, rail/maritime transport, and export clearing (Mintz 2013, chap. 4).
- Industrial Recomposition in Catalonia: In Barcelona, the CNT's industrial unions (Sindicatos Únicos de Ramo) consolidated hundreds of small, unsanitary workshops in woodworking, baking, tanning, and optic/glass production into modern integrated plants, eliminating redundant middlemen while preserving full employment (Souchy 1974).
- Accounting and the "Family Wage": Many Aragonese villages abolished official pesetas internally, distributing abundant staples free according to need (Maxim 4) and issuing a family consumption booklet (libreta de consumo) or local points calibrated to family size for scarce goods (Leval 1975, 165–210). Inter-collective trade relied on standardized valuation tables (indexed in wheat equivalents or shadow pesetas) cleared through cantonal ledgers.

2.5.3 Historical Limitations and Structural Failures

Despite remarkable wartime productivity in munitions, transport, and agriculture, the Spanish collectives suffered four critical weaknesses that directly inform DPEP's design:

1. "Collective Egoism" (Neocapitalismo Colectivo) Between Rich and Poor Enterprises: In urban Catalonia, prior to the socialization decrees of late 1937, collectives remained financially autonomous inside a quasi-market environment. Wealthy collectives that inherited modern machinery or lucrative monopolies (such as the Barcelona tramways, power utilities, and luxury hotels) enjoyed large surpluses, while raw-material-starved textile and construction collectives struggled to pay base wages (Peirats 1974, chap. 6; Mintz 2013). Protocol Lesson: This failure is why Albert and Hahnel (1991a, chap. 5, 84–87) and Sandström (2020) insist that productive assets belong to the Productive Commons, and that every Worker Council be charged an opportunity-cost rental fee ($P_k \cdot k_j$) on its inherited capital stock.

[7] Leval 1975, chap. V (Caspe congress); Peirats 1974, chap. 9 (giving Souchy's estimate); the 80,000 figure appears without a source in the Wikipedia article "Regional Defence Council of Aragon" and is reported only as a range bound. The October 1936 date belongs to the Council, not the Federation.

2. Burnout from Unpaid Evening Governance: Requiring delegates to perform all federation administration after a full day of physical factory labor caused severe exhaustion and favored childless militants over workers with caregiving duties. Protocol Lesson: Governance and facilitation work must be counted inside paid work hours as part of every citizen's Balanced Job Complex.
3. Lack of Unified Shadow Pricing for Complex Intermediate Chains: While cantonal barter cleared simple agricultural staples, urban industries lacked an iterative mechanism to compute social opportunity costs across hundreds of interdependent intermediate inputs once pre-war market prices grew stale.
4. Wartime militia coercion, smallholder conflict, and "workers against work" (Casanova 2005; Seidman 1991): Internal weaknesses were not merely accounting bugs. Casanova (2005) and Seidman (1991, 2002) are the standard critical accounts.[8] These sources describe Aragonese collectives formed under armed militia columns, conflict with "individualist" smallholders, absenteeism in Barcelona factories, credit starvation by the central government, dissolution in Aragon by Líster's 11th Division in August 1937, and Franco's victory in 1939.

2.6 Hungary 1956 and Castoriadis's Self-Managed Society

What people built. During the revolution that began on October 23, 1956, workers elected councils in factories and mines across Hungary. After Soviet troops returned on November 4, the councils federated. The Central Workers' Council of Greater Budapest, formed in mid-November, coordinated the general strike and negotiated with the Kádár government as the effective voice of the working population.[9]

How it was destroyed. On December 9, 1956, the government declared martial law and dissolved all regional and central workers' councils, leaving only those in factories and mines. On December 11 the Central Council's chairman, Sándor Rácz, and its secretary, Sándor Báli, were arrested when they arrived at the government building for talks. The council's 48-hour general strike began the same day with near-unanimous support. In September 1957 the government announced that the remaining councils would be replaced by works councils "under the leadership of the trade unions," and on November 17, 1957, that all remaining workers' councils were abolished.[10]

[8] Neither was read in full for this study. What follows draws on summaries of their work and on other secondary sources, so no page numbers are cited.

[9] Anderson 1964, chapter "The Proletariat Crushed" (read in full); Lomax 1976, cited at bibliographic level. Accounts differ on the exact day the Central Council was formed.

[10] Anderson 1964, "The Proletariat Crushed."

Castoriadis's design. Writing in 1957, just after the Hungarian revolution, Cornelius Castoriadis took the Russian and Hungarian experience as the starting point for a design of a self-managed economy.[11] Four of his conclusions bear directly on DPEP.

1. Direct democracy is the only total form of democracy. "To decide means to decide for oneself. To decide who is to decide is already not quite deciding for oneself." A federation or central assembly is justified only as an instrument of the councils' power, not a delegation of it.
2. Decisions need "full knowledge of the relevant facts." Information must flow in both directions. A center that only receives information and only sends out decisions holds a monopoly of the conditions for exercising power.
3. The "plan factory" computes but does not decide. A highly automated technical body using input–output methods would regroup the councils' proposals, show which are mutually incompatible, and present alternative plans with their consequences for each group (working time, consumption, investment, future growth) before a vote. "The role of the 'plan factory' won't be to decide on the plan," and its staff would not "decide or impose anything—except the organization of their own work." His procedure (proposals from the councils, regrouping, counter-proposals, and repetition, then a vote in the general assemblies) anticipates the iterative structure of participatory planning by more than three decades.
4. Fragmenting power does not abolish the center. Splitting a vital function "only creates an enhanced need for some other institution to reassemble the fragments," and lets a masked, uncontrollable center re-emerge. Central functions must exist, be defined and circumscribed, and be controlled from below.

Castoriadis also proposed absolute wage equality, whereas participatory economics remunerates effort and sacrifice (Section 3.2). This book records the disagreement without resolving it.

Lesson carried forward. DPEP's iteration facilitation board and its deterministic price-update engine are Castoriadis's plan factory, and his rule binds them: they compute and publish, and they never decide. Every council can see the aggregate state, and the consequences of the alternatives, before it votes. Councils formed in a crisis must be able to federate within days without permission from any authority. The first move against them, in 1918 and in 1956 alike, was to cut the federating level off from the base. That is why DPEP keeps each council's state on its own devices and lets councils federate again if a federation is seized (Section 5.1).

[11] Castoriadis (1957) 1972, §§ 3, 7, and 9 and the translators' preface (read in full); also in Castoriadis 1988.

2.7 Chile 1970–1973: Cordones Industriales and Project Cybersyn

What people built. Under the Popular Unity government, workers occupied factories and, from mid-1972, formed cordones industriales: territorial bodies coordinating the workplaces of an industrial district. During the employers' and lorry owners' strike of October 1972, workers kept factories running and organized distribution themselves.[12]

Cybersyn. Separately, the government built Project Cybersyn with Stafford Beer: a telex network linking state enterprises to a central operations room, designed on Beer's Viable System Model to preserve each unit's autonomy. During the October strike its network was used to coordinate the movement of supplies with the few lorries still running. Medina argues that the project embodied an unresolved tension: a system meant to support worker participation could equally serve central control, and worker participation in the project remained limited in practice.[13]

How it was destroyed. The military coup of September 11, 1973, ended both.

Lesson carried forward. Information infrastructure is not neutral. The same network can extend the reach of councils or of a central office, depending on who sets its agenda, who sees its data, and who can switch it off. The cordones show again that the organs of self-management appear from below and in crisis; Cybersyn shows that the computing built for them must belong to them. DPEP's answer is council state kept locally, aggregates that anyone can check, and no central operations room.

2.8 The Zapatistas (1994–): Lead by Obeying

What people built. Since the uprising of January 1, 1994, in Chiapas, Zapatista communities have built autonomous municipalities and, from 2003, regional Good Government Councils (Juntas de Buen Gobierno) based in the caracoles. Offices are unpaid and rotating, and those who hold them are expected to "lead by obeying" (mandar obedeciendo) the assemblies. The Women's Revolutionary Law, published in January 1994, set out women's rights to take part in the struggle and hold office, to choose their partners and the number of children they have, and to be free from violence.[14]

Self-correction. At the end of 2023 the EZLN dissolved the autonomous municipalities and the Good Government Councils. Its communiqués gave two reasons: the growing presence of organized crime, and a self-critique that the structure

[12] Winn 1986, cited at bibliographic level.
[13] Medina 2011, cited at bibliographic level.
[14] Mora 2017; Klein 2015; both cited at bibliographic level.

had become pyramidal, with authorities deciding for themselves and proposals no longer flowing between them and the communities. The new base is the Local Autonomous Government in each community, subject to its assembly. Regional collectives and zone assemblies above it meet and act only at the request of the local governments and remain under their command.[15]

Lesson carried forward. Rotation and unpaid office stop a stratum of permanent officials from forming, at a cost in continuity and expertise. Even a movement committed to leading by obeying found that its intermediate bodies drifted upward, and had to turn them back over deliberately. Direct democracy is not achieved once: the pyramid has to be inverted again and again. DPEP's counterparts are delegate certificates revocable by a single council vote, federation bodies that act only on councils' mandates, sortition, and constitutional governance (Section 7.7).

2.9 Argentina 2001–2003: Assemblies, Recovered Workplaces, and the Barter Networks

What people built. The financial collapse of December 2001 and the uprising of December 19–20, under the slogan ¡Que se vayan todos! ("They all must go!"), produced three forms of self-organization.[16]

- Neighborhood assemblies (asambleas barriales) met in the streets and squares of Buenos Aires and practiced horizontalidad: decision by open assembly without leaders.
- Recovered workplaces (empresas recuperadas): workers occupied closed or bankrupt firms and ran them themselves, among them the Zanón ceramics plant (later FaSinPat, Fábrica Sin Patrones, "factory without bosses") and the Hotel Bauen.
- Barter networks (the Red Global de Trueque and others) grew into a parallel economy with its own scrip (créditos). Estimates of participation at the 2002 peak range from about 2.5 million to more than 7 million, depending on what is counted.

What wore it down. Most assemblies dwindled once the immediate crisis eased. Recovered workplaces lasted longer, under legal insecurity and market pressure. The barter networks collapsed in 2002–2003: créditos were counterfeited and overissued, some were handed out as political favors, and once members could not tell genuine credit from fake, trust and then the networks disappeared.

[15] Moisés 2023; summarized in Baschet 2024 (checked November 2026).
[16] Sitrin 2006; Gómez 2009; both cited at bibliographic level.

Lesson carried forward. Direct democracy erupts when the legitimacy of rulers collapses, and it fades unless it takes on lasting tasks in daily life, such as production and provisioning, as the recovered workplaces did. The collapse of the barter networks is the clearest historical case for DPEP's cryptographic layer. A parallel economy of millions failed because no one could verify that a credit was genuine or that issuance stayed within agreed limits. Signed records, one report per person per epoch, and public budget checks address exactly that failure (Sections 7.3 and 7.4). The recovered workplaces are the closest existing match to DPEP's first pilots (Chapter 11). Section 11.4 discusses complementary currencies as allies and bridges.

2.10 The Uprisings of 2010–2011, Occupy, and Their Successors

What people built. The uprisings that began in Tunisia on December 17, 2010, and spread across North Africa and the Middle East brought down Ben Ali on January 14, 2011, and Mubarak on February 11. They also produced forms of self-organization that are less remembered than the occupied squares. When the Egyptian police withdrew on January 28, 2011, neighborhood popular committees (lijan sha'biyya) formed in Cairo, Alexandria, Suez, and many Delta towns to protect streets and property and to direct traffic. In Syria, local coordination committees organized the protests, and Omar Aziz's discussion paper of October 2011 proposed local councils that would support people in "managing their own lives independent of institutions and state agencies," with horizontal links between neighborhoods and between regions. Aziz died in Adra prison in February 2013.[17]

How it was lost. In Egypt the army removed Mubarak and, in July 2013, the elected government that followed, and the security state returned. In Tunisia constitutional democracy lasted a decade before the president suspended parliament in July 2021. Syria's local councils were ground down between the regime's war and armed factions. Bayat calls these "revolutions without revolutionaries." He argues that they carried radical impulses from below but no articulated vision of an alternative social order, and so ended in what he calls "refolution": changes of ruler within largely unchanged institutions.[18]

Occupy. Occupy Wall Street, which began in New York in September 2011, drew on the example of the occupied squares and ran its camps through open general assemblies. Its experience restates an older warning. In small groups without formal rules, Jo Freeman (1972) showed that power concentrates in unaccountable informal friendship cliques ("the tyranny of structurelessness")—a sociological dynamic that asynchronous software alone does not eliminate (Graeber 2013, chaps. 1–3; Freeman 1972, 151–164; Mansbridge 1980; Dahl and Tufte 1973; Albert 2003,

[17] Aziz 2011, English translation published 2013, https://theanarchistlibrary.org/library/omar-aziz-a-discussion-paper-on-local-councils-in-syria (read in full; checked November 2026).
[18] Bayat 2017, cited at bibliographic level.

chap. 5). Mansbridge (1980) and Dahl & Tufte (1973) are the classic treatments of assembly scale; they were not consulted directly, and no specific size threshold is attributed to them. The ∼50–200 neighborhood-council bound and the 50–150 speaker figure in Table 2.2 are working design heuristics from speaking-time arithmetic (one hour shared among 60 speakers is one minute each), not sourced empirical findings. Albert (2003, chap. 5) argues why unanimity vetoes fit small-team hiring while majority/supermajority rules with minority reconsideration delays fit workplace investments.

Sudan. The second wave, in Sudan and Algeria in 2018–2019, built on that experience. Sudan's neighborhood resistance committees helped bring down al-Bashir in April 2019. After the military coup of October 25, 2021, they drafted charters from below: committees in Wad Madani issued a Revolutionary Charter for People's Power in January 2022, and the Khartoum committees a Charter for the Establishment of the People's Authority in February 2022. In September 2022, 55 committees and revolutionary bodies signed a merged Revolutionary Charter for the Establishment of the People's Authority, which rejected negotiation with the coup leaders and proposed decentralized government.[19] Since war broke out in April 2023, Emergency Response Rooms rooted in the same neighborhood networks have organized food, water, and medical aid.

Lesson carried forward. Removing a ruler does not change who decides. Aziz named the risk precisely: a revolution that stays separate from daily life (housing, food, care, work) leaves the old power in charge of everyday existence. The impulse survives only if self-organization takes over daily provisioning and decision-making, and if a worked-out picture of how a self-managed economy would function exists before the moment of rupture. This book is an attempt at such a picture. DPEP is meant to be usable now, for mutual aid and provisioning by councils that already exist, not only after a transition (Chapter 11). These movements also worked under internet shutdowns, surveillance, and the arrest of organizers, which is why DPEP works offline and protects participants' identities (Section 5.1).

2.11 Rojava (2012–2026): Communes and Women's Councils

In north and east Syria, the Autonomous Administration (DAANES) implemented nested communes, co-chair gender parity, and autonomous women's councils (Kongra Star),[20] before military and geopolitical pressure led to the SDF–Damascus integration agreement of January 30, 2026 merging DAANES civil and security institutions into the Syrian state.

[19] Radio Dabanga, "55 Groups Sign New Charter for Sudan's Resistance Committees," September 30, 2022, https://www.dabangasudan.org/en?p=170001 (checked November 2026).
[20] Hahnel, Bohmer, and Chowdhury, "Reproductive Labour," chap. 10 in Hahnel 2021, 200.

Lesson carried forward. Co-chairs and autonomous women's councils protect a group that a simple majority could outvote. Table 2.2 (row 6) carries this forward as impact-weighted and dual-key governance rules, bounded so that a veto cannot deadlock the federation.

2.12 Granted from Above, and Long-Lived Institutions

The experiences in this section did not arise from a rupture. The first five were initiated or granted by governments and are taken in date order: Yugoslav self-management, participatory budgeting in Porto Alegre, Kerala's planning campaign, Ireland's Citizens' Assembly, and Decidim. The last two are institutions that have lasted for decades inside market and state economies: the Mondragón cooperatives and the self-governed commons studied by Elinor Ostrom. All are included for what they teach about carrying self-management toward direct democracy.

2.12.1 Yugoslavia (1950–1990): Self-Management Granted from Above

What was built, and by whom. After its break with the Soviet Union, the Yugoslav ruling party introduced workers' councils by law in 1950. Enterprises were formally managed by councils elected by their workers, and after the 1965 reform they operated increasingly within a market. Unlike the cases in the first group, self-management here was granted by a ruling party rather than created by workers.

What went wrong. The standard critical accounts[21] discuss three tendencies: effective control drifting to directors, technical staff, and the party inside enterprises; persistent unemployment and labor emigration; and widening inequality between enterprises and between the richer and poorer republics.

Lesson carried forward. Councils placed inside a market reproduce the "collective egoism" of Barcelona in 1936 (Section 2.5). Self-management granted from above can be withdrawn from above, and formal council rights do not stop managers and experts from holding the real decisions. This is the empirical case for participatory planning rather than markets, and for balanced job complexes rather than workers' councils alone.

2.12.2 Porto Alegre Participatory Budgeting (1989–2017)

Porto Alegre's participatory budget combined regional assemblies and recallable delegates with an objective Quality of Life Need Matrix to redirect municipal

[21] Estrin 1983; Woodward 1995. Neither book was read in full for this study; the summary draws on descriptions of them, so no page numbers are cited.

investment to poor districts, before being suspended in 2017 following municipal coalition shifts, fiscal austerity, and ∼2,400 unexecuted citizen project requests.

2.12.3 Kerala's People's Campaign for Decentralized Planning (1996–)

The state government elected in 1996 earmarked 35–40% of the outlay of the Ninth Five-Year Plan for projects drawn up by local bodies through village and ward assemblies (grama sabhas).[22] It is the largest real case of plans drafted from below, and it shows that real resources draw people into assemblies. It also shows the limit of planning granted by a party in office: what one government devolves, the next can take back. Allocation authority has to rest with the assemblies by constitution, not by grant.

2.12.4 Ireland's Citizens' Assembly (2016–2018)

Ninety-nine citizens selected at random, with a chair, deliberated on the Eighth Amendment, and their recommendation led to the 2018 referendum (Farrell, Suiter, and Harris 2019). It is strong modern evidence that people chosen by lot deliberate competently on hard questions. But parliament set its agenda and its role was advisory. In DPEP sortition serves the councils (auditing, juries, facilitation oversight); it does not replace them.

2.12.5 Decidim (Barcelona, 2016–)

Decidim is a free-software platform for participatory democracy, governed by its own community of users and adopted by many cities and organizations.[23] It shows that civic software can be held in common. Where it is run by a municipal government, however, the government still sets the agenda. The tools have to belong to the assemblies that use them.

2.12.6 Mondragón Cooperative Corporation (1956–)

Across roughly 90 cooperatives and a group workforce commonly reported at 70,000–80,000, of whom a substantially smaller number are cooperative worker-owners (socios) because much of the retail (Eroski) and overseas workforce are non-member employees (membership figures vary by year and source and are not independently checked here), Mondragón separates the elected Audit / Watchdog Committee (Comisión de Vigilancia) from the frontline workers' Social Council (Consejo Social),[24] illustrating both federated industrial self-management and the erosion of cooperative membership under global market competition.

[22] Isaac 2002, cited at bibliographic level.
[23] Barandiaran et al. 2024, cited at bibliographic level.
[24] Whyte and Whyte 1991, consulted at bibliographic level.

2.12.7 Ostrom's Common-Pool Resource Institutions (1990)

Elinor Ostrom's study of enduring self-governed commons (e.g., Valencia's Tribunal de las Aguas) established eight design principles: defined boundaries, proportional equivalence, collective-choice arrangements, accountable monitoring, graduated sanctions, rapid local conflict resolution, right to organize, and nested enterprises (Ostrom 1990, chap. 3, 90).

2.13 Synthesis: Institutional Mechanisms and Enduring Lessons

2.13.1 Comparative Institutional-to-Protocol Matrix

Table 2.2 synthesizes the eight core institutional mechanisms analyzed across Sections 2.2–2.12, mapping their historical strengths and limitations directly to concrete capabilities in the Distributed Participatory Economics Protocol (DPEP).

Table 2.2: Comparative matrix: institutional mechanisms mapped to DPEP protocol capabilities

Mechanism & Precedents	Strengths & Advantages	Historical Limitations	Concrete DPEP Protocol Capability
1. General Assembly Spain 1936; Athens Ekklesia; Occupy; Argentina 2001; Mondragón	Builds face-to-face solidarity; surfaces qualitative context.	Strains as attendance grows (working heuristic: ∼50–150 speakers; Section 2.10); meeting fatigue.	Scoped strictly to work teams (10–30) and neighborhood councils (50–200) with async pre-reading queues.
2. Revocable Delegation Paris Commune 1871; Spain CNT; Hungary 1956; Porto Alegre PB	Enables multi-tier federation scaling while tethering delegates to base.	Delegates ossify if recall is procedurally slow or difficult.	Scoped CapCert Delegate Certificates: Issued by council k-of-n multi-signature; revocable in a single council vote.
3. Sortition + Rotation Classical Athens (Boule, Logistai, Dikasteria); Zapatistas; Ireland 2016–18	Prevents factional elections, charismatic capture, and coordinator elites.	Unsuitable for rare specialist tasks unless drawn from qualified pool.	Sortition (beacon open): Intended to select rotating IFB oversight, audit panels, and dispute juries (Section 7.7).
4. Mandatory Audit Athens Euthynai (Logistai); Mondragón Comisión de Vigilancia	Systematically detects bias or unauthorized overrides.	Requires tamper-evident records and independent auditors. Mondragón's elected Social Council is a workers' voice body, not the audit committee.	Deterministic WASM + Two-Root Audit Logs: Any node with the leaf archive can recompute IFB price updates; mobile light clients verify headers only.
5. Endowment Equalization Aragon Depots (1936); Mondragón Funds	Prevents collectives with superior machinery from extracting rents.	Failed in 1936 Barcelona where rich collectives hoarded revenues.	Stand-Alone Capital Asset Ledger: Automatically debits each WC_j's SC_j for capital/resource user rights (P_k, P_r).
6. Proportional & Caucus Rojava Co-Chairs & Women's Councils	Protects disproportionately affected groups from majority tyranny.	Can deadlock if scope of veto is unbounded.	Impact-Weighted & Dual-Key Governance Rules (PEPE Ch. 2 p. 24; DEP Ch. 10 p. 200).
7. Iterative Tâtonnement IETF/Git; Lange-Arrow-Hurwicz; Parecon IFB	Scales quantitative clearing to millions asynchronously.	Pure prices strip out human context unless paired with qualitative text.	Two-Root BLAKE3 Aggregation + Content-Addressed Qualitative Lists.
8. Graduated Sanctions Ostrom CPRs (Valencia Tribunal de las Aguas)	Intended to keep routine disputes local (Ostrom 1990); no sourced "95%" rate is claimed here.	Requires clear invariants and fast local mediation.	Tiered Variance Alerts → UFB Mediation → Industry/Environmental Court.

2.13.2 Enduring Lessons: Carrying the Impulse Forward

Read together, these experiences are not a catalog of better or worse governments. Each began when people took decisions into their own hands, and each was destroyed, captured, or worn down. The question this book takes from them is what must be carried forward so that the impulse to freedom, self-management, and solidarity ends in direct democracy, and not in better government by others. Brinton gave the test that this book applies:

> Meaningful action, for revolutionaries, is whatever increases the confidence[,] the autonomy, the initiative, the participation, the solidarity, the equalitarian tendencies and the self-activity of the masses and whatever assists in their demystification.[25]

The same passage names its opposite: whatever reinforces passivity, apathy, cynicism, hierarchy, alienation, and "reliance on others to do things for them."

Design Principle 2.1 (Brinton's Test). A feature of DPEP is justified only if it increases the confidence, autonomy, initiative, participation, solidarity, egalitarian tendencies, and self-activity of the people who use it, and helps them understand the system they govern. A feature that increases their passivity, their dependence on experts or officials, or their differentiation through hierarchy is rejected, however efficient it is.

Table 2.3 sets out ten lessons and what each requires of the protocol. They are requirements: some are met by mechanisms specified in later chapters, and others are open.

Every case in this chapter that stopped at better government by others (a party, a planning board, a reforming administration) lost the impulse that created it. Castoriadis put the end point plainly: "The only total form of democracy is therefore direct democracy" (Castoriadis (1957) 1972, § 3b). DPEP is worth building only to the extent that it helps people decide for themselves, in full knowledge of the relevant facts, and keep on deciding.

The technology stack and cryptographic primitives proposed in Chapters 5–7 are this book's answer, to the best of present knowledge, to the lessons in Table 2.3: each mechanism is there because one of these failures happened, and each is meant to protect what people win for themselves. They are means, not ends, and they are not final. Early implementations will be immature (Section 6.3), and what they defend against will change: states, capital, and capture from within will adapt, and so will cryptanalysis and computing, including quantum computing. Protecting the social gains will therefore need vigilant review, independent audit, and timely upgrades, which is why the design is crypto-agile (Section 7.2.1) and the code is open. That maintenance is itself a collective, open-source commons. It should become easier to sustain as a self-managed society matures: as balanced

[25] Brinton 1967, point 7, https://www.marxists.org/archive/brinton/1967/04/as-we-see-it.htm (checked November 2026); reprinted in Brinton 2004.

Table 2.3: Enduring lessons and what they require of DPEP

Lesson	Where it was learned	What it requires of DPEP
1. Who manages matters, not who owns.	Russia 1917–21 (Brinton); Yugoslavia	Proposals originate only in councils; no higher body drafts, alters, or replaces them.
2. Self-organization appears from below, quickly, in crisis.	Paris 1871; Russia 1917; Hungary 1956; Chile 1972; Argentina 2001; Egypt and Syria 2011; Sudan 2019	Councils can form, issue credentials, and federate without permission from any authority, and can work offline.
3. Incorporation is the main danger, from allies as much as enemies.	Factory committees into unions (1918); Hungarian councils into works councils (1957); Rojava (2026)	Higher bodies hold only enumerated powers, delegated revocably; councils keep their own state and can federate again.
4. The center must be an instrument: visible, circumscribed, and controlled.	Castoriadis; Freeman; Occupy; Zapatistas 2023	Facilitation computes and publishes but never decides; central functions are named and audited; federations act only on mandate.
5. Decide in full knowledge of the relevant facts.	Castoriadis; Kerala	Aggregates are public; alternative plans and their consequences for each group are published before votes.
6. Turn the pyramid over, again and again.	Athens; Spain 1936; Zapatistas 1994–2023	Rotation, sortition, revocable delegates, periodic constitutional review.
7. Trust in shared accounts is a commons.	Argentine barter networks (2002)	Signed records; one report per person per epoch; public issuance and budget checks.
8. Build the new inside daily life, before the rupture.	Aziz (Syria 2011); Bayat; Argentina's recovered workplaces	Usable now for provisioning, mutual aid, and pilots (Chapter 11), not only after a transition.
9. Councils are crushed by force.	Paris 1871; Hungary 1956; Chile 1973; Syria; Sudan	Privacy, no single point of seizure, offline operation (Section 5.1). No software stops tanks; only solidarity can.
10. Equality decays unless it is renewed.	Barcelona 1936; Yugoslavia	Endowment equalization; effort-based remuneration; public data on inequality between councils.

job complexes spread the skills to read, audit, and change the code, and as fewer concentrated interests remain with the means and the motive to attack it.

CHAPTER 3

Source-Grounded Requirements of Participatory Economics

> Until now, however, descriptions of alternatives have been unconvincing. Here Michael Albert and Robin Hahnel support the libertarian socialist tradition by presenting a rigorous, well-defined model of how producers and consumers could democratically plan their interconnected activities.
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Front Flap / JSTOR Abstract[1]

3.1 Source Provenance and Classification Methodology

Before formulating mathematical equations or distributed protocols, we establish strict traceability to the foundational texts. Every requirement in Table 3.1 is classified into one of four provenance tiers:

- [ORIGINAL-1991]: Explicitly specified in Michael Albert and Robin Hahnel's two 1991 volumes, The Political Economy of Participatory Economics (PEPE) and Looking Forward (LF).[2]
- [INTERPRETATION-LATER]: Institutional elaborations, formalizations, and accounting procedures introduced in Michael Albert's Parecon: Life After Capitalism (PLAC, 2003), Robin Hahnel's Democratic Economic Planning (DEP, 2021) and A Participatory Economy (2022), Anders Sandström's Anarchist Accounting (2020, 2023), Mitchell Szczepanczyk's simulation work (2023), and the Participatory Economy Project's User Story Mapping (2023).[3]
- [TECH-EXTENSION]: Technological extensions required to execute the model over a modern, decentralized, post-quantum, privacy-preserving computational network.
- [UNRESOLVED]: Open theoretical or institutional questions requiring simulation or empirical human research (examined in full in Chapter 10).

[1] Albert and Hahnel 1991a, book overview (https://www.jstor.org/stable/j.ctv131bvr2).

[2] Albert and Hahnel 1991a, print pagination verified via library TOC scan (research/sources/pepe_dandelon.pdf, 1–132) and full text analyzed via ZNetwork (intro.htm, 1.htm–6.htm); Albert and Hahnel 1991b, full text analyzed via ZNetwork (lfIntroduction.htm, lf1_1.htm–lf10_1.htm).

[3] Hahnel 2021, 2022; Sandström 2020; Szczepanczyk 2023; Participatory Economy Project 2023, https://participatoryeconomy.org/project/participatory-planning-user-story-mapping/.

3.2 Normative Foundations and Endogenous Preference Theorems

In Chapter 1 of The Political Economy of Participatory Economics (pp. 8–11), Albert and Hahnel evaluate economic institutions against five valuative criteria: Equity, Self-Management, Solidarity, Variety, and Efficiency (Albert and Hahnel 1991a, chap. 1, 8–9, 11). For distributive equity, they distinguish four distinct maxims (PEPE Ch. 1, p. 8; Ch. 3, p. 44; DEP Ch. 1, pp. 16–22):

1. Maxim 1 (Capitalism): Payment according to the value of the contribution of one's personal labor and productive property owned (rejected due to inheritance—the "Rockefeller grandson problem"—luck, and compounding wealth accumulation; Hahnel 2021, chap. 1, 16–18; Albert and Hahnel 1991a, chap. 3, 45–46).
2. Maxim 2 (Market Socialism / Coordinatorism): Payment according to the value of one's personal labor contribution alone (rejected because differences in marginal productivity stem overwhelmingly from the "genetic lottery," luck, job assignment, and publicly funded training rather than personal sacrifice—illustrated by their "Mozart vs. Salieri" and "Dave Butz" examples; Albert and Hahnel 1991a, chap. 3, 46–47).
3. Maxim 3 (Participatory Economics): Payment according to effort and personal sacrifice expended in socially valued labor (longer hours, higher intensity, or more onerous/dangerous conditions prior to full BJC equalization).
4. Maxim 4 (Humane Solidarity): Payment according to need—combined with Maxim 3 in Parecon via universal free public goods (healthcare, education) and council-approved special-need allowances (PEPE Ch. 3, p. 47).

Crucially, drawing on the formal results of Quiet Revolution in Welfare Economics (1990), consulted here through PEPE's summary,[4] Albert and Hahnel argue that because human preferences are endogenous (people rationally mold their future characteristics ΔC in anticipation of future terms of supply), any institution that systematically mischarges for activities relative to true social opportunity cost causes: (1) a snowballing divergence from Pareto optimality over time, (2) systematic warping of human development trajectories, and (3) ex post invisibility of the bias to participants. Markets systematically underprice negative externalities and overprice public goods (warping preferences toward anti-social individualism), while central planning systematically under-supplies self-managed work roles (warping workers toward apathy and coordinator submission; Albert and Hahnel 1991a, chap. 1, 12–17, and chap. 5, 98–105).

3.3 Formal Traceability Matrix of Requirements

[4] Hahnel and Albert 1990, chaps. 4, 6, and 7; summarized in Albert and Hahnel 1991a, chap. 1, 9–11.

Table 3.1: Master requirements matrix derived from primary and secondary literature

Req ID	Requirement Specification	Classification	Primary Citations
REQ-NORM-01	Productive Commons & User-Right Fees: Abolish private ownership of productive property; all natural resources and capital stocks belong to society. Worker councils acquire temporary user rights during annual planning and pay opportunity-cost charges (P_r, P_k, P_l).	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 1 (pp. 17–20), Ch. 5 (p. 85); DEP Ch. 6 (pp. 98–102), Ch. 12 (p. 244); Sandström (2020).
REQ-NORM-02	Proportional Self-Management: Decision-making input proportional to the degree affected by an economic choice.	[ORIGINAL-1991]	PEPE Ch. 1 (p. 8), Ch. 2 (p. 24); LF Ch. 1; PLAC Ch. 5; DEP Ch. 1 (p. 14).
REQ-NORM-03	Effort & Sacrifice Remuneration: Consumption budgets proportional to effort/sacrifice (Maxim 3) tempered by universal free goods and need exceptions (Maxim 4), with council average effort ratings capped by SB_j/SC_j.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 3 (pp. 44–55); DEP Ch. 6 (pp. 108–117); Sandström (2023).
REQ-NORM-04	Balanced Job Complexes (BJCs): Tasks partitioned into subsets $\{PIPA\}_i$ ordered by empowerment/desirability and combined into balanced complexes within and across workplaces via Job Complex Committees.	[ORIGINAL-1991]	PEPE Ch. 2 (pp. 25–30), Ch. 5 (pp. 82–83); LF Chs. 1–2.
REQ-ACT-01	Nested Consumer Councils & Federations: Individuals/households → Neighborhood CC_h → Ward → City/County → State/Region → National Federation, requesting public goods matched to spatial scale.	[ORIGINAL-1991]	PEPE Ch. 3 (pp. 40–42), Ch. 5 (pp. 81–82); LF Chs. 3, 4, 8.
REQ-ACT-02	Worker Councils & Dual Federations: Democratic workplace councils federated both by industry (scale economies, R&D, SB/SC benchmarking) and by region (local production externalities).	[ORIGINAL-1991]	PEPE Ch. 2 (pp. 23–25), Ch. 4 (p. 62), Ch. 5 (p. 88); LF Chs. 1, 2, 7.
REQ-ACT-03	Six Functional Facilitation Boards: Iteration (IFB), Consumption (CFB), Production (PFB), Employment (EFB), Household (HFB), and Updating (UFB) facilitation boards performing purely informational/mediating tasks.	[ORIGINAL-1991]	LF Ch. 6 (lf6_1.htm), Ch. 9 (lf9_2.htm); PEPE Ch. 4 (p. 62).
REQ-ACT-04	Communities of Affected Parties (CAPs) & Environmental Courts: Airshed/watershed collectives participating in the Pollution Demand-Revealing Mechanism (PDRM) below hard biophysical caps.	[INTERP-LATER]	DEP Ch. 7 (pp. 138–149), Ch. 14 (pp. 261–267); PEP (2023).
REQ-ACT-05	Stand-Alone Capital Asset Entities & Society Account: Double-entry social accounting entities tracking capital production costs, user-right fees, CAP flows, and GDP division between consumption and investment.	[INTERP-LATER]	Sandström (2020, chaps. 3–5; 2023).

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Continued from previous page

Req ID	Requirement Specification	Classification	Primary Citations
REQ-DATA-01	Seven Priced Economic Categories: Private goods, public goods, intermediate inputs, natural resources, labor categories, capital stocks, and regulated pollutants.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 5 (pp. 78–88); DEP Ch. 7, Ch. 9; Szczepanczyk (2023).
REQ-DATA-02	Coarse Planning Taxonomy vs. Granular SKUs: Plan over coarse resource-equivalent product classes; extrapolate specific sizes/colors statistically and resolve at intra-year fulfillment.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 4 (p. 64); LF Ch. 6 (lf6_2.htm); DEP Ch. 8 (pp. 163–167).
REQ-DATA-03	Public Books of Blueprints & Qualitative Lists: Publish all technical input-output recipes $[A, K, R, L]$ and attach qualitative descriptive Lists to quantitative council vectors.	[ORIGINAL-1991]	PEPE Ch. 4 (pp. 61–62), Ch. 5 (p. 84), Ch. 6; PLAC Chs. 11, 17.
REQ-DATA-04	Protected Private Consumption Sphere: Personal consumption within warranted budget B_u cannot be vetoed (PEPE). Cryptographic invisibility of the bundle is a DPEP extension, not Albert and Hahnel's 1991 design (Section 7.1).	[ORIGINAL-1991] (non-veto) + [TECH-EXT] (invisibility)	PEPE Ch. 3 (p. 41), Ch. 5 (p. 80): non-veto of in-budget requests. LF Ch. 3 subjects consumption to collective criticism. PLAC Chs. 12, 18: anonymous submission against "busybody" intrusion, not cryptographic aggregation.
REQ-PLAN-01	Multi-Horizon & Collective-First Sequencing: Long-run (education, environment) and aggregate investment plans precede annual planning; federation collective goods are settled before final private consumption bundles.	[ORIGINAL-1991] + [INTERP-LATER]	LF Ch. 3 (lf3_2.htm), Ch. 8 (lf8_1.htm), Ch. 9 (lf9_1.htm); DEP Chs. 11–15.
REQ-PLAN-02	Slack Planning & Hierarchical Intra-Year Clearing: Include 3–5% planned excess supply buffer (SL^E); cancel routine consumer swaps inside neighborhood/ward/city federations; trigger UFB replanning only when net shocks exceed slack.	[ORIGINAL-1991]	PEPE Ch. 4 (pp. 66–67), Ch. 6 (p. 113, 122); DEP Ch. 8 (pp. 167–170).

CHAPTER 4

Formal Economic Model, Accounting Identities, and Algorithms

> It is impossible to show that a feasible nonmarket system at least approaches the productivity of the market unless (1) a rather well-developed theoretical model of the nonmarket system is available, and (2) it is demonstrated that a sufficiently productive approximation of the ideal… system described in the theoretical model is practically possible.
>
> — Allen Buchanan, Ethics, Efficiency, and the Market (1985), p. 29, quoted in Albert and Hahnel, The Political Economy of Participatory Economics (1991), Ch. 4, p. 58, and Ch. 5, p. 73[1]

Core Mathematical Notation

The symbols below are used throughout this chapter; councils and federations are written WC_j, CC_h, and IF_k as in Appendix B.

Symbol	Mathematical Meaning
Ω	Master catalog of priced items, of size $G = \lvert\Omega\rvert$: $\mathcal{G}^{\text{priv}} \sqcup \mathcal{G}^{\text{pub}} \sqcup \mathcal{G}^{\text{int}} \sqcup \mathcal{R} \sqcup \mathcal{L} \sqcup \mathcal{K} \sqcup \mathcal{Z}$.
$P^{(t)} \in \mathbb{R}_+^{\lvert\Omega\rvert}$	Vector of indicative social opportunity-cost prices at planning iteration t.
$\bar{a}_j, \underline{a}_j, r_j, e_j$	Worker council WC_j's proposed outputs ($\bar{a}_j$), intermediate inputs ($\underline{a}_j$), natural resources (r_j), and emissions (e_j).
$l_j^0, k_j^0, \Delta l_j, \Delta k_j$	Initial labor (l_j^0) and capital (k_j^0) endowments of WC_j, and proposed net additions/releases ($\Delta l_j, \Delta k_j$).
$r^*, l^*, k^*, E_z^{\max}$	Economy-wide available natural resources (r^*), labor supplies (l^*), capital stocks (k^*), and hard biophysical emission caps ($E_z^{\max}$).
$w_{\text{base}}, \epsilon_u, B_u(P)$	Base hourly remuneration rate (w_{base}), worker effort multiplier (ϵ_u), and citizen u's warranted annual consumption budget (B_u).
$S_i^{(t)}, D_i^{(t)}, SL_i^{(t)}$	Aggregate supply ($S_i^{(t)}$), aggregate demand ($D_i^{(t)}$), and planned slack buffer ($SL_i^{(t)} = \eta_i S_i^{(t)}$) for category i.
$\rho_i^{(t)}, \tau$	Normalized relative excess demand imbalance $\rho_i^{(t)} = \frac{2\lvert S_i^{(t)} - D_i^{(t)}\rvert}{S_i^{(t)} + D_i^{(t)}}$ and two-sided clearing tolerance $\tau \in [0.01, 0.05]$ (Definition 4.2).

4.1 Ontology of Economic Actors and Priced Objects

Let the participatory economy $\mathcal{E}$ over annual planning period T consist of the following finite sets of actors and institutional entities:[2]

[1] Buchanan 1985, 29; cited in Albert and Hahnel 1991a, chap. 4, 58n3, and chap. 5, 73.

[2] Formalized from Albert and Hahnel 1991a, chap. 4 (62–68) and chap. 5 (77–95); Hahnel 2021, chap. 7 (130–156); Sandström 2020, chap. 3; and Participatory Economy Project 2023.

- Individuals ($\mathcal{I} = \{1, \ldots, N_{\text{ind}}\}$): Natural persons participating as consumers, workers, and citizens. Each individual $u \in \mathcal{I}$ belongs to a primary neighborhood consumer council $h(u) \in \mathcal{H}$ and, if working, a primary worker council $j(u) \in \mathcal{J}$ (plus optional secondary council $j'(u)$ for cross-workplace Balanced Job Complex rotation) (Albert and Hahnel 1991a, chap. 2, 27–30, and chap. 5, 83).
- Neighborhood Consumer Councils ($\mathcal{H} = \{CC_1, \ldots, CC_H\}$) & Nested Consumer Federations ($\mathcal{F}^{\text{C}} = \bigcup_{\ell=1}^{L} \mathcal{F}_\ell^{C}$): A rooted geographic hierarchy of depth $L = 5$ (Level 1 = Ward, Level 2 = City/County, Level 3 = State, Level 4 = Region, Level 5 = National Federation; Albert and Hahnel 1991a, chap. 3, 40–42, and chap. 5, 81–82).
- Worker Councils ($\mathcal{J} = \{WC_1, \ldots, WC_J\}$) & Dual Worker Federations ($\mathcal{F}^{W} = \mathcal{F}^{W,\text{ind}} \cup \mathcal{F}^{W,\text{reg}}$): Democratic workplace collectives federated both by industry ($IF_k \in \mathcal{F}^{W,\text{ind}}$) and by geographic region ($\mathcal{F}^{W,\text{reg}}$) (Albert and Hahnel 1991a, chap. 2, 23–25, and chap. 5, 88, 96–97).
- Communities of Affected Parties ($\mathcal{C}^{\text{CAP}} = \{CAP_1, \ldots, CAP_Z\}$): Ecological collectives of residents $\text{pop}(CAP_z) \subseteq \mathcal{I}$ impacted by pollutant $z \in \mathcal{Z}$ in a specific airshed or watershed.[3]
- Six Facilitation Boards ($\mathcal{B} = \{\text{IFB}, \text{CFB}, \text{PFB}, \text{EFB}, \text{HFB}, \text{UFB}\}$), Stand-Alone Capital Asset Entities ($\mathcal{A}^{\text{cap}}$), and the Society Account ($\mathcal{A}^{\text{soc}}$).[4]

Definition 4.1 (Master Catalog of Seven Priced Economic Categories). Following Albert and Hahnel (1991a, chaps. 5–6), Hahnel (2021, chaps. 7–9), and Szczepanczyk (2023, sec. 4), the economy tracks supply, demand, and indicative prices across seven disjoint categories indexed by the master catalog Ω:

$$\Omega = \mathcal{G}^{\text{priv}} \sqcup \mathcal{G}^{\text{pub}} \sqcup \mathcal{G}^{\text{int}} \sqcup \mathcal{R} \sqcup \mathcal{L} \sqcup \mathcal{K} \sqcup \mathcal{Z} \tag{4.1}$$

whose size we write $G = |\Omega|$ (the number of priced items), with complete indicative price vector at iteration t given by:

$$P^{(t)} = \left(p^{\text{priv}}, p^{\text{pub}}, p^{\text{int}}, p^{\text{res}}, p^{\text{lab}}, p^{\text{cap}}, p^{\text{pol}}\right) \in \mathbb{R}_+^{|\Omega|} \tag{4.2}$$

4.2 Primal-Dual Reference Model and Sandström's Society Account

In Chapter 5 of The Political Economy of Participatory Economics (pp. 78–79), Albert and Hahnel introduce a primal-dual linear programming reference model to clarify the meaning of shadow prices before presenting their decentralized iterative procedure (Albert and Hahnel 1991a, chap. 5, 78–79). Let x be the vector of production activity levels, A be the produced intermediate input matrix, K be the capital stock coefficient matrix (with available stock k^*), R be the non-producible

[3] Hahnel 2021, chap. 7 ("The Pollution Demand Revealing Mechanism"), 139–148.
[4] Albert and Hahnel 1991b, chap. 6 (lf6_1.htm); Sandström 2023.

natural resource input matrix (with available ecological supply r^*), L be the direct labor input matrix (with available labor l^*), and v be the row vector of relative social values of net outputs. The economy-wide primal and dual reference problems are:

$$\text{Primal:} \quad \max_{x \geq 0} \quad v(I - A)x \quad \text{subject to} \quad Kx \leq k^*, \;\; Rx \leq r^*, \;\; Lx \leq l^* \tag{4.3}$$

$$\text{Dual:} \quad \min_{P_k, P_r, P_l \geq 0} \quad P_k k^* + P_r r^* + P_l l^* \quad \text{subject to} \quad P_k K + P_r R + P_l L \geq v(I - A) \tag{4.4}$$

As Albert and Hahnel explicitly note (PEPE Ch. 5, p. 79), the shadow prices P_l solved from the static dual (4.4) reflect only the marginal physical productivities of different labor categories, not the differential disutilities or desirabilities of work from the workers' point of view; whereas the indicative prices generated by Parecon's iterative council procedure incorporate worker well-being and effort disutility directly (Albert and Hahnel 1991a, chap. 5, 79).

4.2.1 Closing the Macro-Accounting Loop (Sandström 2020, 2023)

Anders Sandström's Anarchist Accounting (2020) and 2023 technical specification formalize how opportunity-cost user fees paid by Worker Councils reconcile with effort-based consumer incomes inside the Society Account ($\mathcal{A}^{\text{soc}}$) (Sandström 2020, chap. 3; Sandström 2023, diagram 1 (arrows 1–14)):

1. Social Value of Final Output (Gross, GDP) at Indicative Prices P:

$$Y^{\text{GDP}}(P) = p^{\text{priv}} \cdot D^{\text{priv}} + p^{\text{pub}} \cdot D^{\text{pub}} + p^{\text{cap,new}} \cdot D^{\text{inv}} \tag{4.5}$$

   In the Society Account, credits consist of all user-right fees charged to Worker Councils for access to society's labor categories ($p^{\text{lab}} \cdot L$), natural resources ($p^{\text{res}} \cdot R$), and productive capital assets held in Stand-Alone Capital Asset Entities ($p^{\text{cap}} \cdot K$), plus Consumer Council depreciation/discount charges on public consumer assets and net WC operating surpluses.

2. Aggregate Consumption Pool, Effort-Weighted Hours (H^{eff}), and Numéraire ($w_{\text{base}} \equiv 1.0$): Let $\sigma_{\text{inv}} \in (0,1)$ be the share of GDP allocated to aggregate investment (DEP Ch. 11),[5] Y^{allow} be the ratified budget for non-working cash allowances (Maxim 4 transfers A_u^{social}), $Y^{\text{CAP}}(P) = \sum_{z \in \mathcal{Z}} p_z^{\text{pol}} E_z$ be total CAP pollution compensation (funded by WC emission charges $\sum_j p^{\text{pol}} \cdot e_j$), and $\Delta S^{\text{agg}} = \sum_{u \in \mathcal{I}} \Delta S_u$ be net aggregate household borrowing. Universal public services already appear as $p^{\text{pub}} \cdot D^{\text{pub}}$ inside Y^{GDP}; they must not also be subtracted as Y^{allow}, or the consumption pool is double-counted (Table 4.2). Effort-weighted hours $H^{\text{eff}} = \sum_{u \in \mathcal{I}^{\text{work}}} h_u \epsilon_u$ convert the worker pool into a

[5] Hahnel and Kerkhoff, "Aggregate Investment Planning," chap. 11 in Hahnel 2021, 219–237.

unit wage. The identity $w_{\text{base}} \equiv 1$ is a numéraire normalization after clearing, not an independent price:

$$Y^{\text{work}}(P) = (1 - \sigma_{\text{inv}})Y^{\text{GDP}}(P) - Y^{\text{allow}} - Y^{\text{CAP}}(P) - \Delta S^{\text{agg}}$$
$$w_{\text{base}}(P) = \frac{Y^{\text{work}}(P)}{\sum_{u \in \mathcal{I}^{\text{work}}} h_u \epsilon_u} \quad \text{(then rescale all prices so } w_{\text{base}} = 1) \tag{4.6}$$

3. Capping Both Mean Workplace Effort Rating and Total Credited Hours (DEP Ch. 6, p. 117): To prevent either collective "grade inflation" or phantom-hours inflation (INV-4B), each worker council WC_j's mean effort multiplier and total credited effort-hours are bounded by its accredited labor for the plan year—the initial endowment plus ratified net hiring, $\sum_\ell (l^0_{j,\ell} + \Delta l_{j,\ell})$, the same quantity charged in Eq. (4.12)—and by its social benefit-to-cost ratio:[6]

$$\frac{\sum_{u \in WC_j} h_u \epsilon_u}{\sum_{u \in WC_j} h_u} \le \min\left(\bar{\epsilon}_{\max},\ \frac{SB_j(P)}{SC_j(P)}\right)$$
$$\sum_{u \in WC_j} h_u \epsilon_u \le \left(\sum_{\ell \in \mathcal{L}} (l^0_{j,\ell} + \Delta l_{j,\ell})\right) \min\left(\bar{\epsilon}_{\max},\ \frac{SB_j(P)}{SC_j(P)}\right) \tag{4.7}$$

Here $l_{j,\ell}$ is measured in accredited hours. Because Δl_j is itself part of the proposal, the cap is checked against the ratified roster at plan ratification and against the proposed roster in each round.

4. Individual Warranted Annual Consumption Budget (B_u): Combining Sandström's four income flows (Sandström 2023, arrows 1–4), citizen u's warranted budget is:

$$B_u(P) = \underbrace{w_{\text{base}}(P) h_u \epsilon_u}_{\text{1: Work Effort}} + \underbrace{A^{\text{social}}_u}_{\text{2: Benefits}} + \underbrace{\sum_{z : u \in CAP_z} \frac{p^{\text{pol}}_z E_z}{|\text{pop}(CAP_z)|}}_{\text{3: CAP Compensation}} + \underbrace{\Delta S_u}_{\text{4: Loan/Need}}$$
$$\sum_{u \in \mathcal{I}} B_u(P) = Y^{\text{work}}(P) + Y^{\text{allow}} + Y^{\text{CAP}}(P) + \Delta S^{\text{agg}} = (1 - \sigma_{\text{inv}})Y^{\text{GDP}}(P) \tag{4.8}$$

These last two displays are identities once Y^{allow} is defined as cash transfers only and A^{social}_u exhausts Y^{allow}. They hold at every price vector, not only at equilibrium. Relations such as $D^{\text{priv}} = S^{\text{priv}}$ are equilibrium conditions, not identities.

[6] Hahnel 2021, chap. 6 ("Capping average effort ratings"), 117. Using the endowment l^0_j alone would leave newly hired workers uncredited. The Phase 1 default for $\bar{\epsilon}_{\max}$ is a political choice set out in Section 9.1.4. Vintage-equipment plants are equalized via lower $p^{\text{cap}} \cdot k^0_j$ rental charges. INV-4B does not help if the labor roster itself is falsified—that remains a physical euthynai problem.

Simultaneity of budgets and demand. Y^{GDP} is valued at the demanded quantities D, so budgets depend on demand and demand depends on budgets: $B_u = B_u(P, D)$. Within an iteration DPEP breaks the loop with a one-round lag, $B_u^{(t)} = B_u(P^{(t)}, D^{(t-1)})$, so councils always optimize against a published budget. The lag is consistent only at a fixed point, where $B_u^{(t)} = B_u(P^{(t)}, D^{(t)})$; whether the lagged process reaches that fixed point is part of open question P2 and is not implied by any convergence result for fixed budgets. The investment share and the investment bundle are not independent choices: at a cleared plan $\sum_u B_u = p^{\mathrm{priv}} D^{\mathrm{priv}} + p^{\mathrm{pub}} D^{\mathrm{pub}}$, so the identities force $\sigma_{\mathrm{inv}} Y^{\mathrm{GDP}} = p^{\mathrm{cap,new}} \cdot D^{\mathrm{inv}}$. Either the federations ratify σ_{inv} and D^{inv} adjusts in value, or they ratify a physical investment bundle and σ_{inv} is implied. DPEP takes the first option (DEP Ch. 11 treats aggregate investment as a prior collective decision), so D^{inv} is scaled each round to match.

Table 4.2: Macro balance sheet (identities vs. equilibrium conditions)

Relation	Content	Status
$Y^{\mathrm{GDP}} = p^{\mathrm{priv}} D^{\mathrm{priv}} + p^{\mathrm{pub}} D^{\mathrm{pub}} + p^{\mathrm{cap,new}} D^{\mathrm{inv}}$	Final uses at indicative prices.	Identity (definition).
$Y^{\mathrm{work}} = (1 - \sigma_{\mathrm{inv}}) Y^{\mathrm{GDP}} - Y^{\mathrm{allow}} - Y^{\mathrm{CAP}} - \Delta S^{\mathrm{agg}}$	Worker remuneration pool. Y^{allow} is cash A_u^{social} only.	Identity, given that split.
$\sum_u B_u = Y^{\mathrm{work}} + Y^{\mathrm{allow}} + Y^{\mathrm{CAP}} + \Delta S^{\mathrm{agg}}$	Warranted budgets exhaust the consumption pool.	Identity at every P.
$p^{\mathrm{lab}} \cdot L + p^{\mathrm{res}} \cdot R + p^{\mathrm{cap}} \cdot K + p^{\mathrm{pol}} \cdot E =$ WC user-right charges	Society Account credits.	Accounting identity.
$D_g \le S_g$, $D_r \le r^*$, $D_z \le \min(E_z, E_z^{\max})$	One-sided feasibility.	Equilibrium / halt condition, not an identity.
$p^{\mathrm{lab}} \ge w_{\mathrm{base}}$	Labor price floor (INV-4B).	Policy constraint; can prevent labor-market clearing in surplus.

4.3 Formal Optimization Problems of Councils and CAPs

4.3.1 Consumer Councils and Nested Federations (PEPE Ch. 5, pp. 79–82, 88)

Each consumer council $CC_h \in \mathcal{H}$ seeks to maximize its members' total economic well-being $W_h(U_h, \Delta C_h)$—incorporating both immediate preference fulfillment U_h and endogenous human characteristic development ΔC_h—subject to the equitable budget constraint that its total social cost $SC(CC_h, P)$ does not exceed its members' warranted budget $B(CC_h, P) = \sum_{u \in \mathrm{pop}(CC_h)} B_u(P)$ (Albert and Hahnel 1991a,

chap. 5, 77–80, 88):

$$\max_{y_h^{\text{priv}},\, y_h^{\text{pub,loc}} \geq 0} \quad W_h\Big(U_h(y_h^{\text{priv}}, y^{\text{pub}}),\ \Delta C_h(y_h^{\text{priv}}, y^{\text{pub}})\Big) \tag{4.9}$$

$$\text{subject to} \quad SC(CC_h, P) \triangleq p^{\text{priv}} \cdot y_h^{\text{priv}} + p^{\text{pub}} \cdot y_h^{\text{pub,loc}} + |\text{pop}(CC_h)| \sum_{f \succ CC_h} \frac{p^{\text{pub}} \cdot y_f^{\text{pub}}}{|\text{pop}(f)|} \leq B(CC_h, P)$$

where $f \succ CC_h$ denotes every higher-level ward, city, state, or national consumer federation containing CC_h. Albert and Hahnel state (PEPE Ch. 3, p. 41; Ch. 5, p. 80) that for any individual or council whose request satisfies $SC \leq B$, other councils may advise but cannot veto the request; veto power applies only to exception requests exceeding warranted allowances ($SC > B$). That is a normative design rule, not a theorem.

4.3.2 Communities of Affected Parties (CAPs) (DEP Ch. 7, pp. 139–148)

For each regulated pollutant $z \in \mathcal{Z}$ with hard biophysical sustainability ceiling $E_z^{\max}$ (established by Long-Run Environmental Planning; DEP Ch. 14; Hahnel 2021, chap. 7, 139–148, appendix 159–162, and chap. 14, 261–267), given indicative damage price p_z^{pol}, CAP_z chooses the total permissive emission supply $E_z \in [0, E_z^{\max}]$ to maximize member well-being:

$$\max_{0 \leq E_z \leq E_z^{\max}} \quad W_{CAP_z}\Big(p_z^{\text{pol}} E_z,\ -D_{CAP_z}(E_z)\Big) \tag{4.10}$$

At an interior optimum below $E_z^{\max}$ the first-order condition is $p_z^{\text{pol}}\, \partial_1 W = D'_{CAP_z}(E_z)\, \partial_2 W$. Only if W_{CAP_z} is linear in compensation and damage (quasi-linear well-being, damages measured in the same units as compensation) does this reduce to the familiar condition that the summed marginal damage to members equals p_z^{pol}, the Pigouvian rule for a public bad. With more general W, or with a CAP that sets E_z by vote rather than by summing members' damages, the condition holds only approximately; how closely the Pollution Demand-Revealing Mechanism achieves it, and whether a CAP can gain by restricting supply, is analyzed in DEP Ch. 7 and its appendix and is not re-derived here.

4.3.3 Worker Councils (PEPE Ch. 5, pp. 85–88; DEP Ch. 7, pp. 152–156)

Each worker council $WC_j \in \mathcal{J}$ begins the planning cycle with initial labor endowment l_j^0 ($\sum_{j=1}^{J} l_j^0 = l^*$) and initial capital stock endowment k_j^0 ($\sum_{j=1}^{J} k_j^0 = k^*$), for which it is automatically charged $p^{\text{lab}} \cdot l_j^0 + p^{\text{cap}} \cdot k_j^0$ (Albert and Hahnel 1991a, chap. 5, 85–87). In iteration t, WC_j proposes output vector $\bar{a}_j = (\bar{a}_j^{\text{priv}}, \bar{a}_j^{\text{pub}}, \bar{a}_j^{\text{int}}, \bar{a}_j^{\text{inv}})$, intermediate inputs $\underline{a}_j$, natural resources r_j, net labor additions/releases Δl_j, net capital additions/releases Δk_j, and pollutant emissions e_j

from its feasible technology set $\mathcal{T}_j$. Defining Social Benefit (SB_j) and Social Cost (SC_j):

$$SB_j(P) = p^{\text{priv}} \cdot \bar{a}_j^{\text{priv}} + p^{\text{pub}} \cdot \bar{a}_j^{\text{pub}} + p^{\text{int}} \cdot \bar{a}_j^{\text{int}} + p^{\text{cap,new}} \cdot \bar{a}_j^{\text{inv}} \tag{4.11}$$

$$SC_j(P) = p^{\text{int}} \cdot \underline{a}_j + p^{\text{res}} \cdot r_j + p^{\text{lab}} \cdot (l_j^0 + \Delta l_j) + p^{\text{cap}} \cdot (k_j^0 + \Delta k_j) + p^{\text{pol}} \cdot e_j \tag{4.12}$$

Equation (4.12) correctly charges $l_j^0 + \Delta l_j$ (endowment plus net hiring). The floor $p^{\text{lab}} \geq w_{\text{base}}$ in INV-4B, however, prevents labor user-rights from falling when labor is in surplus. If prices are to be opportunity costs, a surplus of a labor category should lower p_ℓ^{lab} below w_{base}; the floor converts that surplus into open underemployment rather than a market-clearing wage cut. An underemployment rebate (paying idle accredited hours from the Society Account) can restore household budgets, but it does not restore the opportunity-cost reading of p^{lab}, and it is funded by reducing Y^{work} or σ_{inv}. This trade-off is unresolved.

WC_j maximizes its workers' fulfillment and development $W_j(U_j, \Delta C_j)$ subject to the Socially Non-Abusive Production Constraint (PEPE Ch. 5, pp. 87–88):

$$\max_{\pi_j \in \mathcal{T}_j} \; W_j\Big(U_j(\pi_j),\, \Delta C_j(\pi_j)\Big) \quad \text{subject to} \quad SB_j(P) - SC_j(P) \geq 0 \tag{4.13}$$

that is, $SB_j(P)/SC_j(P) \geq 1$.

4.4 Critical Algorithmic Audit of pequod-cljs and Convergence Rules

In Chapter 9 of Democratic Economic Planning (2021, pp. 173–194) and Szczepanczyk (2023),[7] the authors simulated 40 experiments (dep1ex61..100.csv, generated via bin/bigger_gen.sh) with 30,000 Worker Councils, 30,000 Consumer Councils, and 500 categories (100 private, 100 public, 100 intermediate, 100 natural resources, 100 labor categories), alongside 50 smaller 3,000-council runs (dep1ex01..50.csv).

4.4.1 The Hahnel-Weisdorf-Szczepanczyk Price Rule and CSV Audit

In csvgen.clj (upstream lines 391–392 and 463–469), let $S_i^{(t)}$ be total supply, $D_i^{(t)}$ be total demand, and $\text{surplus}_i^{(t)} = S_i^{(t)} - D_i^{(t)}$. Defining the two-sided relative imbalance $\rho_i^{(t)}$:

$$\rho_i^{(t)} = \frac{2\left|S_i^{(t)} - D_i^{(t)}\right|}{S_i^{(t)} + D_i^{(t)}}, \qquad \tilde{\delta}_i^{(t)} = 1.05 - 0.5^{\rho_i^{(t)}} \tag{4.14}$$

[7] Hahnel, Weisdorf, and Szczepanczyk, "Computer Simulations of Participatory Planning," chap. 9 in Hahnel 2021, 173–194; Szczepanczyk 2023, 43–54; audited against upstream msszczep/pequod-cljs at commit 8bd46b5c233f0e48c767ad6d5069b9437a67c38b (src/clj/pequod_cljs/csvgen.clj, 836 lines; trimmed copy in research/sources/pequod_cljs_csvgen_apache2.clj, 799 lines).

the damped percentage price delta $\delta_i^{(t)}$ and multiplicative price update are:[8]

$$\begin{aligned}\delta_i^{(t)} &= \max\left(0.001,\ \tilde{\delta}_i^{(t)} \cdot \min\left(0.25, \rho_i^{(t-1)}\right)\right)\\ P_i^{(t+1)} &= P_i^{(t)}\left(1 + \operatorname{sign}\left(D_i^{(t)} - S_i^{(t)}\right)\delta_i^{(t)}\right)\end{aligned} \tag{4.15}$$

The upstream code computes $\max(0.001, \min(\tilde{\delta}, \tilde{\delta} \cdot \min(0.25, \rho^{(t-1)})))$. Because $\tilde{\delta} \in [0.05, 0.80]$ is positive and $\min(0.25, \cdot) \leq 0.25$, the inner min always returns its second argument, so the simplified form above is identical. (A commented-out variant in the same file, $\min(\tilde{\delta}, \rho^{(t-1)})$, is the one in which the outer min would bind.) Our programmatic audit of the committed CSV files[9] reports the subsets the book actually cites. Across dep1ex61–100.csv ($n = 40$), Year 1 reaches the 5% threshold (:green) in 11.85 rounds on average and the 3% threshold (:blue) in 19.20 rounds; Year 2 reaches 5% in 6.58 rounds. Averaging all 99 files other than dep1ex100.csv yields 11.505 Year-1 5% rounds and must not be quoted as 11.85. File dep1ex100.csv is a committed large-run log; there is no matching dep1ex100.clj. The 50 dep2ex*.csv files have no corresponding .clj generators in the clone; a code comment in csvgen.clj associates a second experiment family with toothaches, but committed dep2ex CSVs do not show the large between-section exponent jump that would mark a toothache run. Because no wall-clock timing log exists in the upstream repository, any per-iteration wall-clock figure (~ 2.5 s) is an unverified estimate. Hahnel (2021) Ch. 9 reports the Cobb–Douglas simulations; that chapter was consulted only through secondary notes and does not claim a page-level reading of increasing-returns results.

4.4.2 Audit of pequod-cljs and the "Toothache" Construction

Direct inspection of wcs.clj, ccs.clj, csvgen.clj, and all 70 dep1ex*.clj generators establishes four structural properties of pequod-cljs. Fifty files (dep1ex01–50) are stored directly; twenty (dep1ex51, dep1ex60–78) are Git LFS objects of about 160 MB each, fetched from the repository's media endpoint and checked against the SHA-256 in their pointers. Together they hold 750,000 Worker Council records.[10]

1. Unconstrained Closed-Form Cobb-Douglas Optimization: Each WC_j maximizes $\Pi(x, e) = \lambda a e^c \prod_{m=1}^{n} x_m^{b_m} - \sum_{m=1}^{n} p_m x_m - s e^k$ via symbolic Mathematica first-order conditions (solution-1..8, upstream lines 221–401) without enforcing $SB_j/SC_j \geq 1$ or labor stock conservation $\sum_j \Delta l_j = 0$.
2. "Increasing returns" (create-toothaches) and unboundedness: In wcs.clj (lines 31–70), each WC_j has $n \in \{3, \ldots, 8\}$ inputs with baseline exponents $\sum_{m=1}^{n} b_m \in [0.75, 0.85]$, effort elasticity $c \in [0.05, 0.10]$, and effort disutility

[8] Szczepanczyk 2023, eq. 9, 48; upstream csvgen.clj lines 391–392 (get-deltas) and 463–469.
[9] Performed by research/analysis/verify_r2.py in the project repository.
[10] Checked by research/analysis/verify_pequod_datasets.py in the project repository.

exponent $k \in [3,4]$ (:du). Along the ray $x_m(t) = t$, $e(t) = t^{1/k}$, revenue scales as $t^{\sum b_m + c/k}$ against linear cost, so Π is bounded with an interior maximum along that ray iff $\sum b_m + c/k < 1$. (Joint concavity of Π on (x,e) is a stronger, distinct condition and is not claimed.) create-toothaches in csvgen.clj selects 20% of councils and adds 0.1 to each b_m. Across all 750,000 records the baseline index $\sum b_m + c/k$ lies in $[0.7696, 0.8781]$; with the toothache increment it would lie in $[1.0696, 1.6652] > 1$ for every record, and hence for every selected council, so the FOC is a saddle along the scale ray and the formula predicts downward-sloping supply. Whether the committed CSVs contain runs with toothaches could not be established. -main currently sets toothaches? true, but the CSVs do not record production exponents, so the increment cannot be seen directly; and the indirect sign one would expect, a change in convergence between the Year-1 and Year-2 sections of a file, is small (the first round reaching 5% changes by at most 0.03 rounds on the files inspected). This is a question about which runs were committed, not a finding about Hahnel (2021) Ch. 9, which describes the experiments and was not read at page level for this study; it should be put to the simulation's authors.

3. Exogenous Consumer Budgets: In consume (upstream lines 188–204), each Consumer Council's budget (:income cc) is fixed at $5,000$ across all iterations.

4. Two-Sided Resource Tolerance Without Pollutants: Natural resources are given a fixed supply of $1,000$ (upstream line 168) and priced within the two-sided $\tau = 5\%$ tolerance (permitting up to $\sim 5.1\%$ overshoot), without CAP pollutant pricing or strict one-sided ceiling enforcement.

4.4.3 Why General Convergence Cannot Be Guaranteed by Heuristic Tâtonnement

Simply adding heuristic proximal regularization and trust regions to Eq. (4.15) cannot guarantee convergence when incomes feed back on prices or when technologies are non-convex, on both theoretical and empirical grounds:

- Income feedback: Scarf's (1960) example and trust-region stalling. By the Sonnenschein–Mantel–Debreu theorem and Herbert Scarf's (1960) counterexample (see also Sonnenschein 1973), price tâtonnement can cycle indefinitely in an exchange economy, where each person's income is the market value of what they own and so moves with relative prices. Running Eq. (4.15) on Scarf's three-good economy[11] confirms that over 3,000 rounds $\max_i \rho_i^{(t)}$ never drops below 10.54% ($\beta \in \{\text{None}, 0.25, 0.10, 0.02\}$), while a decaying trust region ($\beta^{(t)} = 0.25 \cdot 0.99^t$) freezes proposals at a 56.60% imbalance: shrinking trust regions stall, they do not clear the cycle.

  The cause is the income feedback, not the complementarity of the goods. In a control run with the same Leontief consumers, starting prices, and price rule,

[11] Run with research/analysis/verify_economics.py in the project repository, which also runs the fixed-budget control.

but with each budget fixed at one third of the value of total supply (so relative incomes do not move with relative prices), the same rule reaches 5% in round 9 and stays below 0.02%. This agrees with Cheung, Cole, and Devanur (2020), who show that tâtonnement converges in Fisher markets (fixed budgets) with Leontief and all complementary CES utilities, because it is then a form of gradient descent on a convex program.[12] Parecon incomes are set by effort, not by ownership, so Scarf's mechanism does not carry over directly. The relevant risk for DPEP is narrower: budgets depend on prices and plans through the effort cap (SB_j/SC_j in Eq. 4.7), credited hours and CAP compensation (Eq. 4.8). Whether that feedback is strong enough to cause cycling is untested; it is the endogenous-income question that Algorithm 3d and open question P2 address. Scarf's economy is therefore kept as a stress test of income feedback, not as evidence that complementary goods defeat council iteration.

- Indivisible Plants vs. One-Good Shapley–Folkman Smoothing: For a single indivisible facility with $S(p) = 100 \cdot \mathbb{I}_{p \geq 10}$ and demand $D(p) = 70 - p$, no clearing price exists: the theoretical minimum imbalance is $\rho = 50.0\%$, and over 500 rounds Eq. (4.15) never falls below 50.37%. Disaggregating that capacity across $K = 20$ plants of size $100/K$ with staggered thresholds $8.5 + 3i/(K-1)$ yields a minimum imbalance of 0.40% across 500 rounds. That is the best round, not a stable gap. The last-100-round maximum at $a = 70$ is 8.4%; a sweep $a \in [60, 80]$ in steps of 0.5 has median last-100-round max 6.0%, worst 9.5%, and only 24% of cases stay $\leq 5\%$ in the last 100 rounds.[13] Identical thresholds at 10 give no smoothing ($\min \rho \approx 50\%$). Starr (1969) bounds the Hausdorff gap between the sum of non-convex sets and its convex hull; this one-good staggered-threshold test does not instantiate that theorem for interdependent Leontief markets (Starr 1969).

4.4.4 Research-Candidate Solvers (Algorithms 3a–3d)

DPEP separates four candidate coordination algorithms to be benchmarked in Phase 1. None is a production commitment. Table 4.3 places them among the method families under consideration. All four share one definition of success.

Definition 4.2 (Feasible, cleared, and socially acceptable plans). Let $u^{\text{out}} = \sum_g p_g \max(0, S_g - D_g) / \sum_g p_g S_g$ be the value-weighted unused-output share, $u^{\text{need}} = \sum_g p_g \max(0, D_g - S_g) / \sum_g p_g D_g$ the unmet-demand share, u^{lab} and u^{cap} the shares of accredited labor hours and capital services left idle, and ΔW the relative welfare or need-satisfaction gap to a reference oracle (a central LP or MILP on the same instance, Table 4.3) where one exists.

1. A plan is feasible if $D_g \leq S_g$ for every good and every hard ecological or resource cap holds ($D_r \leq r^*$, $D_z \leq E_z^{\max}$). The one-sided rule belongs here:

[12] Cheung, Cole, and Devanur 2020 (conference version STOC 2013). The result covers multiplicative tâtonnement with suitably bounded steps, not Eq. (4.15) specifically.

[13] Computed by research/analysis/verify_r2.py in the project repository.

caps must never be exceeded, and slack against a cap is not waste.

2. A plan is cleared if it is feasible and, for produced goods, labor, and capital, $\max_i \rho_i \le \tau$, $u^{\text{out}} \le \bar{u}^{\text{out}}$, $u^{\text{lab}} \le \bar{u}^{\text{lab}}$ and $u^{\text{cap}} \le \bar{u}^{\text{cap}}$, net of the planned slack buffer SL^E.
3. A plan is socially acceptable if it is cleared, $u^{\text{need}} \le \bar{u}^{\text{need}}$ for priority (Maxim 4) categories, and $\Delta W \le \bar{\Delta}$ where an oracle exists. These thresholds are ratified politically, not derived.

Proposed defaults for Phase 1 (for author ratification): $\tau = 5\%$, $\bar{u}^{\text{out}} = \bar{u}^{\text{lab}} = \bar{u}^{\text{cap}} = 5\%$ beyond planned slack, $\bar{u}^{\text{need}} = 2\%$, $\bar{\Delta} = 5\%$ median and 15% at the 90th percentile.

The distinction matters. In the single-plant case of Section 4.4.3, Eq. (4.15) reaches one-sided feasibility at round 3 with $S = 100$ and $D = 58.48$: 42% of output is unused, and the case has no clearing price. Under the definition above the run is reported as feasible but not cleared in every one of 500 rounds. With $K = 20$ staggered plants the first feasible round (round 3) still leaves 8.4% unused, the first cleared round is round 5, and 57% of rounds are cleared. Scarf's cycle (with endowment incomes) is never even feasible: some good is over-demanded in every one of 3,000 rounds.[14]

Algorithm 3a—Proximal damped tâtonnement (heuristic).

1. Input: $P^{(0)}$, $\mu > 0$, thresholds of Definition 4.2, $T_{\max}$, caps $r^*, E_z^{\max}$.
2. For $t = 1, \dots, T_{\max}$: each council solves $\max_{\pi \in \mathcal{T}_c} W_c(\pi; P^{(t)}) - \frac{\mu}{2}\|\pi - \pi^{(t-1)}\|_2^2$ subject to $SB/SC \ge 1$ or $SC \le B$ as appropriate; IFB updates P by Eq. (4.15) and rescales so $w_{\text{base}} = 1$.
3. Halt with success when the plan is cleared. If $T_{\max}$ is reached, publish the best feasible plan with its u^{out}, u^{lab}, u^{cap} and u^{need}, labeled feasible, not cleared, or report infeasibility. No rate is claimed.

Algorithm 3b—Proximal Jacobian ADMM (Deng, Lai, Peng & Yin 2017, Algorithm 4; fixed-budget convex case only). Deng et al. treat $\min \sum_{i=1}^N f_i(x_i)$ subject to $\sum_i A_i x_i = c$ with each f_i closed proper convex. DPEP's coupling constraints are inequalities $\sum_c A_c x_c \le b$ (demand within supply and caps), so a slack block s with $f_s = \iota_{\{s \ge 0\}}$ (zero on the non-negative orthant, $+\infty$ elsewhere) is added and the constraint becomes $\sum_c A_c x_c + s = b$; the number of blocks is $N = N_{\text{active}} + 1$. With $f_c = -W_c + \iota_{\mathcal{T}_c}$:

1. Input: closed convex $\mathcal{T}_c$, fixed B_u, penalty $\rho > 0$, dual damping $\gamma \in (0, 2)$, proximal matrices $P_i \succ \rho\left(\frac{N}{2-\gamma} - 1\right) A_i^\top A_i$ (Deng et al. eq. 2.11; for $P_i = \tau_i I$ this is $\tau_i > \rho(\frac{N}{2-\gamma} - 1)\|A_i\|^2$, and for the prox-linear choice $P_i = \tau_i I - \rho A_i^\top A_i$ it is $\tau_i > \frac{\rho N}{2-\gamma}\|A_i\|^2$).

[14] Computed by research/analysis/verify_r3.py in the project repository.

2. In parallel for every block i (councils and slack), with $r^k = \sum_j A_j x_j^k - b$:

$$x_i^{k+1} = \arg\min_{x_i} \; f_i(x_i) + \tfrac{\rho}{2}\left\| A_i x_i + \textstyle\sum_{j \neq i} A_j x_j^k - b - \lambda^k/\rho \right\|^2 + \tfrac{1}{2}\|x_i - x_i^k\|_{P_i}^2.$$

3. Dual step: $\lambda^{k+1} = \lambda^k - \gamma\rho\,(\sum_i A_i x_i^{k+1} - b)$. With the slack block the dual needs no projection; indicative prices are $-\lambda$.
4. Halt when the plan is cleared (Definition 4.2) and $\|u^k - u^{k+1}\|$ is below tolerance; otherwise fail at $T_{\max}$.

Deng et al. prove convergence and an $o(1/k)$ rate on the successive-iterate quantity $\|u^k - u^{k+1}\|_{G'}^2$ (primal and dual together), not a rate to optimal welfare. The theorem does not cover endogenous budgets, ε-caps that depend on prices, renormalizing duals so that $w_{\text{base}} \equiv 1$ mid-run, or non-convex $\mathcal{T}_c$ (convergence results for ADMM on non-convex problems, such as Hong, Luo, and Razaviyayn 2016, need structural conditions that DPEP's lumpy technologies do not meet). Writing the objective as $\sum_c W_c$ assumes cardinal, interpersonally comparable well-being. Because N includes every active council, τ_i grows with N_{active}; whether that makes convergence too slow at $N = 10^6$ is untested.

A simpler-looking iteration, $x_c^{t+1} = \arg\min -W_c(x) + \langle \lambda^t, A_c x\rangle + \frac{1}{2}\|x - x_c^t\|_{P_c}^2$ with a projected dual ascent, is easily mistaken for ADMM. It is an Arrow–Hurwicz-type primal–dual proximal iteration: it omits the augmented residual term, and the Deng et al. theorem does not apply to it. In a two-council test with linear welfare and a shared capacity of 1, it oscillates indefinitely (residual ≈ 1.0, price swinging between 0 and 2 over 20,000 rounds), while Algorithm 4 with a slack block converges to residual 0 and price 1; both converge when welfare is strictly concave.[15]

Algorithm 3d—Damped budget fixed point around a fixed-budget inner solve (candidate for endogenous income). Endogenous income makes full DPEP an equilibrium problem: budgets $B_u(P, D)$ (Eq. 4.8) depend on the prices and quantities they help determine, and the effort cap couples incomes to SB/SC. The natural formulation is a mixed complementarity problem (MCP) in (x, P, B). Phase 1 needs a solver it can actually run, so the proposed candidate is an outer loop:

1. Hold budgets B^k fixed and solve the inner fixed-budget problem with Algorithm 3b (convex instances) or 3c (lumpy instances), giving (x^k, P^k).
2. Recompute warranted budgets $\hat{B}^k = B(P^k, D^k)$ from the Society Account identities and update $B^{k+1} = (1-\alpha)B^k + \alpha\hat{B}^k$ with damping $\alpha \in (0, 1]$.
3. Stop when $\|B^{k+1} - B^k\| / \|B^k\| \leq 10^{-3}$ and the inner plan is cleared; report cycling or divergence otherwise.

[15] Computed by research/analysis/reference_tests_r3.py and research/analysis/verify_r3.py in the project repository.

Assumptions and limits. Each inner solve inherits 3b's or 3c's assumptions. No convergence theorem is claimed for the outer loop; a fixed point need not be unique, and the loop can cycle when income effects are strong (the same mechanism that defeats tâtonnement on Scarf's example). As a cross-check on small instances, the MCP can be passed to a PATH-type complementarity solver; PATH is proprietary and its free license is limited to small problems, so licensing terms must be checked before relying on it, and no open-source MCP solver of comparable robustness is assumed. Lemke's method applies only to the linear complementarity case and is used as a diagnostic.

Interaction with Algorithm 3c. The 3c objective below is written at the current budgets B_u. Under endogenous income a lumpy plant decision changes incomes, and so the demand it is meant to meet; 3c is therefore run as the inner solve of 3d, and binary decisions taken at one budget vector are re-checked after each budget update. Phase 1 tests 3c first with fixed budgets and then inside 3d.

Algorithm 3c—Two-tier federation MILP + continuous ADMM (undefined until the following are fixed).

1. What councils reveal: each WC_j posts a discrete menu of plant/shift options $z_{j,k} \in \{0,1\}$ with publicly listed input–output rows (REQ-DATA-03), not its full private technology set $\mathcal{T}_j$. Continuous operating bands around each chosen z remain local.
2. Objective (provisional): lexicographic (1) one-sided ecological and resource feasibility, (2) minimize unmet consumer demand at current B_u, (3) minimize unused output and idle labor and capital (Definition 4.2), (4) minimize a public lumpy-cost index. This objective is a federation proposal, not a social-welfare theorem. Under endogenous income 3c runs inside Algorithm 3d.
3. Decision rights: binary z are ratified by the relevant industry and regional federations plus affected CAPs; a central HiGHS MILP is an aid, not an authority (Kuch 2024).
4. Infeasibility: if the MILP is infeasible, return the IIS (irreducible infeasible subsystem), raise the relevant cap or investment debate, and do not silently relax $E_z^{\max}$. Halt and publish the IIS after $T_{\max}$ unsuccessful repairs.

4.5 Protocol State Machine and Canonical CBOR Schemas

To prevent IEEE-754 floating-point non-determinism across CPU architectures (x86_64, ARMv8 smartphones, and WebAssembly), all quantities and prices in DPEP are encoded as deterministic 64-bit fixed-point integers (FixedDecimal, default scale 10^{-6} micro-units):
Canonical wire encoding. Listings 4.1 and 4.2 are an IDL sketch (proto3-like field names for readability). It is not sent as protobuf. All signed payloads use RFC

Table 4.3: Ten method families by problem class (research map, not a ranking)

Family	Problem class	DPEP status
Unconstrained Cobb–Douglas FOC	Smooth convex, fixed income	pequod-cljs baseline; rejected for production.
Damped tâtonnement (3a)	Heuristic, any $\mathcal{T}_c$	Candidate; fails Scarf (endowment income) and single-plant tests; clears Scarf's consumers with fixed budgets.
Proximal Jacobian ADMM (3b)	Convex, fixed budgets	Candidate; Deng et al. Algorithm 4 with slack block; successive-iterate rate only.
Arrow–Hurwicz primal–dual	Convex, fixed budgets	Resembles 3b but lacks the augmented term; oscillates on linear welfare; not used.
Fisher / Eisenberg–Gale	Linear utilities, fixed budgets	Fits fixed B_u; not endogenous income.
Negishi welfare weights	Convex equilibrium	Requires comparable W_c; open normatively.
Variational inequality / MCP	Endogenous income + prices	Leading formulation. Phase-1 candidate: Algorithm 3d (damped outer loop); PATH-type solver as small-instance cross-check, subject to license.
Federation MILP (3c)	Discrete lumpy $z \in \{0,1\}$	Candidate; needs ratification and IIS exit.
Scarf / Lemke path-following	Linear complementarity	Diagnostic for cycles; not a planning procedure.
Central LP (Cockshott-style)	Known A, K, R, L	Rejected as workplace authority; usable as Phase-1 oracle.

8949 §4.2 Canonical CBOR (CDE) with a fixed global scale of 10^{-6} (int64 micro-units). Multiplication uses 128-bit intermediate products, round-half-to-even, saturating on overflow (the round fails closed rather than wrapping). Division by zero is a fault. The map 0.5^{ρ_i} is a degree-6 minimax polynomial on $\rho \in [0, 2]$ with published coefficients; outside that interval the update is clipped. Protocol checks verify the rounded inequalities $SB/SC \geq 1$ and $SC \leq B$ after this arithmetic. They do not prove that a council maximized a truthful W_c, and they do not exist as KKT certificates for MILPs.

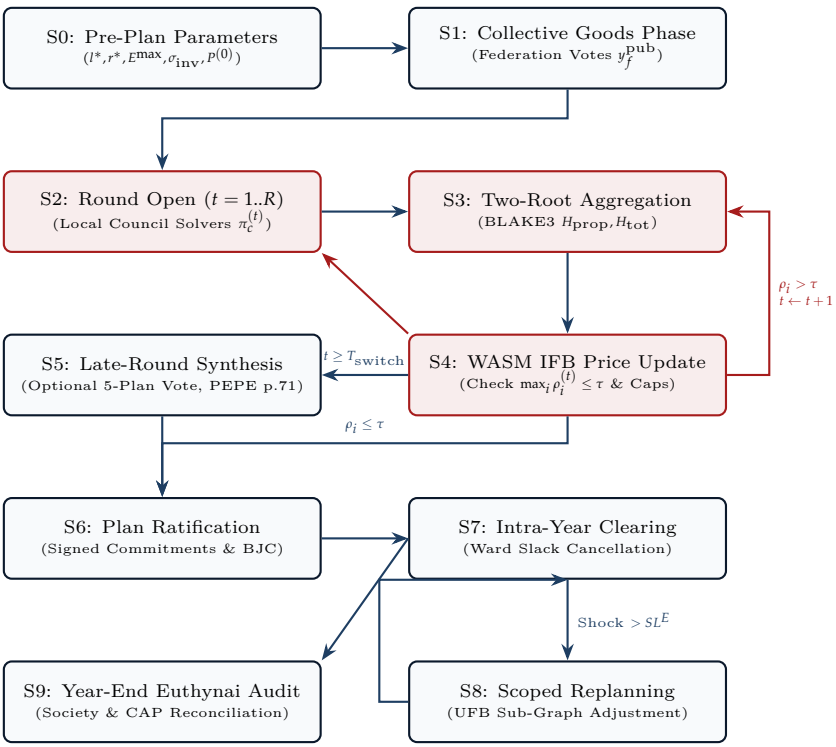


Figure 4.1: Complete 10-state DPEP annual planning and intra-year operational state machine

Listing 4.1: IDL sketch of DPEP item vectors and consumer-council proposals. Signed encoding is RFC 8949 Canonical CBOR, not protobuf.

```
// IDL sketch only --- not the wire format
package dpep.v1;

message SparseItemVector {
  repeated uint32 item_ids = 1;          // Strictly ascending catalog indices in Omega
  repeated int64 micro_quantities = 2;   // Strictly positive quantities (> 0) in 10^-6 micro-units
}

message QualitativeAnnotation {          // Implements PEPE Ch. 6 'Lists' with GDPR redactability
  uint32 item_id = 1;
  string summary_code = 2;
  string narrative_text = 3;
  bytes redactable_pointer_cid = 4;      // Mutable pointer supporting tombstone erasure
}

message ConsumerCouncilProposal {
  string domain_tag = 1;                 // Fixed: "DPEP-WD-v0-CC-PROPOSAL"
  uint32 plan_year = 2;
  uint32 round_number = 3;
  bytes council_id = 4;
  uint32 accredited_member_count = 5;
  SparseItemVector private_demand = 6;   // Coarse-category totals (HRU x s_g) from the K Prio3 aggregators
  SparseItemVector local_public_demand = 7;
  int64 total_social_cost_micro = 8;
  int64 warranted_budget_micro = 9;
  bool requests_exception = 10;
  repeated QualitativeAnnotation qualitative_list = 11;
  bytes vdaf_transcript_root = 12;       // Commits Prio3 aggregate-share commitments
  bytes signer_bitmap = 13;
  repeated bytes council_signatures = 14; // k-of-n, suite per CryptoSuiteId (Ed25519 under tiered policy)
}
```

Listing 4.2: IDL sketch of worker-council proposals and IFB round checkpoints (continuation of Listing 4.1).

```
message WorkerCouncilProposal {
  string domain_tag = 1;                    // Fixed: "DPEP-WD-v0-WC-PROPOSAL"
  uint32 plan_year = 2;
  uint32 round_number = 3;
  bytes council_id = 4;
  uint32 industry_federation_id = 5;
  SparseItemVector proposed_outputs = 6;
  SparseItemVector intermediate_inputs = 7;
  SparseItemVector natural_resources = 8;
  SparseItemVector total_labor_requested = 9;
  SparseItemVector total_capital_requested = 10;
  SparseItemVector pollutant_emissions = 11;
  int64 effective_effort_hours_micro = 12; // INV-4B: sum(h_u*eps_u) <= (1.0+dl)*
      min(eps_max,SB/SC)
  int64 social_benefit_micro = 13;
  int64 social_cost_micro = 14;
  int64 sb_sc_ratio_micro = 15;
  bool requests_appeal = 16;                // Subject to soft-budget cap at IF_k
  repeated QualitativeAnnotation qualitative_list = 17;
  bytes signer_bitmap = 18;
  repeated bytes council_signatures = 19;
}

message RoundAggregationAndPrices {
  string domain_tag = 1;                    // Fixed: "DPEP-WD-v0-ROUND-
      CHECKPOINT"
  uint32 plan_year = 2;
  uint32 round_number = 3;
  repeated int64 indicative_prices_micro = 4; // Normalised so w_base_micro == 1
      _000_000
  repeated int64 total_supply_micro = 5;
  repeated int64 total_demand_micro = 6;
  repeated int64 planned_slack_micro = 7;
  repeated int64 one_sided_shortage_micro = 8; // max(0, D_i - S_i)
  int64 base_hourly_remuneration_micro = 9;    // Fixed numeraire: 1_000_000
  bool strict_one_sided_feasible = 10;
  bytes proposals_merkle_root = 11;            // 32-B BLAKE3 root over sorted
      council proposals (512-B proof)
  bytes category_totals_merkle_root = 12;      // 32-B BLAKE3 root over category
       sums
  bytes council_participation_bitmap = 13;     // 2 bits/council: 01=Active, 10=
      Partitioned, 00=Excluded
  bytes deterministic_wasm_code_hash = 14;
  bytes validator_quorum_bitmap = 15;          // or witness bitmap (witnessed-
      log option)
  repeated bytes validator_signatures = 16;    // aggregated BLS or Ed25519
      under tiered policy
  repeated bytes pq_anchor_signatures = 17;    // k-of-n SLH-DSA (BFT) or ML-DSA
      -44 cosigs (log)
}
```

CHAPTER 5

Distributed Systems Architecture, Consistency, and Low-Power Participation

> Planning is computerized. All councils can access data banks of all facilitation boards and all other councils. The only thing demanding about flexible information access is that the computer system not only automatically accept but also sum and compare proposals to ascertain the status of any good.
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 6, "Preparing First Proposals"[1]

5.1 Why These Properties: Architectural Commitments and Their Limits

Architecture D borrows six properties from the vocabulary of distributed systems: decentralization, trust minimization, censorship resistance, open source, tamper-evident history, and disintermediation. Chapter 2 warned that none of them makes an economy democratic (Section 2.1). This section says why the architecture adopts them anyway. Each answers a specific way in which participatory planning could be captured, falsified, or repressed, and each has a limit beyond which institutions, not software, must carry the load. Table 5.1 summarizes the argument.

Decentralization. Participatory economics is already a decentralized planning procedure: no central agency composes the plan, and councils and federations revise their own proposals (epigraph to Chapter 1). The architecture applies the same rule to computation. No single server, operator, or vendor holds the plan, the price-update rule, or the record of who proposed what. This is the protocol's answer to the coordinator-class critique (Section 2.1): a planning system that runs on one institution's machines gives that institution's technicians the power the participatory institutions are meant to deny them. Decentralization does not mean the absence of structure. Federations, facilitation boards, and a validator committee remain (Section 5.4). What it rules out is a single point at which the plan can be seized, switched off, or quietly edited.

Trust minimization, not trustlessness. The design does not assume good people. It assumes the people who will use it: people formed by capitalism, racism, sexism, and a lifetime without democratic control over their work, some of whom will cheat,

[1] Albert and Hahnel 1991b, chap. 6 (lf6_2.htm).

collude, discriminate, or reach for power when they can. This is an old principle of constitutional design. Hume called it "a just political maxim, that every man must be supposed a knave," while noting that the maxim is false as a description of people (Hume 1741); Madison wrote that "if men were angels, no government would be necessary" (Madison 1788); and Ostrom found that durable commons rely on monitoring and graduated sanctions rather than on virtue (Chapter 2 and Table 2.2). DPEP makes part of this principle mechanical. Where mathematics can check a claim, it is checked; where it cannot prevent misconduct, the design tries to make misconduct visible; where neither is possible, it names the institution that carries the risk (Figure 7.5).

"Trustless" overstates what this achieves. Privacy of household reports needs at least one honest aggregator, correct totals need all aggregators honest and independent, and physical output is checked by people, not proofs (Section 7.6). The accurate term is trust-minimized. Trust minimization is also consistent with the endogenous-preference argument of Section 3.2. If institutions shape the people who live within them, a participatory economy cannot wait for people formed by cooperative institutions before it builds those institutions. Verification is scaffolding for the transition: it lets cooperation begin among people who do not yet trust one another, without first requiring them to become better people.

Censorship resistance and cryptography as protection. Rogaway argues that "cryptography rearranges power," that it can be developed in directions that favor either the weak or the powerful, and that mass surveillance, by suppressing dissent, makes social progress unlikely (Rogaway 2015, 1, 18–19, 29). DPEP's cryptography is chosen to shield people, communities, and movements, and it faces two kinds of adversary. The first is external: states and private actors able to surveil, coerce, or shut down a cooperative economy and the movements around it. Against them, household consumption stays hidden from councils, federations, and anyone who seizes some but not all of the aggregators (Section 7.4); membership credentials do not link an epoch's report to a person (Section 7.3); and channels use post-quantum key exchange now, because recorded traffic can be decrypted later (Section 7.2.1). The second adversary is internal: a federation that quietly drops a dissident council's proposal, a facilitation board that adjusts a total, or a majority that singles out a minority. Against these, signed receipts and the Two-Root inclusion path let an excluded council prove that it was left out (Section 5.3.1). The source tradition worries about capture from within as much as repression from without, and the protocol treats both as threats. Two external vectors sit outside the protocol and must be designed around: the two mobile platform owners can remove an app from their stores, and push notifications routed through their services reveal to them who uses the app and when. Section 5.5 answers both with distribution outside the stores, reproducible client builds, and notifications that carry no content.

Disintermediation, not the absence of intermediaries. Participatory planning has intermediaries: iteration facilitation boards, federations, aggregators, and validators. The claim is therefore narrower than "no middlemen." DPEP removes intermediaries that are unaccountable, irreplaceable, or able to extract rent, of the kind platform firms place between producers and users. The intermediaries it keeps compute nothing that cannot be recomputed (prices come from a ratified deterministic rule that every node runs), report nothing that cannot be checked (totals are committed in roots that councils verify), and hold their position by sortition or by revocable delegation, so they can be replaced (Sections 5.4 and 7.7).

Open source. Publishing the code is a democratic requirement, not a development preference. Lessig's observation that in networked systems code acts as law (Lessig 1999, 2000) implies that the code must be as public as law: anyone must be able to read the price-update rule, the aggregation protocol, and the eligibility checks. Open source also gives a right of exit: if a faction captures the protocol's governance, others can fork it and leave. And it makes change visible: a rule that every node executes and that is ratified in public cannot be altered by a vendor or a state without the alteration being seen.

Tamper-evident history, not immutable personal data. Immutability applies narrowly. Commitments, checkpoints, ratified rules, and plan history are append-only and tamper-evident, so no one can rewrite what was proposed, aggregated, or agreed after the fact. Personal data is treated in the opposite way: profiles stay on the participant's device, Prio3 input shares are deleted when the challenge window closes (Section 7.4.3), and qualitative records sit behind mutable pointers that can be erased (Section 11.2). The two rules serve one purpose: public decisions stay accountable, and private lives are not permanently recorded.

Design Principle 5.1 (Design for People as They Are). No safeguard in DPEP may depend on participants, facilitators, validators, or operators acting altruistically. Each protection is either guaranteed by cryptography under stated assumptions, made detectable, or assigned to a named institution with consequences for failure. A safeguard that rests on good conduct alone is treated as unfinished: it is listed as an open question or a weakness (Chapters 10 and 12), not claimed as a protection.

Limits common to all six. Three limits apply across the table.

1. Protection is symmetric. The tools that hide a dissident's consumption from a state also hide a household's misreporting from its neighbors. DPEP draws the line between public decisions, which are attributed and auditable (proposals, receipts, council votes), and private life, which is protected (household consumption). Where the line falls, for example which ballots must be secret (Section 7.6), is a constitutional choice, not a technical one.

Table 5.1: The six architectural properties: what each answers, how DPEP provides it, what it rests on, and its limit

Property	Answers	DPEP mechanism	Rests on	Limit
Decentralization	A single operator, vendor, or state seizing, editing, or switching off the plan; a coordinator class	Local-first council state; federated aggregation; rotating validator committee drawn by sortition	Architecture and node-placement rules	Informal cliques can still dominate (Freeman).
Trust minimization	Cheating, collusion, and discrimination by people as they are	Signatures, inclusion proofs, deterministic recomputation; detection where proof is impossible	Cryptography, detection, and institutions (Fig. 7.5)	Correct totals and physical output still rest on institutions.
Censorship resistance (external)	Surveillance and coercion by states and private actors	Prio3 household privacy; unlinkable credentials; post-quantum key exchange; distribution outside app stores; content-free notifications	Cryptography (≥ 1 honest aggregator)	Fails if every aggregator is compromised; credentials interim classical; covers annual requests, not consumption at the point of use.
Censorship resistance (internal)	A federation or board omitting or altering a council's proposal	Signed receipts; Two-Root inclusion path; challenge window	Detection	Detects omission; does not prevent it.
Disintermediation	Unaccountable, irreplaceable, or rent-extracting intermediaries	Ratified deterministic rules; verifiable roots; sortition and revocable delegation	Detection and institutions	Facilitators and solver experts can still set defaults.
Open source	Hidden rules; silent change; capture of the protocol	Public code and specification; public ratification of rule changes; forking	Institutions	Few people read code; forking splits coordination.
Tamper-evident history	Rewriting what was proposed or agreed	Append-only roots and checkpoints under signature	Cryptography, given an archive	Needs the leaf archive to be kept.
Erasable personal data	Permanent records of private life	Local profiles; deletion of input shares; erasable pointers	Design and operational practice	Deletion is an operational promise, not a proof.

2. Decentralization can hide power. Freeman (1972) showed that groups without formal structure concentrate power in unaccountable informal cliques. A protocol with no center can still be dominated by the facilitators, solver experts, and articulate members who set its defaults (Chapter 12). Participatory economics pairs decentralization with balanced job complexes, sortition, and explicit rules for this reason. The software inherits that need; it does not meet it.
3. Cryptography does not create solidarity. A verifiable record of a decision says nothing about whether the decision was democratic (Section 2.1). The six properties are safeguards that a participatory economy at scale needs, not conditions that make one.

5.2 Why Participatory Economics Is Not a High-TPS Blockchain Problem

When we examine the computational structure formalized in Chapter 4, three architectural truths emerge:

1. Annual Planning Is an Epoch-Synchronized Bulk-Synchronous Parallel (BSP) Process: Millions of participants and tens of thousands of councils draft proposals locally, aggregate them up nested geographic and industrial federation trees, and compute updated indicative prices $P^{(t+1)}$ once per planning round (10–20 rounds per year). Global state finality is required only at iteration boundaries and at plan ratification.
2. Intra-Year Execution Is Hierarchically Scoped: As Albert and Hahnel (1991a, chap. 4, 66–67) specify, routine intra-year consumer preference shifts cancel out at the neighborhood, ward, or city federation level within planned slack buffers ($SL^E \approx 3$–5%) without altering aggregate production plans.
3. Participation Must Work Offline and on Resource-Constrained Devices: Workers and consumers in rural collectives or intermittently connected networks must be able to deliberate, optimize local proposals, and sign ballots offline (Kleppmann et al. 2019).

5.3 The Four-Layer Hybrid Architecture (Architecture D)

Figure 5.2 traces one planning round message by message across the layers, and Figure 5.4 (Section 5.4) shows where each node role sits.

5.3.1 Two-Root Authenticated Aggregation, Completeness Bitmaps, and Physical Boundaries

Naively embedding an 80 KB dense 10,000-category vector inside every internal node of a depth-16 Sparse Merkle-Sum Tree would cause a single mobile inclusion

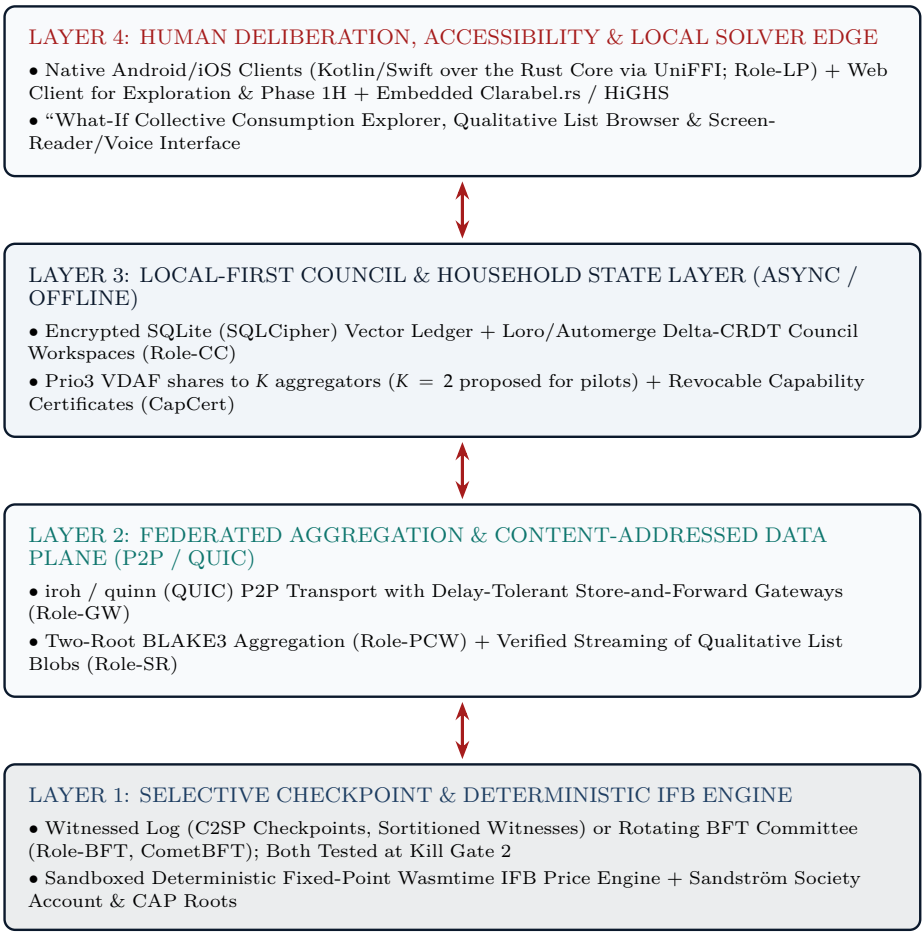


Figure 5.1: Four-layer architecture of the Distributed Participatory Economics Protocol (Architecture D)

path to bloat to 16×80 KB = **1.28** MB—and still could not prevent a dishonest aggregator from fabricating off-path sibling subtrees unless auditors recomputed the entire tree from the leaves up. Moreover, a Merkle inclusion proof for council c proves membership, not completeness (whether dissident council c' was omitted) or physical truth (whether reported output physically exists).

A uniform 5.2 kB proposal size is too small: 5,200 B less one 3,373 B hybrid Ed25519 + ML-DSA-65 signature (3,309 B for ML-DSA-65 plus 64 B for Ed25519, concatenated as in RFC 10073) leaves 1,827 B, about 150 sparse items at 12 B each (4-byte id, 8-byte quantity), while a neighborhood's summed demand over 3,000 private categories is about 36 kB, and every extra multi-signature signer adds a further 3,373 B. Table 5.2 reworks sizes by council type under two signature suites: hybrid throughout, and the classical Ed25519 suite that the proposed tiered policy

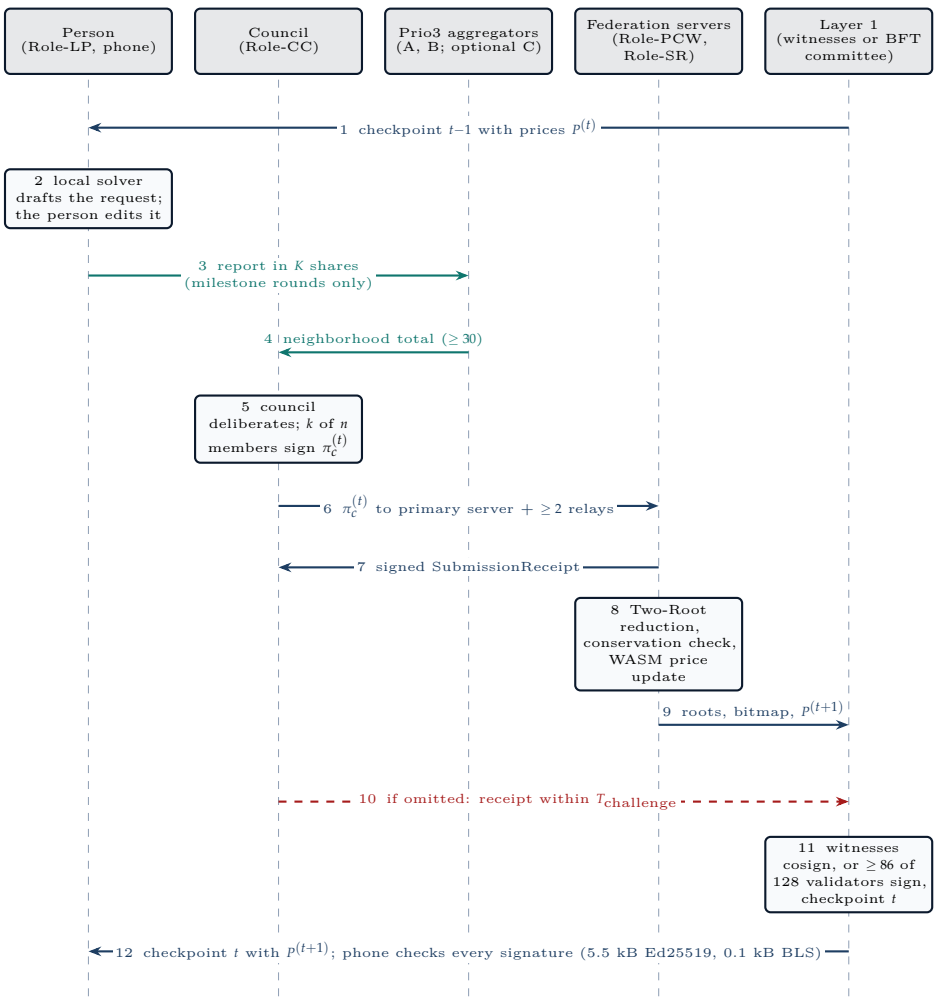


Figure 5.2: One planning round as a message sequence. Teal steps carry private household data and run only at milestone rounds; the dashed crimson step is the censorship challenge (Section 5.3.1).

(Section 7.2.1) uses for per-round objects. Council threshold signatures are not specified: until a standard post-quantum threshold scheme exists (open question P12), a council signs with a k-of-n multi-signature, i.e., k separate signatures plus a signer bitmap.

Table 5.2: Council proposal sizes and round archive at $N = 10^6$ by signature suite (raw canonical CBOR; analytical)

	Hybrid (3,373 B)		Ed25519 (64 B)	
Council type	1 signer	5 signers	1 signer	5 signers
Worker council, 150 items	5.6 kB	19.1 kB	2.3 kB	2.5 kB
Consumer council, 200 coarse categories	6.2 kB	19.7 kB	2.9 kB	3.1 kB
Consumer council, 3,000 categories	39.8 kB	53.3 kB	36.5 kB	36.7 kB
Round archive, 20k WC + 20k CC				
Coarse	235 MB	775 MB	103 MB	113 MB
Dense	907 MB	1.45 GB	775 MB	785 MB
Raw archive per 10-round year, coarse	2.3 GB	7.7 GB	1.0 GB	1.1 GB
Signature checks per round	40k	200k	40k	200k
Time to verify them, one desktop core (measured)	2.0 s	9.9 s	0.3 s	1.7 s

Header 400 B (estimate) + 12 B per item + one signature per signer. Signatures are incompressible, so compression helps little where they dominate. Ed25519 verification times use batches of 64.[2]

At either suite, a round's proposals are small enough that full replication is cheaper than heavy internal-vector proofs. DPEP therefore uses a Two-Root Authenticated Aggregation Protocol instead of internal-vector Merkle-sum proofs:

$$H^{(t)}_{\text{proposals}} = \text{BLAKE3-Root}\Big(\big\{\big(\text{id}_c,\ \text{BLAKE3}(\text{CBOR}(\pi_c^{(t)}))\big)\big\}_{c=1}^{M}\Big) \tag{5.1}$$

$$H^{(t)}_{\text{totals}} = \text{BLAKE3-Root}\Big(\big\{\big(i,\ S_i^{(t)},\ D_i^{(t)},\ P_i^{(t+1)}\big)\big\}_{i\in\Omega}\Big) \tag{5.2}$$

- 512-Byte Mobile Inclusion Proofs: Any mobile client (Role-LP) verifies that its council's exact proposal was included in proposals_merkle_root using a standard 16-level hash path of 16×32 B = **512** bytes.
- Full Leaf Data Availability (DA) & Conservation Verification: Auditing servers (Role-SR, Role-PCW, Role-BFT) download the round's leaf archive (0.10–1.45 GB raw; Table 5.2), verify the council signatures against the accredited roster, and check that the exact fixed-point sum equals category_totals_root before signing or cosigning the checkpoint. That is 40,000–200,000 signature verifications per round. Measured on one desktop core they take 2–10 s if every signature is hybrid and 0.3–1.7 s with batched Ed25519, so on a desktop verification load is not a reason to choose one suite over the other; archive size and phone traffic are. Raspberry Pi and community-server timings remain a Kill Gate 2 measurement, and could change that conclusion for small federations.

[2] Sizes from research/analysis/verify_signature_suites.py; timings from research/analysis/bench_signatures/ (Apple M4 Pro, one core; ed25519-dalek 3.0.0 batch verification, libcrux-ml-dsa 0.0.11), run November 7, 2026. A community server core is slower by an unmeasured factor.

- Distinguishing Censorship from Network Partitions (council_participation_bitmap): Each checkpoint commits a 2-bit-per-council bitmap (01 = Fresh Round-t Proposal, 10 = Timeout/Partition Carry-Forward, 00 = Excluded). Councils submit proposals to both their primary aggregator and ≥ 2 independent Role-SR relays, receiving timestamped signed SubmissionReceipts. Any council falsely marked 10 or 00 broadcasts its receipt during the pre-finality challenge window $T_{\text{challenge}}$ to force inclusion and fault the censoring aggregator.
- Explicit Cryptographic vs. Physical Boundary: Cryptography verifies digital authenticity, inclusion, and arithmetic conservation; it cannot verify that physical grain was harvested or that emissions sensors were not bypassed, which remain governed by federated physical euthynai spot-audits and counterparty delivery reconciliation.

Figure 5.3 shows the two trees, the phone's inclusion path, the participation bitmap, and the challenge that turns an omission into a public fault.

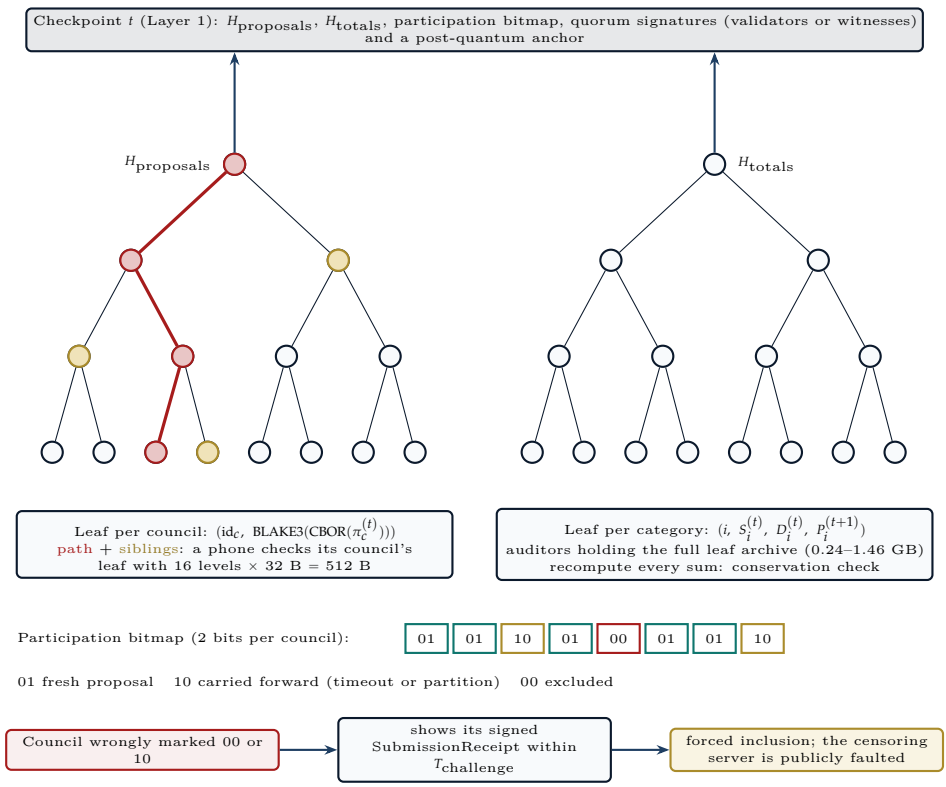


Figure 5.3: Two-Root authenticated aggregation. A phone verifies its council's leaf with a 512-byte path; auditors with the leaf archive recompute every category sum; the participation bitmap and signed receipts separate censorship from partition.

5.3.2 Layer 1: BFT Checkpoints or a Witnessed Log

Every Layer-1 block is a result that any node can recompute from published leaves under a ratified rule. Layer 1 therefore does not have to order transactions or execute contracts. It must provide four things: (a) a tamper-evident, append-only history; (b) one view for everyone, so that no council is shown a different price vector from its neighbors (split-view resistance); (c) a deadline-bound decision on each round's inclusion set; and (d) attribution of misbehavior. A Byzantine-fault-tolerant committee provides all four. So does a witnessed transparency log, the design behind Certificate Transparency and the Go checksum database, and specified for general use in the C2SP tlog-checkpoint, tlog-cosignature and tlog-witness specifications.[3]

The witnessed-log design. In each round a log operator, drawn by sortition from the federations' Planning Compute Workers, appends the round's roots (and the hash of its price and total arrays) to a log and publishes a signed checkpoint: the log's size and Merkle root. A set of witnesses, also drawn by sortition (illustratively 16, with a quorum of 11), cosigns the checkpoint only if two checks pass: the operator's consistency proof shows that the new checkpoint extends the last one the witness cosigned, as tlog-witness requires; and the witness's own recomputation of the Two-Root totals and of $P^{(t+1)}$ from the leaf archive matches. A phone accepts a checkpoint carrying a quorum of cosignatures. Requirement (c) comes, as in the BFT design, from the submission deadline, signed receipts, and the challenge window (Section 5.3.1); a witness refuses to cosign a checkpoint whose participation bitmap omits a council that has presented a valid receipt.

What each design gives up. The log design trades liveness for simplicity. A malicious or failed operator can delay a round, and only the federations' replacement procedure ends the delay; a BFT committee keeps going as long as two-thirds of its members are online. The BFT design pays for that liveness with a large always-on committee, a second language in the stack, and, at present, a choice between classical keys and pure post-quantum keys. Neither design is needed for total ordering, because the round's result is deterministic.

Hash functions. RFC 6962 Merkle trees, and therefore C2SP checkpoints and existing witnesses, use SHA-256. If the log design is adopted, checkpoint and log roots would use SHA-256 under CryptoSuiteId for compatibility with existing witnesses, monitors, and tools, while content-addressed blobs keep BLAKE3, which iroh uses. The cost is two hash functions in the protocol instead of one; SHA3-256 remains the registered alternative for the Two-Root trees (Table 7.2).

[3] C2SP, https://c2sp.org/tlog-checkpoint, https://c2sp.org/tlog-cosignature, https://c2sp.org/tlog-witness (versions 1.1.0 cited by tlog-witness; checked November 7, 2026). tlog-cosignature defines Ed25519 and ML-DSA-44 cosignatures and states that "ML-DSA-44 SHOULD be used for new deployments."

Table 5.3: Layer-1 options compared against the four requirements and operating criteria (judgment; both are tested at Kill Gate 2)

Criterion	BFT checkpoint committee (CometBFT reference)	Witnessed log (C2SP checkpoints)
History and attribution, (a) and (d)	Signed chain of checkpoints; equivocation detectable.	Append-only log with consistency proofs; witnesses refuse to cosign forks, so a fork is attributable to the operator.
Split view, (b)	Prevented while at most 42 of 128 validators are faulty.	Prevented while fewer than a quorum of witnesses collude with the operator.
Inclusion decision, (c)	Deadline, receipts, and challenge window.	The same.
Liveness	Continues while 86 of 128 are online.	One operator per round can stall it. Recovery is a replacement operator taken from a pre-drawn order: an institutional fallback, not a protocol one.
Operator burden	128 validators, 24/7 routing, consensus gossip.	One operator; witnesses are small services that need to be online only to cosign.
Hardware	Community servers.	Witnesses fit a Raspberry Pi; the operator needs a server.
Language fit	CometBFT is Go beside a Rust core; Malachite (Rust) is alpha and unaudited.	Mature logs and witnesses are Go (Tessera, Trillian, Sigsum); Rust libraries exist but are young.
Phone check per checkpoint	86 signatures: 290 kB hybrid, 5.5 kB Ed25519, 0.1 kB aggregated BLS (an engine that aggregates; CometBFT does not).	11 cosignatures: about 0.9 kB Ed25519 or 30 kB ML-DSA-44.
Post-quantum path	Engine-dependent: CometBFT 0.40 has pure ML-DSA-65 keys; BLS keys exist only behind a build flag, without vote aggregation.	ML-DSA-44 cosignatures already specified and recommended.
Post-quantum history anchor	3-of-5 federation SLH-DSA signatures: 48.7 kB per block, 20.7 MB/yr.	11 ML-DSA-44 witness cosignatures: 26.8 kB per block, 11.4 MB/yr, plus an annual SLH-DSA backstop (Table 7.3).
Maturity	M4–M5 algorithm; DPEP use M2.	M4 pattern (Certificate Transparency); DPEP use M1.

Recommendation (proposed, for author ratification). Test both designs at Kill Gate 2 on the same workload and fault-injection suite (Table 9.2). CometBFT remains the reference BFT engine, provisionally and without lock-in. The weighted comparison (Table 6.1) cannot separate the two: they score 4.07 and 4.10, and under random perturbation of the scores each ranks first about half the time. The deciding evidence is operational: round latency under a stalled operator, recovery time, and the effort of running each in a co-operative federation.

5.4 Granular Consistency Mapping and Node Role Taxonomy

[4] Shapiro et al. 2011.
[5] Buchman, Kwon, and Milosevic 2018; Yin et al. 2019.

Table 5.4: Consistency models and partition trade-offs across economic subsystems

Economic Subsystem	Consistency Model	Latency Target	CAP Partition Behavior	Architectural Justification
1. Proposal Drafting (S1–S2)	Strong Eventual Consistency (Delta-CRDT + SQLite)[4]	< 50 ms local	AP (Offline-First): Drafts continue on local LAN/device.	Unsubmitted drafts affect no external council; offline usability is paramount.
2. Council Ratification Vote	Council-Scoped Quorum (k-of-n multi-signature)	Seconds to hours	Locally CP on Council LAN: Independent of WAN.	Only members of WC_j or CC_h hold authority over their council's proposal.
3. Federated Round Aggregation (S3)	Deterministic Two-Root Reduction	1–10 min per round	Bounded-Wait CP + Fallback: Carries forward $\pi_c^{(t-1)}$ if WAN severed past grace window.	Prevents a single disconnected hamlet from halting national planning while preserving verifiable inclusion.
4. IFB Price Update & Final Plan (S4, S6)	Single agreed view: epoch BFT[5] or a witnessed log (Section 5.3.2)	2–15 s per block	CP (Safety over Liveness): BFT needs $\geq 2f+1$ of $3f+1$ validators; a log needs its operator and a witness quorum.	Every council must optimize round $t+1$ against the exact same indicative price vector $P^{(t+1)}$.
5. Intra-Year Fulfillment (S7)	Depot records + daily federated rollup (Section 7.5)	< 500 ms (offline-capable)	AP with Bounded Credit: Online check for rationed goods; offline only at the home depot, up to one credit cap per person per day.	Point-of-delivery pickups at neighborhood depots must never block on global consensus. Privacy and double-spending design still open.

Figure 5.4 places the six node roles of Table 5.5 in the federation structure.

5.5 Client Platform, Distribution, and Notifications

The participant's client holds keys, drafts proposals, splits household reports into shares, and verifies checkpoints. Three platform facts, checked on November 7, 2026, rule out a progressive web app as the primary client. First, a web app cannot reach the hardware keystores (Android Keystore and StrongBox, Apple's Secure Enclave) on which split hybrid custody depends (Section 7.6): keys created through WebCrypto are held by the browser in software. Secondly, browsers do not broadly expose ML-KEM or ML-DSA: the WICG "Modern Algorithms" specification is a draft community report, and Chrome's support is an experiment

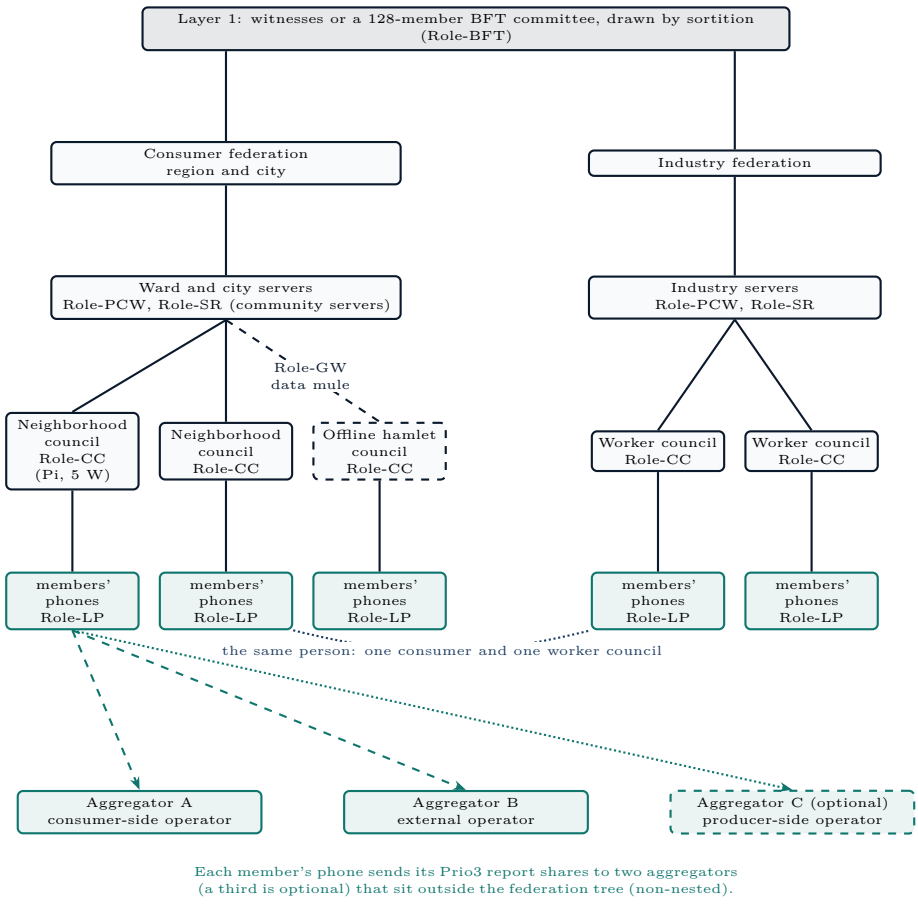


Figure 5.4: Federation topology and node placement (illustrative). Councils and servers nest by geography and industry; the Prio3 aggregators (two proposed for pilots, a third optional) sit outside that nesting, and the Layer-1 witnesses or BFT committee are drawn by sortition across federations.

Table 5.5: Six-role hardware realism matrix (supporting legacy smartphones and Raspberry Pis)

Node Role	Core Responsibilities	Entry Smartphone (3GB RAM ARM)?	Raspberry Pi 4/5 (4–8GB RAM)?	Community Server (8+ Cores)?
1. Lightweight Participant (Role-LP)	Native app (Kotlin/Swift over the Rust core): local budget QP solver, Prio3 share generation, CRDT sync, checkpoint verifier, ballot signer, keys in the platform keystore.	YES (Primary) (RAM unverified)	YES	YES
2. Council Coordinator (Role-CC)	Hosts WC_j or CC_h local Wi-Fi CRDT relay, runs workplace LP/QP solver, assembles the council k-of-n multi-signature.	YES (Failover Hotspot)	YES (Ideal $50 Appliance at 5W)	YES
3. Storage Replica (Role-SR)	Pins regional content-addressed BLAKE3 blobs for qualitative Lists and audit logs.	NO (Flash wear & cell data limits)	YES (with USB SSD)	YES (Primary)
4. Planning Compute Worker (Role-PCW)	Computes Two-Root reductions, deterministic WASM price updates, and input-output analytics.	NO	PARTIAL (Up to 5,000 councils)	YES (Primary IFB Node)
5. Validator or Witness (Role-BFT)	BFT option: member of the sortitioned committee finalizing checkpoints. Log option: witness that recomputes and cosigns each checkpoint.	NO (BFT needs 24/7 fixed routing)	YES (witness; BFT for local federations)	YES (Primary)
6. Delay-Tolerant Gateway (Role-GW)	Bridges offline mesh/LoRa/USB sneakernet bundles to federation relays.	YES (As "Data Mule")	YES (Mesh Box)	YES

behind a flag.[6] Thirdly, Safari, on iOS as elsewhere, supports neither the Background Sync nor the Periodic Background Sync API, so a web app on an iPhone cannot reliably sync or submit while closed.[7]

Recommendation (proposed, for author ratification). Make native Android and iOS apps the primary participant client: thin Kotlin and Swift interfaces around the shared Rust core (dpep-core) through UniFFI, with keys in the platform keystore. Keep the web client for read-only exploration of plans and prices, for

[6] WICG, "Modern Algorithms in the Web Cryptography API," Draft Community Group Report, October 23, 2026, https://wicg.github.io/webcrypto-modern-algos/; Chromium, "Intent to Experiment: Algorithm Updates in WebCrypto" (blink-dev, June 2026). No Safari implementation was found.

[7] MDN browser compatibility data for SyncManager and PeriodicSyncManager (https://github.com/mdn/browser-compat-data, checked November 7, 2026): Chrome only.

accessibility work, and for the Phase 1H probe, where Architecture A's trusted server makes hardware custody unnecessary.

Distribution and notifications as censorship vectors. The two platform owners can remove an app from their stores, and push notifications sent through Firebase Cloud Messaging or the Apple Push Notification service tell those companies which devices use the app and when. DPEP therefore requires: (i) distribution of the Android app through F-Droid and as a directly downloadable package, besides the stores; (ii) reproducible builds of the client apps, so that anyone can check that a distributed binary matches the public source, just as the WASM rules are checked (Section 7.7); (iii) notifications that carry no content, only "something changed," so that the client then fetches the update over its own channel, with self-hosted push or polling as alternatives to the platform services. iOS distribution outside the App Store is limited and depends on the jurisdiction, so iPhone users remain exposed to store removal; the web client is their fallback for reading, not for signing.

CHAPTER 6

Weighted Architecture and Implementation Stack Comparison

> Devising and choosing from among these and other possibilities is a practical issue in implementing any actual participatory economy. Assuming the procedures chosen do not violate principles essential to participatory planning, considerations include (1) the extent to which iteration workers could bias outcomes, (2) the extent of reductions in the number of iterations required to reach a plan…
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 4, p. 68[1]

6.1 Weighted Comparison of the Four Architectural Paradigms

We compare the four candidate architectures across all 14 required criteria: Architecture A (Conventional Federated SQL/REST Services), Architecture B (Pure P2P Gossip + CRDTs without finality), Architecture C (Full Per-Transaction BFT Blockchain), and Architecture D (Hybrid Local-First + Two-Root Aggregation + Selective Epoch-BFT Checkpoints). Scores are expert judgements of the current design, not measured performance. Rows 4, 5, 6, 9, and 12 are scored against the measured Prio3 costs, the reworked proposal sizes (Table 5.2) and the carry-forward constraint in Section 7.4.3. Column D assumes BFT checkpoints and the proposed tiered signature policy (Section 7.2.1); column D-log is the same architecture with a witnessed log at Layer 1 (Section 5.3.2). A leads D on six criteria (throughput, bandwidth, hardware, post-quantum readiness, implementation simplicity, energy) and D equals or beats A on the other eight. With every signature hybrid, D scores the same 4.07: the tiered policy gains on bandwidth what it loses on post-quantum readiness, so its case rests on phone traffic and archive size, not on this table.

[1] Albert and Hahnel 1991a, chap. 4, 68.

Table 6.1: Weighted 14-criterion comparison (1–5 scale): A federated SQL/REST, B pure P2P gossip, C per-transaction blockchain, D hybrid with BFT checkpoints, D-log hybrid with a witnessed log

Criterion	Wt.	A	B	C	D	D-log	Rationale for D and D-log (judgment)
1. Decentralization	9%	2.0	5.0	3.0	4.5	4.0	Local autonomy plus rotating BFT; a log has one operator per round.
2. Trust Assumptions	10%	2.0	3.0	4.5	4.0	4.0	One honest aggregator for privacy, all for correct totals (VDAF-22 §9).
3. Censorship Resistance	8%	2.0	4.5	4.0	4.5	4.0	Signed receipts prove omission; a log operator can stall a round.
4. Throughput & Latency	8%	5.0	4.0	2.0	4.0	4.0	10^6 report verifications cost each aggregator 4–61 desktop core-minutes.
5. Network Bandwidth	7%	4.5	2.5	1.5	4.0	4.5	Phone checkpoint check 5.5 kB (0.9 kB for log cosignatures); 103–785 MB round archives.
6. Hardware Requirements	8%	4.5	3.5	1.5	4.0	4.5	Verification load light on a desktop; witnesses fit small boards; no device benchmark yet.
7. Offline Capability	9%	1.5	5.0	1.0	5.0	5.0	SQLite + CRDTs + DTN store-and-forward.
8. Byzantine Fault Tolerance	8%	1.0	2.5	5.0	4.0	3.5	Detects leaf tampering given data availability; a log's liveness rests on one operator.
9. Privacy	9%	2.5	2.0	1.0	3.5	3.5	Coarse Prio3 for annual requests; point-of-use privacy unspecified.
10. Post-Quantum Readiness	6%	4.0	3.5	2.0	3.5	4.0	Confidentiality post-quantum now; round signatures classical until a trigger; ML-DSA-44 log cosignatures are specified.
11. Implementation Simplicity	6%	5.0	3.0	2.5	3.5	4.0	Far more code than A; a log avoids running a consensus engine.
12. Operational Cost	4%	4.0	4.5	2.0	4.0	4.5	$\approx$ 177 MB/yr checkpoints plus 11–21 MB/yr of anchors, 1.0–7.9 GB/yr raw archive; audits dominate (note).
13. Energy Efficiency	4%	4.5	4.0	2.5	4.0	4.0	No proof-of-work; no sourced energy ratio claimed.
14. Maintainability	4%	4.0	3.5	2.0	4.0	4.0	Economic rules as upgradable, ratified WASM modules.
Weighted score	100%	3.10	3.59	2.54	4.07	4.10	D and D-log lead; they are not separable (Section 6.1.1).

Note on row 12 (research/analysis/parameters.py, verify_r3.py; SI units, 1 kB = 1,000 B): at $N = 10^6$, $G = 10{,}000$, $n = 128$, a Layer-1 block is 5 FixedDecimal arrays (400,000 B), $\lceil 2n/3 \rceil = 86$ signatures, a 10,000 B participation bitmap and two 32 B roots. With Ed25519 signatures (5,504 B) the block is 415,568 B $\approx$ 416 kB; with one aggregated BLS signature and a signer bitmap (112 B), 410,176 B; with hybrid signatures (290,078 B), 700,142 B. A 10-round plan has $10 + 365 + 50 = 425$ blocks/year: $\approx$ 177, 174 and 298 MB/year respectively. Layer-2 archive sizes come from Table 5.2; index, receipt, qualitative-List and replica overhead is unspecified. Hosting 100 federation servers at \$30–\$60/month is \$3,000–\$6,000/month and excludes audits, facilitation labor, and research. All figures are unbenchmarked.

6.1.1 Sensitivity Analysis of Architecture Rankings

All figures below are computed from the cells of Table 6.1 by a script.[2]

- Baseline weights: D-log 4.10, D 4.07, B 3.59, A 3.10, C 2.54. Equal weights ($1/14$ each): D-log 4.11, D 4.04, B 3.61, A 3.32, C 2.46.
- Weight on simplicity, throughput, and cost. Put a share w of the weight on criteria 4, 11, and 12 equally and spread the rest in proportion to the baseline. A overtakes D at $w \approx 0.63$ and D-log at $w \approx 0.74$; at $w = 0.70$ the scores are D-log 4.15, A 4.08, D 3.92, B 3.75, C 2.30. The ranking of A against D still depends on how much a community values operational simplicity, which is a legitimate reason to use A for Phase 1H and the earliest pilots; the log design narrows that gap.
- Harsher D cells (trust 3.5, bandwidth 2.5, hardware 3.0, privacy 3.0, applied to both D columns): D 3.79, D-log 3.75, still ahead of B 3.59 and A 3.10.
- Score uncertainty. Perturbing every cell of every column independently by $U(-1, 1)$ points (clipped to $[1, 5]$; 20,000 draws) puts D-log first in 55% of draws, D in 44% and B in 0.1%.

Architecture D's lead over A, B, and C is robust to plausible errors in individual scores, but not to a weighting that puts simplicity first. The comparison cannot choose between BFT checkpoints and a witnessed log; Kill Gate 2 has to.

6.2 Two-Stage Language Strategy: Python-First Research to Rust Protocol

Rather than locking the project prematurely into systems-level Rust before the non-convex economic coordination algorithms are validated, we adopt a two-stage implementation strategy:

1. Phase 1 (Economic Coordination Research)—Python Scientific Stack (NumPy/SciPy, highspy MILP/LP, Clarabel Python bindings): Minimizes iteration time and maximizes reproducibility for testing candidate ADMM, MILP federation decomposition, and Scarf/Leontief stress tests.
2. Phase 2 & 3 (Validated Distributed Protocol & Mobile Engine)—Rust + WebAssembly (dpep-core, dpep-node): Translates the Phase-1-validated clearing rules into memory-safe Rust (Clarabel.rs,[3] FIPS 203/204/205 implementations with independent assurance (libcrux, formally verified with hax and F*; aws-lc-rs, FIPS 140-3 validated; the unaudited RustCrypto crates only as a WASM fallback),[4] quinn QUIC with rustls and its hybrid X25519MLKEM768

[2] The script is research/analysis/verify_r3.py (section H) in the project repository.
[3] Goulart and Chen 2026; software at https://github.com/oxfordcontrol/Clarabel.rs.
[4] The RustCrypto ml-kem documentation states that the implementation "has never been independently audited" (https://docs.rs/ml-kem); libcrux-ml-kem is verified with hax

key exchange, Wasmtime, and Mozilla UniFFI for native Kotlin and Swift client interfaces, the proposed primary client; Section 5.5). Local council solvers use IEEE-754 floating-point heuristics to draft proposals. Protocol-level checks ($SB_j/SC_j \geq 1$, $SC_h \leq B_h$, IFB price updates) are specified in deterministic 64/128-bit fixed-point arithmetic (Section 4.5). Those checks verify constraint satisfaction after rounding, not a truthful objective or MILP optimality.

Table 6.2: Programming language comparison across core protocol requirements (1–5 scale)

Technical Dimension	Rust	Go	TypeScript	Kotlin	Swift	WASM (from Rust)
Memory Safety & Zero GC Pauses	5.0	3.5	2.5	3.0	4.0	5.0 (Sandboxed linear memory)
NIST PQC (libcrux, aws-lc-rs, RustCrypto)	5.0	4.0	2.0	2.5	2.5	5.0 (Direct compilation)
Local-First & P2P (iroh, CRDT, SQLite)	5.0	4.5	4.0	3.0	3.0	4.5 (Browser worker target)
Operations Research (Clarabel.rs, HiGHS)	4.5	2.5	2.0	2.5	2.5	5.0 (Runs on every phone/PC)
Cross-Platform Mobile Reuse (UniFFI)	5.0	3.0	4.5	3.0	1.5	5.0 (Universal execution)
Hardware Keystore & Background Work	3.0 (via UniFFI)	1.5	1.0 (web app)	5.0	5.0	1.0 (no keystore access in a browser)
Accessible UI & Screen-Reader Tooling	2.5	1.5	5.0	4.5	4.5	Paired with native UI; web client for exploration

6.3 Component Maturity

A component can be mature at one level and not at another, so Table 6.3 rates three levels separately: the algorithm or protocol, the library this book names, and the DPEP combination (the way DPEP uses it). The scale runs from M5 (ubiquitous: final standard, several independent implementations, billions of users, ten or more years of public analysis) through M4 (production-proven by several major operators, with audits or formal verification), M3 (early production: a new standard, an Internet-Draft, or few operators), M2 (pre-production: unaudited, pre-1.0 or beta), M1 (research) to M0 (unspecified). Ratings are judgements as of November 2026, not measurements. Evidence, production users, the researchers behind each component, and sources are in the project's maturity dossier and

and F* (https://docs.rs/libcrux-ml-kem) and its generated C is vendored in Mozilla NSS; AWS-LC FIPS 3.0 is a FIPS 140-3 validated module including ML-KEM (https://aws.amazon.com/blogs/security/aws-lc-fips-3-0-first-cryptographic-library-to-include-ml-kem-in-fips-140-3-validation). Checked November 2026.

register.[5]

Table 6.3: Component maturity at three levels (M0–M5; judgment as of November 2026)

Component	Algorithm	Library	DPEP use	Main gap
X25519 / Ed25519	M5	M4–M5	M4	None material.
Hybrid X25519 + ML-KEM-768	M4	M4	M2	Default in major browsers, OpenSSH 10, and Apple OS 26 TLS; the gap is DPEP's own use.
Hybrid Ed25519 + ML-DSA-65	M3	M2–M3	M2	Composite format now RFC 10073; 3.4 kB per signature; proposed for long-lived objects only.
Aggregated BLS12-381 (blst)	M4	M4	M1	Not post-quantum; CometBFT does not aggregate votes.
SLH-DSA-SHA2-192s	M3	M2–M3	M3	Little production use; acceptable for rare root signing.
Post-quantum history anchor (k-of-n SLH-DSA; ML-DSA-44 cosignatures)	M3	M2–M3	M1	No threshold SLH-DSA, so several independent signatures; anchor-key custody unspecified.
BLAKE3	M3	M4	M3	Not standardized; SHA3-256 registered as alternative.
Prio3 VDAF (libprio 0.18.1)	M3	M3	M1	Budget circuit unspecified; library tracks draft 18; custom validity circuits possible through its public Type interface.
Aggregation protocol, $K = 2$ (DAP; Janus)	M3	M3	M2	DAP draft 20 requires exactly two aggregators; a custom budget circuit means modifying the server.
Aggregation protocol, $K = 3$	—	—	M0	DPEP-original: upload, aggregation, collection, replay, and retention would all be new code.
Post-quantum HPKE for shares	M2	M2	M1	No standard suite yet.
Two-Root aggregation	M5	M4	M1	Merkle logs mature (Certificate Transparency); DPEP protocol original.
Witnessed log (C2SP; Tessera, Sigsum)	M4	M4 / M2[d]	M1	Go implementations mature; Rust libraries young; operator replacement procedure unspecified.
Intra-year fulfillment (S7)	M4 / M2[e]	—	M0	Unspecified: privacy at the point of use and offline double spending (Section 7.5).
One-report-per-person credentials	M4 / M1[a]	—	M1	Interim classical (Section 7.2.1); B_u binding open.
Sortition beacon	M4 / M1[a]	—	M1	Interim threshold beacon; key generation and custody unspecified.
Capability certificates (CapCert)	M2	M2	M0	Format not chosen (UCAN 1.0 release candidate, Biscuit).
QUIC: quinn + rustls	M5	M4	M3	Hybrid key exchange must be enabled explicitly.
iroh	M3	M3	M2	1.0 released June 2026; relays must be self-hosted.

Continued on next page

[5] reports/STACK_MATURITY.md and research/stack_maturity_register.csv in the project repository.

Continued from previous page

Component	Algorithm	Library	DPEP use	Main gap
SQLite / SQLCipher; CRDTs	M5 / M4	M3–M5	M3	CRDT schema evolution and metadata growth.
CometBFT	M4–M5	M4 / M2[b]	M2	Post-quantum keys are pure ML-DSA, not hybrid; BLS keys only behind a build flag.
Fixed-point WASM on Wasmtime	M4	M4	M2	Pin a long-term-support release; fix NaN and fuel settings.
HiGHS / Clarabel	M4	M4 / M2[c]	M2	No phone benchmark.
Mozilla UniFFI	M3–M4	M3–M4	M2	Interface still changing; carries the native clients.
Formal model (Quint / TLA$^+$)	M4	M3–M4	M0	Not yet written; required before Phase 2 Rust work (Section 9.1.2).

[a] Classical / post-quantum. [b] Ed25519 keys / ML-DSA-65 keys (v0.40). [c] Native / WASM on a phone. [d] Go (Tessera v1.0.4, Trillian v1.8.0) / Rust (tlog_tiles 0.2.0, sigsum 0.4.0), checked November 7, 2026. [e] Attributed receipts / offline anonymous tokens.

Ten DPEP combinations have no deployment precedent, so they inherit no maturity: the Prio3 budget circuit, Two-Root aggregation, the tiered signature policy with per-round post-quantum anchoring by several parties, CapCerts, credentials, the beacon, the light-client policy (Table 7.10), non-nested aggregator governance (and a three-aggregator protocol, if chosen), ratified WASM economic rules, and intra-year fulfillment. Two-Root aggregation, WASM rules, the Layer-1 design, and server-side signature verification are tested at Kill Gate 2. The budget circuit, signature policy, CapCerts, credentials, light-client policy, aggregator governance, and fulfillment are tested at Kill Gate 3, whose external audit covers them. The beacon has an interim construction and stays an open question. Post-quantum and transparency-log deployment is moving quickly, so the hybrid-signature, Prio3, aggregation-protocol, share-encryption, credential, beacon, witnessed-log, iroh and CometBFT rows are re-checked before each gate.

CHAPTER 7

Post-Quantum Cryptography, Democratic Membership, and Privacy

> Decisions about what individuals wish to consume will be subject to collective criticism by fellow council members, though with specific guarantees for preserving individual freedoms and privacy.
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 3, "Consumption Norms"[1]

7.1 What the Source Tradition Requires of Consumption Privacy

The epigraph is the 1991 design: consumption is subject to collective criticism, with unspecified "guarantees" for freedom and privacy. PEPE Ch. 3 (p. 41) and Ch. 5 (p. 80) add a non-veto rule for in-budget requests. Albert (2003, chaps. 12, 18) argues for anonymous submission against busybody intrusion. None of those texts specifies cryptographic aggregation. DPEP's Prio3 invisibility is a technical extension: neighbors cannot see or comment on the anonymous bundle, so the 1991 criticism channel is lost. That is a genuine normative cost, not a proven improvement. REQ-DATA-04 therefore splits: non-veto is [ORIGINAL-1991]; invisibility is [TECH-EXT].

7.2 Post-Quantum Cryptography (NIST FIPS 203, 204, 205)

To protect multi-decade household records against harvest-now-decrypt-later adversaries, DPEP uses cryptographic agility (CryptoSuiteId) and hybrid classical + NIST PQC primitives. This is not a uniform NIST Level-3 claim, and it is not a "zero lattice" claim: ML-KEM and ML-DSA are lattice schemes. BLAKE3's specification targets 128-bit security for all goals. VDAF-22 makes no post-quantum claim. Table 7.2 states the status of each component.[2]

One format is settled and one is not. Hybrid signatures follow RFC 10073 (Ounsworth et al. 2026), which defines the Ed25519 combination id-MLDSA65-Ed25519-SHA512 as the ML-DSA-65 signature followed by the Ed25519 signature,

[1] Albert and Hahnel 1991b, chap. 3 (lf3_2.htm); Albert 2003, chap. 12 ("Consuming") and chap. 18 ("Privacy / Frenzy: A Busybody Economy?"), where Albert specifies that individual consumption proposals within warranted effort-rated budgets are submitted anonymously to prevent "busybody" intrusion; see also Dapprich 2024.

[2] NIST 2024a, 2024b, 2024c. Byte sizes in Table 7.1 are specification constants. The latency columns are placeholders carried from an unsourced estimate; in particular $\sim 80\ \mu$s for a hybrid X25519+ML-KEM-768 operation looks optimistic for an entry-level phone. They must be replaced by device benchmarks (Kill Gate 3) before use.

3,309 + 64 = 3,373 B. Plain ML-DSA in X.509 certificates is standardized (Massimo et al. 2025). Share encryption: DAP encrypts report shares with HPKE, whose standard suites are classical, so the hybrid share encryption in Table 7.1 needs a post-quantum HPKE suite (ML-KEM or the X-Wing combiner), which is still in IETF drafting.

Table 7.1: Post-quantum primitives: specification byte sizes and placeholder ARMv8 latencies (unbenchmarked)

Primitive & Standard	Security	Pub Key	Cipher/Sig	ARM Sign	ARM Verify	Assigned Role in DPEP
X25519 + ML-KEM-768 (FIPS 203 Hybrid KEM)	NIST L3 + Classical	1,216 B	1,120 B	$\sim 80\ \mu s^{P}$	$\sim 85\ \mu s^{P}$	Point-to-point QUIC channels & Prio3 aggregator share encryption.
Ed25519 + ML-DSA-65 (FIPS 204 Hybrid Sig)	NIST L3 + Classical	1,984 B	3,373 B	$\sim 230\ \mu s^{P}$	$\sim 175\ \mu s^{P}$	Long-lived objects: accreditation, CapCerts, roster roots, annual ratification. Under the proposed tiered policy, per-round objects use Ed25519 (Section 7.2.1).
SLH-DSA-SHA2-192s (FIPS 205 Hash-Based)	NIST L3 (Hash Only)	48 B	16,224 B	unverified	unverified	Level-3 Root Anchor for Federation accreditation, WASM rule signing, and k-of-n history anchoring. Signing is slow by design; device timings not benchmarked.
BLAKE3 (BLAKE3 spec) / SHA3-256 (FIPS 202)	BLAKE3: 128-bit (all goals). SHA3-256: 128-bit collision.	N/A	32 B	$< 1\ \mu s/kB^{P}$	$< 1\ \mu s/kB^{P}$	Two-Root Merkle trees & content CIDs. BLAKE3 targets 128-bit security for preimage as well as collision resistance; quantum (Grover/BHT) reductions are not claimed.

[P] Placeholder, not measured on ARM. On an Apple M4 Pro core, ML-DSA-65 verification measured 32 μs and hybrid verification 49 μs (research/analysis/bench_signatures/). Council signatures are k-of-n multi-signatures (3,373 B per signer hybrid, 64 B Ed25519); no post-quantum threshold signature is assumed.

Table 7.2: Component-by-component post-quantum status

Component	Primitive	Claim type
Share encryption	Hybrid X25519 + ML-KEM-768	Standards-based (FIPS 203 + classical). Lattice. NIST L3 for ML-KEM-768.
Long-lived signatures (accreditation, CapCerts, ratification)	Hybrid Ed25519 + ML-DSA-65	Standards-based (FIPS 204 + classical; RFC 10073 format). Lattice.
Per-round signatures (proposals, receipts, checkpoints)	Ed25519 or aggregated BLS (proposed tiered policy)	Classical, not post-quantum; history protected by per-round k-of-n post-quantum anchoring (Table 7.3).
Root / WASM signatures	SLH-DSA-SHA2-192s	Standards-based (FIPS 205). Hash-based NIST L3.
Transcript hashes / Merkle	BLAKE3 (SHA3-256 registered as alternative)	Spec: 128-bit for all goals. Not NIST Level 3. Collision classically ≈ 128 bits. Not standardized by NIST or the IETF, so SHA3-256 (FIPS 202) is registered under CryptoSuiteId for checkpoint roots.
FLP soundness / joint randomness	Finite-field FLP + XofTurboShake128	Inferred from VDAF-22; no PQ theorem. Joint randomness is a known attack surface if the XOF is broken.
B_u credentials / sortition beacon	Interim classical (Section 7.2.1)	Not post-quantum. Accepted as interim because forgery and bias need a live CRQC; post-quantum replacement is open research.
Prio3 as a whole	Mix of the above	VDAF-22 Informational draft; no PQ claim.

7.2.1 Which Properties Need Post-Quantum Protection Now, and Crypto Agility

Post-quantum protection is urgent for one property and not for the others. Confidentiality must be post-quantum now: ciphertext recorded today can be decrypted once a CRQC exists, so household reports, share encryption, and transport use hybrid ML-KEM from the first deployment and are never downgraded. Authenticity, Sybil resistance and randomness need post-quantum protection before a CRQC exists, not before deployment: a CRQC can forge future signatures and bias future draws, but it cannot alter a checkpoint whose history is already anchored under a hash-based signature, or redo a committee draw already made.

Design Principle 7.1 (Interim Classical Constructions Under Crypto Agility (adopted)). Where a post-quantum construction is standardized and practical, DPEP uses it in hybrid. Where none exists (one-report-per-person credentials, the sortition beacon) or the consensus engine does not yet support the hybrid (Layer-1 validator keys), DPEP uses the most mature available construction, on four conditions:

1. Suite inside the signature. Every signed or encrypted object carries its CryptoSuiteId inside the signed bytes, next to its domain-separation tag, so a suite cannot be stripped or swapped.
2. No downgrade below a ratified floor. The minimum suite for each object type

is fixed in the checkpoint and changed only by the dual-federation ratification used for WASM rules (Section 7.7); objects below the floor are rejected, never negotiated.

3. Published migration triggers. A suite is replaced when the first of these occurs: the replacement is a final standard (composite ML-DSA RFC; FIPS 206; a standardized post-quantum credential or VRF); the software in use supports it; or a credible public warning of CRQC progress. Each migration has a stated window with dual signing during the transition.

4. History survives the switch. Checkpoint history signed under an outgoing suite is re-anchored under SLH-DSA before that suite is withdrawn; per-epoch credentials and the beacon's threshold key are re-issued, not converted.

A tiered signature policy (proposed, for author ratification). The design rule above already lets a function use a classical construction until a trigger fires, provided its history is re-anchored under a hash-based signature. Applying it to every per-round object removes most of the signature bytes in the system. Objects that matter only for one planning round and its challenge window (council proposals, submission receipts, round checkpoints) are signed classically: Ed25519 for councils and receipts, and either one aggregated BLS signature or 86 Ed25519 signatures for validators or witnesses. At the close of each round, that round's roots are given a post-quantum anchor held by several parties rather than one (see "Who anchors history" below). Long-lived objects (accreditation certificates, CapCerts, roster roots, WASM rule hashes, and annual plan ratification) stay hybrid or SLH-DSA, and confidentiality stays hybrid ML-KEM. The effect is large where signatures dominate: a phone's full checkpoint check falls from 290 kB to 5.5 kB (Ed25519) or about 0.1 kB (aggregated BLS, where the engine or log aggregates; CometBFT does not), and a coarse round archive with five signers per council from 775 MB to 113 MB (Table 5.2). Verification load hardly differs on a desktop core, where even all-hybrid verification is fast (Section 5.3.1); Raspberry Pi and community-server timings are a Kill Gate 2 measurement.

What the policy gives up. If a cryptographically relevant quantum computer existed covertly before a migration trigger fired, it could forge current round objects (a proposal, a receipt, or a checkpoint) until the switch. All-hybrid signatures would resist that. Rounds already anchored are protected: a forged signature on an old proposal cannot change a root that the post-quantum anchor has fixed, so exposure is bounded by the time from round close to anchoring (proposed: within 24 hours). The second cost is operational: moving many phones and federation servers to a new suite needs a dual-signing window and working key rotation, which must be rehearsed before it is needed. The policy is therefore a bet that a CRQC will be seen coming. The project's own urgency analysis makes that bet for authenticity already; the tiered policy makes it explicit and gives its cost.

Who anchors history (proposed, for author ratification). Under the tiered policy the anchor is the only post-quantum protection on round history once a quantum computer exists, so it must not rest on one key. SLH-DSA has no threshold form, so the anchor is built from several independent signatures. Table 7.3 compares the options. For the BFT design, the anchor is (i): three of five federations, chosen so that no single federation type holds a majority, each sign the round's roots with their own SLH-DSA key; CometBFT validators sign only classically under the tiered policy and produce no post-quantum cosignature. For the witnessed-log design, the anchor is (ii): each witness adds an ML-DSA-44 cosignature, which the C2SP cosignature specification already recommends, beside its Ed25519 cosignature; phones check only the Ed25519 cosignatures, and replicas archive the ML-DSA-44 ones for auditors. Because ML-DSA-44 rests on lattice assumptions, a once-a-year 3-of-5 federation SLH-DSA signature over the year's final log root adds a hash-based backstop at negligible cost. Anchor keys are held by each federation (or witness) under the custody rules of Section 7.7, never on a shared server, and are rotated by dual-federation ratification: new keys are announced in an anchor signed by the outgoing keys, and old anchors stay verifiable because SLH-DSA is stateless.

Table 7.3: Post-quantum anchoring of round history (425 anchored blocks a year)

Option	Per anchor	Per year	Trust, custody and rotation
One federation SLH-DSA key (not recommended)	16.2 kB	6.9 MB	A single key is the whole post-quantum record; its theft or loss forfeits it.
(i) 3-of-5 federation SLH-DSA signatures (BFT design)	48.7 kB	20.7 MB	Rewriting history after a CRQC needs three federation keys; hash-based assumption only. Each federation holds its own key; rotation by dual-federation ratification. (4 of 7: 64.9 kB, 27.6 MB.)
(ii) ML-DSA-44 cosignatures, 11 of 16 witnesses (log design)	26.8 kB	11.4 MB	Rewriting history needs 11 witness keys; lattice assumption. Keys rotate with the witness draw. Phones still fetch only the 0.8 kB of Ed25519 cosignatures.
Annual backstop: 3-of-5 SLH-DSA over the year's log root (with (ii))	48.7 kB	0.05 MB	Hash-based check on the whole year if lattice assumptions fail.

Sizes from research/analysis/verify_signature_suites.py (section S8): 16,224 B per SLH-DSA-SHA2-192s signature; 2,420 B per ML-DSA-44 signature plus 12 B of signed-note framing; a one-byte signer bitmap.

Figure 7.1 draws the same table as migration lanes.

Credentials. Threshold issuance, in which no single council can mint or link a credential, follows Coconut (Sonnino et al. 2019), a classical pairing-based construction. Lattice alternatives exist as research (Bootle et al. 2023), but reported sizes, under 80 kB per credential in one recent construction,[3] do not fit beside

[3] Argo et al. 2024 (IACR ePrint 2024/131).

Table 7.4: Post-quantum urgency, current construction, and upgrade path by function

Function	PQ needed	Construction now	Upgrade path and trigger
Transport, report-share encryption	Now	Hybrid X25519 + ML-KEM-768	Already post-quantum; add HQC as a backup KEM once standardized (final expected 2027).
Per-round objects: council proposals, receipts, round checkpoints	Before CRQC (live forgery)	Proposed: Ed25519 for councils and receipts; aggregated BLS or Ed25519 for validators or witnesses; each round's roots anchored post-quantum by several parties at round close (Table 7.3)	Hybrid (RFC 10073) or FN-DSA when a trigger fires; dual signing during the window.
Long-lived objects: accreditation, CapCerts, roster roots, annual ratification	Before CRQC (long-lived)	Hybrid Ed25519 + ML-DSA-65 (RFC 10073)	FN-DSA once FIPS 206 is final (smaller signatures).
Root anchors, WASM rules, re-anchoring	Before CRQC (long-lived)	SLH-DSA-SHA2-192s	None needed (hash-based).
One-report-per-person credentials	Before CRQC (forgery)	Classical anonymous credential issued jointly by k-of-n councils, with a per-epoch nullifier	Lattice credentials once they fit the phone budget of Kill Gate 3.
Sortition beacon	Before CRQC (bias)	Threshold beacon run by the federations (threshold BLS)	A post-quantum VRF or VDF beacon once one is standardized.
Binding B_u to the credential	—	Unspecified (Section 7.4.4)	Open research.

a 55 kB report in a 150 kB round. Selection criterion for the interim scheme: its unlinkability should not rest on an assumption a CRQC breaks (statistically zero-knowledge presentation proofs, statistical blinding). This must be verified for the chosen scheme; it is not a property of every classical credential. RFC 9474, for example, attributes the blindness of RSA blind signatures to a separate proof without stating its assumptions, and states that the protocol is not post-quantum ready. If no scheme meets the criterion, a future CRQC could link past credential showings to identities. That reveals who submitted, not what was submitted, since report contents stay secret-shared under ML-KEM.

Beacon. The interim beacon uses threshold-BLS randomness of the kind produced by the open-source drand software,[4] with the threshold key shared among the federations rather than delegated to an outside network. It has no last-revealer problem, and a CRQC could bias only draws made after it exists. Its distributed key generation and share custody are not yet specified.

[4] https://drand.love (checked November 2026).

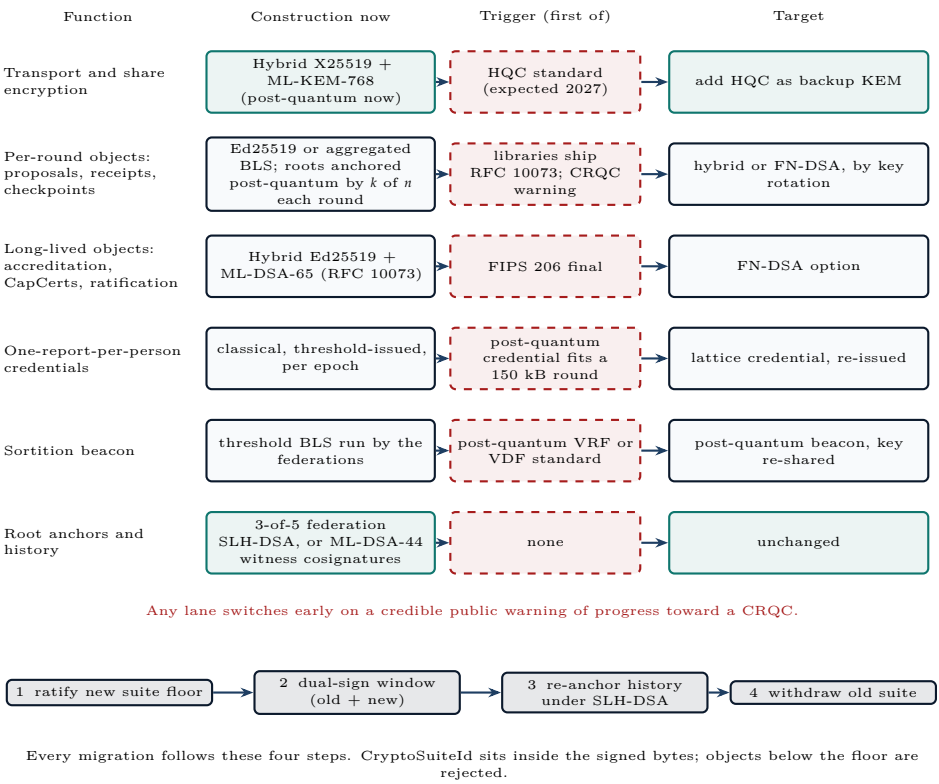


Figure 7.1: Crypto-agility lanes. Each function moves from its current construction to its target when the first trigger occurs, through the same four-step migration.

Layer 1. CometBFT remains the reference BFT engine (provisional), beside the witnessed-log option of Section 5.3.2. Its v0.40 release (July 27, 2026) accepts ML-DSA-65 consensus keys through key rotation, as a pure rather than hybrid scheme, and the Cosmos documentation estimates about 331 kB of signatures per block for 100 validators, consistent with the 290 kB for 86 hybrid signatures in Table 7.10. CometBFT also defines a BLS12-381 key type, but only behind a build flag, and it does not aggregate votes into one signature; an aggregated checkpoint signature would be produced at the application level or by another engine. Under the tiered policy, Ed25519 validator keys are sufficient until a trigger fires.[5] The validator suite therefore moves to ML-DSA by key rotation and to the hybrid when a composite standard and engine support exist, without

[5] Cosmos SDK documentation, "Post-quantum keys" (requires Cosmos SDK 0.55 and CometBFT 0.40), https://docs.cosmos.network/sdk/latest/keys/post-quantum-keys; CometBFT v0.40.0 was released on GitHub on July 27, 2026 (checked November 7, 2026). BLS key type: crypto/bls12381 in the CometBFT repository, built only with the bls12381 tag; vote aggregation exists only as an exploration (RFC 018, 2022).

a change of engine. A Rust engine with a pluggable signing interface (Malachite) is the fallback if a hybrid consensus signature becomes mandatory earlier; it is alpha software without an external audit.

7.3 Democratic Membership, Sybil Resistance, and Civil Liberties

Design Principle 7.2 (Four-Way Separation of Identity, Eligibility, Accountability, and Legitimacy). DPEP strictly separates: (1) Cryptographic Identity (control of a keypair), (2) Institutional Eligibility (accredited membership in a council), (3) Economic Accountability (correspondence of reported claims to physical reality), and (4) Democratic Legitimacy (self-management proportional to degree affected).

Figure 7.2 shows how the separation is carried into membership, delegation, and private reporting.

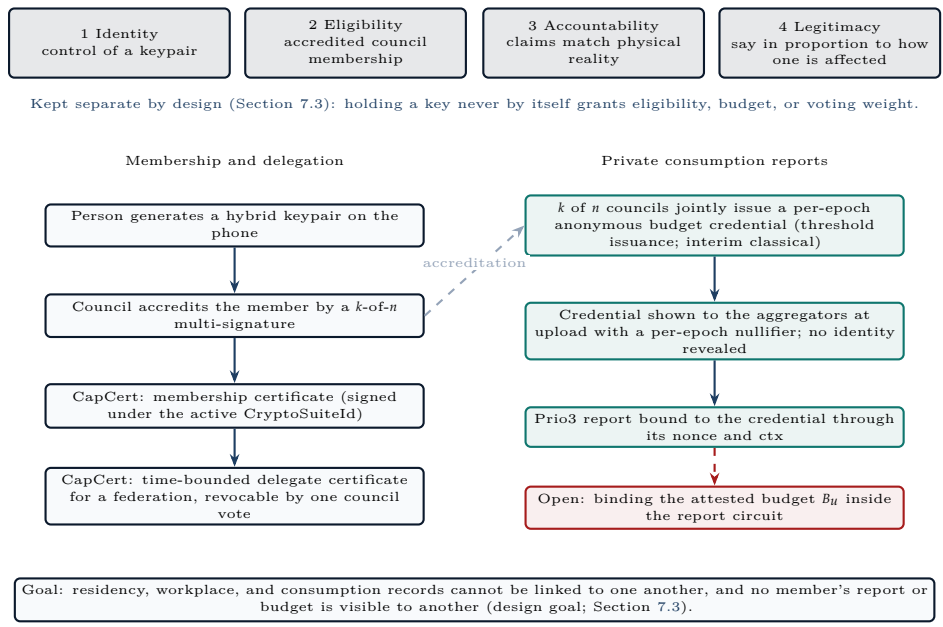


Figure 7.2: Identity, eligibility and the two credential paths. Membership and delegation are attributed and revocable; private consumption reports use a separate, unlinkable credential whose budget binding is still open.

While phantom digital accounts in CC_h cannot mint adult work income without WC_j effort certificates capped by SB_j/SC_j, four deeper Sybil and collusion vectors must be closed explicitly:

1. Closing Workplace Hours Inflation (INV-4B): Capping only the mean effort rating $\bar{\epsilon}_j$ fails if a WC_j fabricates phantom workers to inflate total hours $\sum h_u$. Equation (4.7) closes this by capping total credited effort-hours

$\sum_{u \in WC_j} h_u \epsilon_u \leq \sum_\ell (l^0_{j,\ell} + \Delta l_{j,\ell}) \min(\bar{\epsilon}_{\max}, SB_j / SC_j)$ (accredited hours) and flooring labor user-right charges at $p^{\text{lab}}_\ell \geq w_{\text{base}}$.

2. Detecting Collusive "Carousel Fraud" Between Fake Producer and Consumer Councils: If a colluding WC_j and CC_h sign fictitious delivery receipts for non-existent services to mint effort credits, cryptography alone cannot detect the lie. DPEP pairs automated input-output ratio anomaly flags (detecting closed-loop service claims without corresponding physical/intermediate inputs) with mandatory Industry Federation physical euthynai spot-audits prior to accrediting capacity.

3. Preventing Vote & Sortition Dilution: Sortition tickets and federation referenda are weighted by cross-audited council accreditation quotas rather than raw public keys.

4. Civil-Liberties Protection Against a Coercive Population Register (design goal, not yet constructed): To prevent housing (HFB) and workplace (EFB) registries from forming a surveillance database or trapping victims of domestic coercion, the design aims to let each person turn council attestations into unlinkable per-epoch budget credentials, separating residency from workplace and consumption. No such construction is specified (Section 7.4.4), so this protection does not yet exist. Budgets and reports are individual: the index u in Section 7.4 is one budget-holding person, never a household head. People who share a household and wish to plan jointly each submit their own report against their own budget; no member's report or budget is visible to, or controlled by, another. Unaffiliated migrants and transients receive unconditional Maxim-4 baseline provisioning through community clinics.

7.4 Private Household Aggregation via Prio3 VDAFs

A discarded design in which each household u generates a standalone zero-knowledge budget proof Π_u on a commitment C_u while separately submitting a pairwise-masked vector $\tilde{y}_u$ has three flaws: (1) nothing binds C_u to $\tilde{y}_u$; (2) if even one of many neighbors drops offline, pairwise masks fail to cancel; (3) aggressive FRI/STIR/WHIR parameterizations rely on Reed–Solomon proximity-gap conjectures that Crites and Stewart (IACR ePrint 2025/2046) disproved at up-to-capacity rates (Crites and Stewart 2025). Those constructions are not part of Architecture D.

Catalogs. Ω with $G = |\Omega| = 10{,}000$ is the full priced catalog, and $\mathcal{G}^{\text{priv}}$ with $G_{\text{priv}} = 3{,}000$ the private-consumption categories at national scale. Household reports need not use $\mathcal{G}^{\text{priv}}$ directly. Section 7.4.2 proposes a coarser reporting catalog $\mathcal{G}^{\text{rep}}$ of 100–250 categories, consistent with REQ-DATA-02.

VDAF construction and what it checks. Each budget holder u (one person; Section 7.3) encodes its request as a vector $q_u \in \{0, \ldots, 2^b - 1\}^{|\mathcal{G}^{\text{rep}}|}$ in Household Reporting Units (Section 7.4.1) and splits it into K additive Field128 shares, encrypted with hybrid X25519MLKEM768 to K aggregators ($K = 2$ proposed for pilots, $K = 3$ an option; see below). Standard Prio3SumVec proves only that every coordinate lies in $[0, 2^b - 1]$ (VDAF-22 §7.4.3). The budget predicate

$$\sum_{g \in \mathcal{G}^{\text{rep}}} \pi_g^{(t)} \, q_{u,g} + \text{PubShare}_u \le B_u, \qquad \pi_g^{(t)} = p_g^{(t)} s_g \tag{7.1}$$

needs a custom validity circuit, which is not specified. Two facts constrain it. First, the prices $\pi^{(t)}$ must enter the circuit as public constants, so each price vector defines a different VDAF instance: a report proves the predicate for the round's prices and for no other. The application context string ctx should bind the report to (epoch, t, BLAKE3($P^{(t)}$), aggregator set), but ctx enters only the domain separation of the XOF, not the circuit's arithmetic. Second, B_u must reach the circuit either as a public value (which identifies the person) or as a secret-shared input whose correctness is vouched for by a credential; the second option depends on the open credential problem (Section 7.4.4). Reference: IETF CFRG draft-irtf-cfrg-vdaf-22 (August 14, 2026; Informational), full text.[6]

Privacy versus correctness (VDAF-22 §9). These must not be collapsed. Privacy of a measurement requires at least one honest aggregator. Correctness of the aggregate requires all aggregators: "An Aggregator can, without detection, manipulate the aggregate result by modifying its own aggregate share." Any one aggregator can inflate or deflate demand and so move prices.

Two aggregators or three. The VDAF itself works with any number of aggregators, and libprio 0.18.1 accepts from one to 254 and exposes a public interface for user-defined validity circuits. The protocol that runs a VDAF between clients and aggregators is a different matter. DAP, the IETF protocol for doing so, states: "Some VDAFs may involve three or more Aggregators, but DAP requires exactly two Aggregators" (draft-ietf-ppm-dap-20, §2, October 21, 2026). Production DAP servers such as Divvi Up's Janus (MPL-2.0; release 0.7.146, November 4, 2026) follow it.[7] With $K = 3$, the upload, aggregation, collection, replay, and retention protocol would be DPEP's own code, to be built, budgeted, and audited. With $K = 2$, DPEP can use DAP and an existing server: privacy then needs one of two aggregators honest, and correctness needs both. Either way the budget predicate of Eq. (7.1) needs a custom validity circuit, which for $K = 2$ means modifying the DAP server as well as writing the circuit.

[6] Corrigan-Gibbs and Boneh 2017; Boneh et al. 2019; Barnes et al. 2026. Security analysis: Davis et al. 2023; cite the full version (IACR ePrint 2023/130), whose 2024 revision corrects the proof of Theorem 1 with a weaker bound and a larger PRG seed.

[7] IETF, https://datatracker.ietf.org/doc/html/draft-ietf-ppm-dap-20; Janus, https://github.com/divviup/janus (both checked November 7, 2026).

Figure 7.3 shows the asymmetry.

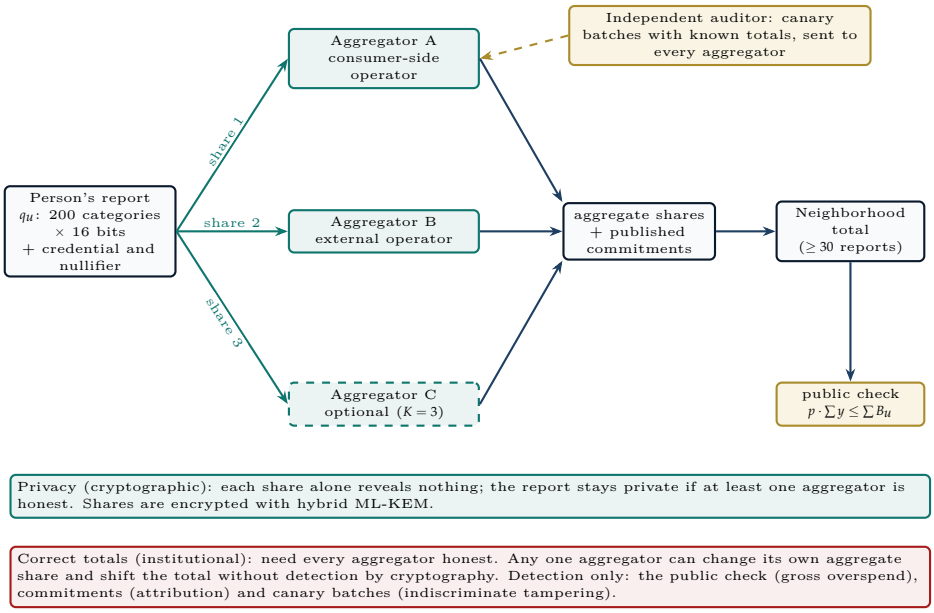


Figure 7.3: Household report flow and the privacy/correctness asymmetry, drawn for the proposed two aggregators with an optional third. Privacy needs one honest aggregator; correct totals need all of them, and the remaining safeguards detect rather than prevent manipulation.

How much the public budget check limits a cheating aggregator. The check $p \cdot \sum_u y_u \le \sum_u B_u$ is cheap and should be run, but it constrains only one number. A malicious aggregator can (i) shift demand between categories while keeping the value total fixed, which is undetectable and moves relative prices; (ii) inflate the total up to the slack between actual spending and $\sum_u B_u$ (if households leave 10% of budgets unspent, about 10% of value can be added undetected); and (iii) deflate demand without limit, since the inequality has no lower bound. It is a tripwire for gross overspend, not a correctness guarantee.

Recommended correctness model for Phase 3 (proposed, for author ratification). Start pilots with two non-nested aggregators on DAP run by organizations with opposed interests (for a co-op pilot: the consumer side and an external civil-society or academic operator), and move to a third, producer-side aggregator only if a pilot shows a governance case for it that justifies a protocol of DPEP's own. Add the public budget check and published commitments to each aggregator's aggregate share, so that a dispute can at least be attributed. Nested CC/Ward/Federation aggregators should not be the default. Add canary batches run by an auditor inde-

Table 7.5: Aggregator-correctness options (experimental except the first row)

Option	Guarantee	Cost / institutional demand
Nested CC / Ward / Federation aggregators	Privacy if ≥ 1 honest; correct totals only if all honest.	The organs belong to one federation, so their independence is doubtful.
Two non-nested aggregators on DAP ($K = 2$)	Privacy if either is honest; correctness needs both.	Existing protocol and servers (DAP, Janus); a custom circuit still needs server changes.
Three non-nested aggregators ($K = 3$)	Privacy if any one is honest; correctness needs all three.	Independence more plausible across three interests; protocol is DPEP's own (+\$40k–\$95k, wider audit).
Replicated aggregators ($K' > 3$)	Privacy if one honest; correctness still needs all shares.	More KEMs and operational load; no correctness gain without robust aggregation.
Robust / maliciously secure MPC	Detects or bounds a cheating aggregator.	Super-linear communication; not Prio3; unaudited for DPEP.
Public check $p \cdot \sum y \leq \sum B_u$	Detects gross aggregate overspend only (see text).	Cheap; complements every option.
Canary batches	Detects indiscriminate tampering with aggregate shares, with probability rising with the canary share.	An independent auditor submits whole batches with known totals. Needs per-batch aggregate shares and batches the aggregators cannot tell apart; does not catch tampering aimed at batches known to be real.
Committee-run aggregation	Ties aggregation to the $n = 128$ BFT set.	Concentrates privacy-breaking power in the body that finalizes prices.
Differential privacy on released totals	Bounds one-household differencing; outside VDAF.	Bias against planning accuracy; parameters unspecified.

pendent of every aggregator. Cryptography in this model guarantees privacy, not correct totals: correctness rests on institutional independence plus detection, and no mature construction removes that trust while keeping Prio3's privacy. Robust MPC is deferred until a pilot shows that aggregator manipulation is a real rather than a hypothetical threat. Trusted execution environments are rejected for this role, because they would make the chip vendor the party trusted for correctness. This choice cannot be made final before Phase 3 because it depends on which organizations exist in a pilot.

7.4.1 Report Sizes, Units, and Measured Costs

Household Reporting Unit (HRU). At the protocol's fixed 10^{-6} scale, a b-bit coordinate can express at most $(2^b - 1)/10^6$ units: 1.05 units at 20 bits, which is useless, and a full 64-bit field makes reports enormous. Household reports therefore use a per-category Household Reporting Unit: the catalog fixes for each reporting category g an integer scale s_g (micro-units per HRU, for example 100,000 for 0.1 kg), and a report carries $q_{u,g} \in \{0, \ldots, 2^{16} - 1\}$. The aggregate in protocol

units is

$$D_g^{\text{micro}} = s_g \sum_u q_{u,g},$$

which is exact integer arithmetic; no rounding occurs after the report is made. Rounding occurs once, when a person's intended quantity is rounded to whole HRUs (round-half-to-even): at most half an HRU per category per person, so at most $Ns_g/2$ micro-units in aggregate and, for independent errors, a standard deviation of $s_g\sqrt{N/12}$ (for $s_g = 0.1$ unit and $N = 10^6$, about 29 units against a total in the hundreds of thousands). Sixteen bits gives 0–65,535 HRU per category per report; categories whose range does not fit choose a larger s_g. With two aggregators instead of three, the coarse upload falls from 55,120 B to 55,024 B (one helper seed fewer), so the choice of K does not affect phone budgets.

Table 7.6: Prio3SumVec with 3 aggregators: measured sizes and desktop timings (libprio 0.18.1)

Length	Bits	Leader share	Client upload	Shard	Verify (leader)	Verify (helper)	Fits 75 kB?
100	20	35.1 kB	35.3 kB	0.55 ms	0.21 ms	0.23 ms	Yes
100	32	54.9 kB	55.1 kB	0.65 ms	0.21 ms	0.22 ms	Yes
100	64	107.7 kB	108.0 kB	1.13 ms	0.30 ms	0.32 ms	No
230	20	78.0 kB	78.3 kB	0.81 ms	0.23 ms	0.25 ms	No
230	32	123.6 kB	123.8 kB	1.28 ms	0.33 ms	0.35 ms	No
230	64	243.3 kB	243.6 kB	2.52 ms	0.82 ms	0.86 ms	No
3,000	20	975.8 kB	976.0 kB	10.6 ms	3.46 ms	3.64 ms	No
3,000	32	1,556.3 kB	1,556.5 kB	16.2 ms	4.85 ms	5.21 ms	No
3,000	64	3,100.4 kB	3,100.7 kB	33.3 ms	12.0 ms	12.7 ms	No
Computed with the same size formula (exact on all nine measured rows); not timed							
150	16	41.7 kB	41.9 kB				Yes
200	16	54.9 kB	55.1 kB				Yes
250	16	68.1 kB	68.3 kB				Yes
10,000	64	10,292.8 kB	10,293.0 kB				No

Medians of 30 runs, one core, Apple M4 Pro, macOS 26.6, rustc 1.98.1, –release, chunk_length chosen by libprio's optimal_chunk_length. "Verify" is verify_init (proof-share computation); combining verifier shares and verify_next take under 5 μs. Helper shares are 64 B seeds; the public share is 96 B. libprio 0.18.1 implements VDAF draft 18, not draft 22; the encoded sizes follow the same Prio3SumVec layout. These are desktop figures: phones and Raspberry Pi boards will be slower by an unmeasured factor. Code and commands: research/analysis/bench_prio3/.

What the measurements change. A dense 3,000-category report is about 1 MB, thirteen times the 75 kB budget. A 50-coordinate "sparse" report cannot be used to send 50 items chosen from 3,000: Prio3SumVec sums fixed coordinates, so a short vector either covers only 50 fixed categories or reveals which categories each coordinate stands for, the same leak that dense reports exist to prevent. No private-sparse construction is specified here; any such design would be experimental. Proof checking is not negligible either: verifying one dense report costs each aggregator, helper as well as leader, about 3.5 ms on a fast desktop core, so 10^6 reports cost about 61 core-minutes per aggregator per fresh-report round (3.8

minutes on 16 cores), before network transfer and on hardware much faster than a neighborhood Raspberry Pi. A coarse 200-category, 16-bit report costs about 0.25 ms, roughly 4 core-minutes per aggregator at 10^6, and a neighborhood of 200 people verifies in about 50 ms of desktop time. The illustrative "1 ms per report" used before measurement was optimistic for dense reports even on a desktop.

7.4.2 A Working Privacy Baseline (proposed)

Table 7.7: Privacy baselines compared at $N = 10^6$ (sizes per fresh report; CPU per aggregator per fresh-report round)

Option	Upload	Agg. CPU (desktop)	Hidden from councils	Costs and risks
(i) Dense Prio3, 3,000 categories, 20 bits	976 kB	≈ 61 core-min	Every category choice	Exceeds mobile budgets; 20 bits too narrow for many goods.
(ii) Coarse Prio3, 100–250 categories, 16-bit HRU	28–68 kB	≈ 4 core-min	Choice among coarse categories; not the product within a category	Loses detail; more reliance on disaggregation to products (P5) and on local intra-year fulfillment; Hayek's objection bites harder.
(iii) Council-level trust, no household cryptography	Plain request to council tallier	Negligible	Bundle hidden from neighbors by policy only; tallier sees it	Trusted tallier can leak or alter; fine for small consenting pilots with data-protection safeguards.
(iv) Pilot-selectable modes	(ii) or (iii)	–	As chosen	More code paths to audit; mode choice must itself be democratic and disclosed.

Recommendation (proposed, for author ratification). Adopt option (ii) with $|\mathcal{G}^{\text{rep}}| = 200$ and 16-bit HRUs (55 kB upload) as the working baseline for household cryptography, and option (iii) for Phase 1H and the first Phase 3 cooperative pilots. Household-level cryptography is therefore not required for early pilots: with 10–30 consenting cooperatives a disclosed tallier, data minimization, and a data-protection impact assessment give adequate protection at a fraction of the audit cost, and the pilot can test whether members value bundle secrecy enough to justify option (ii). Whatever option is chosen, it protects the annual request only: what a person later collects at a depot is handled by intra-year fulfillment (Section 7.5), which is not yet designed to keep it private. Option (i) is not a baseline at any scale considered here. The coarse baseline ties three open questions together: P5 (how well coarse totals disaggregate to products, now with 200

rather than 3,000 reported categories), P13 (the minimum batch $m_{\min} = 30$ now applies per coarse category, which makes small-hamlet anonymity easier), and Kill Gate 3's bandwidth and verification caps (Table 9.2). The resulting mobile budget per fresh-report round is about 55 kB up plus the light-client download in Section 7.7.

7.4.3 Prices, Carry-Forward, and Replay

What an offline household can prove. Eq. (7.1) uses the round's prices. A report made at round t proves the budget predicate at $P^{(t)}$ and nothing about $P^{(t+1)}$; scaling an old report proportionally does not create a new proof. VDAF-22 adds a harder limit: "Prio3 only permits a report to be aggregated once" (§7.2.3), because re-aggregation can reuse correlated randomness and compromise confidentiality (§9.4). Carrying an individual report forward into a later round's batch is therefore not available as specified.

Table 7.8: Handling households between fresh reports

Option	What is verified	Cost
(a) Fresh report every round	Budget at every round's prices	10 uploads and verifications per year; excludes anyone offline in a round.
(b) Fresh reports at milestone rounds only; aggregate carried between	Budget at milestone prices; between milestones, only the aggregate check at current prices	Household demand frozen between milestones; councils adjust at council level.
(c) Bounded overspend	Aggregate $p^{(t)} \cdot \sum y \le (1+\kappa) \sum B_u$	Individual overspend between milestones is not detected.
(d) Exclude stale reports	Nothing stale	Biases totals against people with poor connectivity.

Proposed rule (for author ratification). Combine (b) and (c). Households submit fresh, budget-checked reports at the three milestone rounds of the hybrid schedule ($t = 1, 4, 8$; P3). Between milestones, each neighborhood's demand is held at its last milestone aggregate: aggregators reuse their aggregate shares as constants, with no individual report re-verified or re-aggregated. The public aggregate check is applied at current prices with a tolerance κ (proposed 5%), and price-driven revisions between milestones happen at council level. A person offline at a milestone may submit within a grace window; otherwise they are counted at their council's per-capita average, flagged as imputed. Each released total at a milestone is a fresh aggregate of at least $m_{\min} = 30$ reports, which limits but does not remove differencing between milestones (minimum batches, coarser release, and DP noise remain the mitigations). This raises the per-person upload to three fresh reports a year: about 0.17 MB at the coarse baseline, against 9.8 MB for ten dense reports.

Figure 7.4 places the milestone rounds in the planning season and operating year.

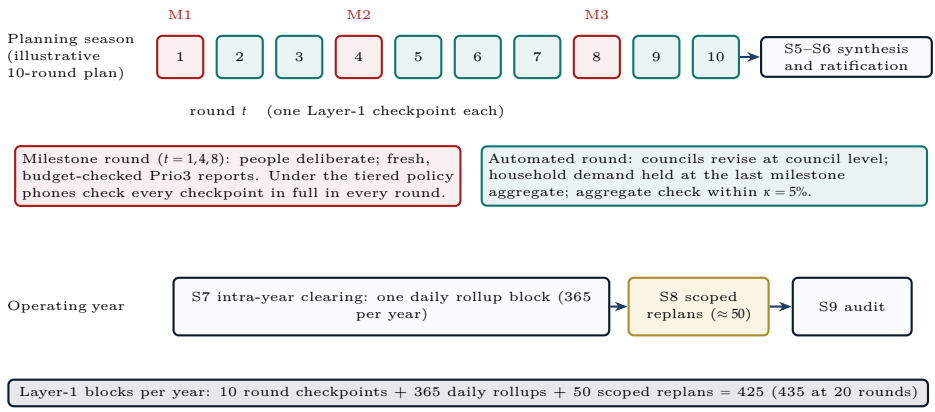


Figure 7.4: Annual calendar for an illustrative 10-round plan: three human milestone rounds with fresh reports, automated rounds in between, then the operating year and its Layer-1 block count.

Replay, re-runs and retention. Aggregators keep the nonce of every accepted report for the epoch and reject duplicates (VDAF-22 §9.2 recommends the nonce for replay protection); ctx binds epoch, round, price-vector hash, and aggregator set, so a report cannot be replayed into another round or instance. A round cannot be re-run from stored reports with a replacement aggregator: the stored shares are encrypted to the failed aggregator, and re-verifying them would aggregate them a second time. Recovery therefore means failing that milestone and requesting fresh reports, or carrying the previous milestone's aggregate. Aggregators retain encrypted input shares, nonces, and aggregate-share commitments only until the challenge window $T_{\text{challenge}}$ closes (proposed 14 days), to settle inclusion disputes, and then delete them; that bounded retention is what "ephemeral shares" means in Section 11.2. People keep their own reports locally.

7.4.4 Budget Credentials and One-Report-Per-Person

The protocol needs each eligible person to submit at most one report per milestone, against their own warranted budget, without a register linking reports to identities. No construction is specified, and Table 7.1 lists no post-quantum blind or anonymous credential. VDAF-22 notes that clients may be anonymous and that Sybil and rate-limit enforcement is outside the VDAF. Three points narrow the problem. (1) BBS-style anonymous credentials are pairing-based and offer no post-quantum security; lattice-based anonymous credentials are an active research area without a standard, and none is named here. (2) Checking the credential inside the FLP is not realistic: the FLP validates low-degree arithmetic over the secret-shared measurement, while verifying a signature or hash chain needs a very large circuit, and the credential itself would have to be secret-shared. The credential should instead be shown to the aggregators at upload, outside the VDAF, with

a per-epoch nullifier to block double submission, and bound to the report through its nonce and ctx. (3) What remains open is binding a secret-shared B_u inside Eq. (7.1) to the credential's attested budget. Until that is solved, one-report-per-person rests on council accreditation and rate limits, which is why option (iii) above is the early-pilot default. Section 7.2.1 adopts a classical credential with threshold issuance as the interim construction for the one-report-per-person part; the binding of B_u remains open.

7.5 Intra-Year Fulfillment (State S7): Consumption at the Point of Use

Once a plan is ratified, people collect goods and services through the year at neighborhood depots, shops, and service points (state S7 of Figure 4.1). This is the highest-volume part of the system, since every person uses it most days and often without a network connection, and it is where product-level detail reappears: the coarse annual report of Section 7.4.2 says "clothing," but a pickup says which garment. A record of pickups linked to a person reveals actual consumption, which is more sensitive than the request Prio3 hides. The household privacy claim of this chapter therefore covers annual requests only; it does not yet cover consumption at the point of use.

Requirements. A design for S7 has to answer six questions: (1) what a depot learns about the person in front of it; (2) what the person proves at pickup (membership, and that the pickup is within an entitlement); (3) how spending against an individual budget is enforced while the depot or the phone is offline; (4) how double spending across depots is detected or bounded; (5) what reaches the daily rollup on Layer 1 (Figure 7.4); and (6) how the rollup reconciles with the annual plan and triggers scoped replans (S8) when net shocks exceed planned slack.

Daily rollup and reconciliation. Under every design, the daily rollup carries depot-level category totals and a Merkle root over that day's receipts or redeemed token serial numbers, not individual records. Federations compare totals with the planned flows and the slack buffer SL^E; when the net shock exceeds slack, the Updating Facilitation Board opens a scoped replan (S8). Receipts held by depots and home councils are kept only until the reconciliation window closes, under the same retention rule as Prio3 shares (Section 7.4.3).

Recommendation for pilots (proposed, for author ratification). Use design (c) for staples and design (a) for rationed or high-value goods, with receipts held only by the person's home council and Ed25519 receipts under the tiered signature policy. Rationed goods are checked online against the home council's ledger whenever a connection exists; when it does not, they can be collected only at the person's home depot, up to an offline credit cap. A person's offline overspend is then

Table 7.9: Candidate designs for intra-year fulfillment (illustrative costs: two pickups per person per day)

Design	Privacy at the depot	Offline behavior and double spending	Cost; maturity
(a) Attributed receipts with a per-person credit cap	The depot and the home council see what each person collects.	Offline overspend is at most the cap times the number of depots visited before the daily rollup; restricting offline use to the home depot makes the bound one cap per person per day.	About 0.5 kB per person per day with Ed25519 (14 kB hybrid); established practice, M4.
(b) Anonymous tokens per category and period (blind-signature e-cash)	The depot learns the category and that the token is valid, not who holds it.	Tokens are unlinkable, so an offline double spend cannot be traced to a person unless the scheme reveals identity on double spending; otherwise only per-depot caps bound it. GNU Taler is explicitly an online system, and its offline design was abandoned.	About 0.4–0.8 kB per person per day; offline form research-grade, M1–M2.
(c) Council-level allocation for staples, tokens or receipts only for rationed goods	Staples are not tracked per person; the depot draws against the neighborhood's allocation.	Nothing to double-spend for staples; rationed goods use (a) or (b).	Lowest cost; needs depot stock accounting, not individual ledgers.

Byte costs from research/analysis/verify_signature_suites.py (section S6) with an assumed 120 B receipt payload. GNU Taler design document 30, "Offline payments," states that the proposal was abandoned and that "Taler is explicitly meant to be an online payment system" (https://docs.taler.net/design-documents/030-offline-payments.html, checked November 7, 2026).

bounded by one cap per day, reconciled at the daily rollup. A spend counter held in the phone's secure hardware would allow offline collection anywhere, but it would make the platform vendors part of the trust base and exclude phones without such hardware, so it is left for later study. State plainly to pilot members that what they collect is visible to their depot and home council. Treat design (b) as research for a later phase. Until a design with privacy at the point of use is specified and tested, the book's household privacy claim is narrowed to annual requests (Kill Gate 3, Table 9.2; open question P16).

7.6 Threat Model, Equivocation, Recovery, and Ballots

Figure 7.5 summarizes what each part of the design rests on; the list below gives the detail.

Holds under stated assumptions. Hybrid KEMs as in FIPS 203; hybrid and hash-based signatures as in FIPS 204/205 for long-lived objects, and Ed25519 or BLS for per-round objects until a migration trigger; BLAKE3 collision/preimage as in the BLAKE3 spec (128-bit target); Two-Root inclusion if a verifier has the leaf archive; Prio3 measurement privacy if ≥ 1 aggregator is honest.

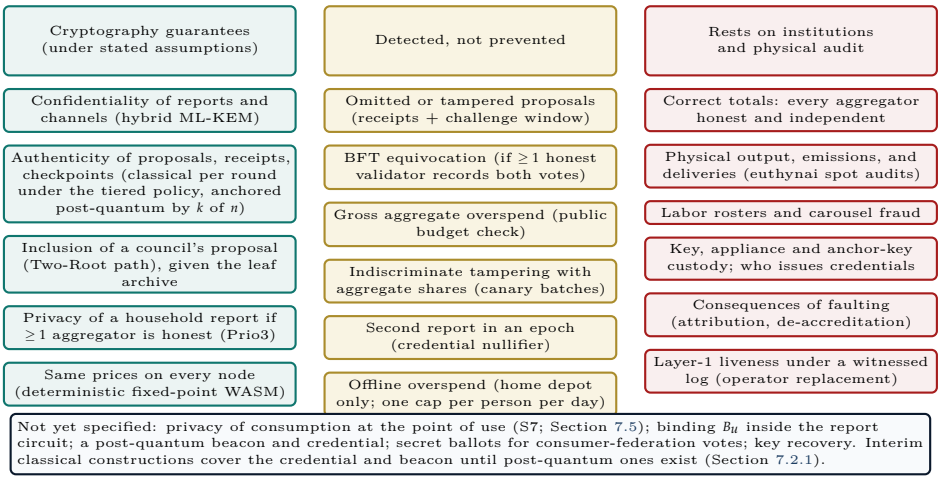


Figure 7.5: Trust map: what cryptography guarantees, what is only detected, and what rests on institutions and physical audit.

Detected, not prevented. Leaf omission given receipts and a challenge window; BFT equivocation given ≥ 1 honest validator recording both votes; aggregate overspend if the public $p \cdot \sum y \leq \sum B_u$ check is implemented.

Depends on social institutions. Physical output, emissions, labor rosters, carousel fraud, aggregator independence, sortition appliance custody, and whether "faulting" an aggregator has consequences. There is no stake to slash; faulting means public attribution, dual-federation de-accreditation, and exclusion from the next sortition pool.

Aggregator compromise. Replace the aggregator and either request fresh reports for the affected milestone or carry the previous milestone's aggregate. Stored reports cannot be re-verified with a replacement aggregator (Section 7.4.3). Old committee keys are a long-range risk: checkpoints must be weakly subjective (users pin a recent checkpoint) or re-signed by the new committee.

Key compromise. Individual keys: revoke by council k-of-n decision and re-issue. No specified recovery UX or secure-element/mobile-storage profile; community-hosted appliances may make the host the real validator unless keys are in a citizen-held token. Hardware custody needs the native client (Section 5.5); a web app cannot reach these keystores. Until Android exposes post-quantum keys in hardware, long-lived hybrid keys use split hybrid custody: the classical half of each hybrid key stays in the platform keystore (Android Keystore or StrongBox, Apple Secure Enclave) and the ML-DSA half in software, encrypted under a hardware-held key. Until a CRQC exists a forger needs both halves, so the hybrid inherits the hardware's protection. Apple's Secure Enclave can

hold ML-DSA keys directly from OS 26.[8] Forward-secure erasure of committee keys is an operational promise until a ratchet (e.g., per-round key derivation with deletion) is specified.

Replay and domain separation. Every signed object includes epoch_id, round, role, and a domain-separation tag of the form DPEP-WD-v0-<OBJECT> (Listing 4.1); the version component changes when the specification leaves working-draft status. Unspecified objects are not to be accepted.

Intra-year fulfillment. Depots and home councils see what each person collects under the pilot design; rationed goods are checked online when possible and otherwise only at the home depot, so offline overspending is bounded by one credit cap per person per day and detected at the daily rollup (Section 7.5).

Distribution and notifications. App-store removal and push-notification metadata are external censorship and surveillance vectors; they are handled by distribution outside the stores, reproducible client builds, and content-free notifications (Section 5.5).

Secret ballot. Multi-signature bitmaps reveal who voted yes. Binding council votes may stay attributed; consumer-federation ballots that must be secret need a separate coercion-resistant scheme (open).

7.7 Constitutional Governance and Sortition

To prevent software maintainers or server-rich councils from becoming a new coordinator class:

1. Dual-federation ratification of WASM rules, catalogs, and caps: Changes to wasm_rule_hash, product categories Ω, or 5-plan voting grids require a $\geq 2/3$ dual supermajority of the National Consumer Federation and Congress of Industry Federations with a 60-day reproducible-build audit window, and cannot be altered mid-plan.

2. Sortition beacon (unresolved). A beacon "anchored" on outgoing SLH-DSA-192s signatures is grindable if those signatures are randomized (FIPS 205 permits this). Commit-reveal has a last-revealer problem. A verifiable delay function could address grinding but is not specified here. A post-quantum beacon remains an open cryptographic problem, and there is no VRF in this draft. The interim construction is a threshold beacon run by the federations (Section 7.2.1); its key generation and custody are unspecified.

3. Custody: Sortition draws human delegates and provisions Role-BFT appliances. If keys live on the appliance, the host is the validator. Required:

[8] Apple, "Enhancing your app's privacy and security with quantum-secure workflows," CryptoKit documentation, https://developer.apple.com/documentation/cryptokit/enhancing-your-app-s-privacy-and-security-with-quantum-secure-workflows (checked November 2026). No equivalent Android hardware profile was found.

citizen-held tokens, or mark custody unresolved. The same applies to post-quantum anchor keys (Table 7.3): each federation's SLH-DSA key, or each witness's ML-DSA-44 key, is held by that body alone, on its own signing hardware or offline appliance, and never on a shared server; its custody rules are as yet unspecified.

4. Committee size $n = 128$: $f = 42$, quorum $\lceil 2n/3 \rceil = 86$. Exact binomial tails[9] for a single annual draw, corrupt share p: at $p = 15\%$, $1.3 \times 10^{-5}\%$ ($n = 128$) vs. $9.5 \times 10^{-3}\%$ ($n = 64$); at $p = 20\%$, 0.022%; at $p = 25\%$, 1.82%; at $p = 30\%$, 21%. Larger committees give little protection at a 30% corrupt pool. An annual draw means a captured committee lasts a year; 1–4 rotations/year reduce but do not eliminate that window.

Light clients. At $n = 128$ a checkpoint carries 86 quorum signatures. If they are hybrid they total about 290 kB, which would dominate a phone's per-round traffic; if they are Ed25519 they total 5.5 kB, and an aggregated BLS signature with a signer bitmap is 112 B, checked with one pairing equation in under half a millisecond on a desktop core, provided the engine or log aggregates signatures (CometBFT does not). Table 7.10 sets out what each option costs and trusts.
Proposed default (for author ratification): under the tiered policy, every phone checks every checkpoint in full, at 5.5 kB with Ed25519 (CometBFT as configured) or 0.1 kB with an aggregated signature where the engine or log supports one, so coordinator attestation and the sync committee are no longer needed. If the all-hybrid policy is kept instead, the earlier default stands: coordinator attestation every round, a full check at milestone rounds and on Wi-Fi, and a sync committee as the planned upgrade once the interim beacon is running.[10] Under the tiered policy a phone's traffic is about 21–46 kB in an automated round and 76–101 kB in a milestone round, including the 55 kB report; with the hybrid full check it would be about 306–386 kB. The mobile target written into Kill Gate 3 (Table 9.2) is therefore $\leq$ 150 kB in every round under the tiered policy, and under the all-hybrid policy $\leq$ 150 kB per ordinary round and $\leq$ 450 kB in a round with a full quorum check.

[9] Computed by research/analysis/verify_r2.py in the project repository.
[10] Ethereum consensus specifications, Altair light-client sync protocol, https://github.com/ethereum/consensus-specs/blob/dev/specs/altair/light-client/sync-protocol.md. Probabilities are exact hypergeometric tails from research/analysis/verify_sync_committee.py; sizes from research/analysis/verify_signature_suites.py.

Table 7.10: Light-client verification options for Role-LP (per checkpoint)

Option	Download	What the phone trusts
All-hybrid signatures		
Fetch and verify all 86 signatures	≈ 290 kB	Only the BFT assumption (≤ 42 faulty validators).
Council coordinator verifies and attests	≈ 3.4 kB	The council's own Role-CC; a corrupt coordinator can feed a false checkpoint until a member checks.
Random sample of 8 signatures	≈ 27 kB	Probabilistic: if only the 42 corrupt validators really signed a checkpoint claiming 86, all 8 samples pass with probability $(42/86)^8 \approx 0.3\%$. Not a proof, and it fails if the sample is predictable to the adversary.
Full check at milestone rounds only	290 kB ×3/yr	BFT at milestones; coordinator attestation in between.
Rotating sync committee: 24 validators drawn by the beacon, quorum 16	≈ 54 kB	That fewer than 16 of the 24 are faulty. With 42 faulty validators among 128, a draw yields a capturing committee with probability 0.017% (16 members, quorum 11: ≈ 37 kB, 0.19%). A captured committee lasts until the next rotation, and the draw is only as good as the beacon.
Tiered signature policy (proposed)		
Verify all 86 Ed25519 signatures	5.5 kB	Only the BFT assumption.
Verify one aggregated BLS signature	0.1 kB	Only the BFT assumption; classical pairing assumption. Needs an engine or log that aggregates; CometBFT does not.
Witnessed log: verify 11 of 16 cosignatures	0.9 kB (Ed25519) or 30 kB (ML-DSA-44)	That fewer than 11 witnesses collude with the log operator.

CHAPTER 8

Accountable AI, Mathematical Optimization, and Failure Modes

> Finally, we point out to those who fear that computers could become the new dictators, the computer programs are human products, socially evaluated and improved each year… In short, neither consumers nor producers need accept computer facilitated alternatives.
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 9, "Revising Data in Subsequent Iterations"[1]

8.1 Architectural Rule: Deterministic Core, Advisory Edge

AI must never become an unaccountable central planner or surrogate coordinator class. DPEP enforces a strict boundary:

- Core Clearing & Pricing Engine (non-AI by specification): All price updates, SB/SC ratios, and Society Account settlements are specified as deterministic fixed-point arithmetic. Local IEEE-754 solver output, including any KKT certificate, checks stationarity of the program the council chose to solve. It does not prove a truthful objective, and MILPs have no KKT certificate of global optimality (Section 4.5; Boyd et al. 2011).
- Edge Advisory & Accessibility Support: Classical hierarchical time-series models (ETS/ARIMA) forecast SKU size/color distributions; local QP solvers draft initial proposals from last year's bundle (eliminating David Schweickart's "consumption list burden" objection; Schweickart 1993, 329–334); and optional local open-weight LLMs are restricted to translation, voice/screen-reader accessibility, and summarizing qualitative Lists with mandatory links to raw council text.

8.2 Substantive Engagement with Major Economic & Democratic Critics

This section engages the main theoretical critiques of Participatory Economics across philosophy, economics, and mechanism design: the Winter 2024 special issue of the Erasmus Journal for Philosophy and Economics (John O'Neill, Hannes Kuch, Jan Philipp Dapprich), read in the open-access issue PDF;[2] David Laibman

[1] Albert and Hahnel 1991b, chap. 9 (lf9_2.htm).

[2] Erasmus Journal for Philosophy and Economics 17, no. 2 (Winter 2024), issue PDF at https://ejpe.org/journal/issue/download/35/67: O'Neill 2024a; Kuch 2024; Dapprich 2024; O'Neill 2024b.

Table 8.1: Evaluation of key decision-support capabilities against verifiability and override criteria

Capability	Best Method (Is AI Needed?)	Local?	Verifiability & Challengeability	Override & Power Concentration Safeguards
1. SKU Demand Forecasting	No LLM needed: Hierarchical ETS/ARIMA + demographics.	Yes (< 50 ms)	Published MAPE error bounds & open parameters.	Non-binding baseline; councils edit freely.
2. Workplace Input Optimization	No AI needed: Conic QP / LP (Clarabel.rs / HiGHS).	Yes (WASM time unverified)	KKT certificates, where they exist, witness stationarity of the local convex program, not a truthful W_j or MILP optimum.	Presents 3 Pareto-frontier options for worker council vote.
3. Household Proposal Drafting	No AI needed: Deterministic diff + constrained utility QP.	Yes (on device)	Line-by-line diff against prior year's bundle.	Mandatory review UI highlights top changes; zero data leaves user's device.
4. Trade-off Explanation	Deterministic Leontief Cost-Push Attribution ($(I-A)^{-1}$).	Yes (< 50 ms)	Exact algebraic decomposition of ΔP_i.	Avoids LLM hallucination by rendering exact resource/labor/pollution cost shares.
5. Qualitative List Translation	Yes: Local Neural Machine Translation + TTS.	Yes (Role-CC)	Original source text & BLAKE3 CID always linked alongside.	Bilingual delegates can post verified overrides; expands cross-regional inclusion.
6. Late-Round 5-Plan Synthesis	No AI needed: Parametric LP over 5 fixed weight vectors.	Yes (< 2 s)	Deterministic WASM LP verifiable by any council.	Councils can vote to reject all 5 plans and continue iterating (LF Ch. 9 lf9_2.htm).

(2014), consulted through the publisher's abstract, with Robin Hahnel's (2014) response;[3] Paul Cockshott and Allin Cottrell (1993); János Kornai (1980, 1986); Martin Weitzman (1974, 1980); and David Schweickart (1993).

1. Dapprich (2024) on democratic theory, liberty, and general procedures: Dapprich asks whether three justifications of democracy (protecting interests, revealing preferences, and liberty in the neo-Roman republican sense) support citizen participation in concrete plans, and argues that they do not: citizens should instead choose general planning procedures, an "algorithmic planning process" in his words, which protects them from the arbitrary power of coworkers and neighbors as well as of central planners. He also argues that most participatory input in ParEcon actually comes from individual consumption requests, leaving consumer councils with little to vote on, and that without

[3] Laibman 2014; Hahnel 2014. Only the abstract of Laibman's article and the bibliographic record of the response were consulted.

constitutional constraints a worker is exposed to the arbitrary will of the majority on their own council.

- Where Dapprich is right: Neighbor veto of in-budget personal consumption is a liberty problem; DPEP's non-veto rule follows PEPE, and cryptographic invisibility goes further (Section 7.1). His point about workplace majorities is also well taken: DPEP needs entrenched individual rights inside councils (safety, exit to another council through the EFB, BJC protections), not only majority procedures.
- Where this monograph disagrees: Workplace pace, safety and emissions, and city-scale public goods, are collective activities whose terms cannot be fixed in advance by a general procedure without someone writing the procedure's parameters, which re-creates a planning authority one level up. DPEP therefore keeps councils as decision-makers for collective activities and uses general procedures (WASM rules ratified by dual supermajority, Section 7.7) as constraints on them. This answers Dapprich only in part.

2. Kuch (2024) on centralization tendencies: Kuch argues that a participatory economy needs more central institutions than Hahnel suggests, for strong normative reasons: industry federations act as gatekeepers, appellate bodies, and adjustment agencies; national federations decide investment and society-wide public goods; disciplinary bodies may be needed to counter the procedure's false incentives; and conflicts between federations may require a national legislature or something like a ministry of planning. He doubts that council-democratic remedies resolve this. DPEP's responses (Algorithm 3c's federation MILP as an aid rather than an authority, sortition, the Innovation Jury below, dual-federation ratification) are institutional proposals that Kuch's argument puts directly in question; they are tested in P7 and are not shown to work.

3. Cockshott & Cottrell (1993) on direct LP and scalar labor-time: Cockshott and Cottrell propose replacing iterative council negotiation with central LP in embodied labor hours. Two objections are raised here (from the bibliographic record and the secondary literature, not a full reading of Towards a New Socialism): (a) scalar labor-time omits the independent scarcity of non-producible natural resources (r^*), ecological sinks ($E_z^{\max}$) and fixed capital (k^*), and differential workplace disutility; (b) a central LP that needs each workplace's full private $\mathcal{T}_j$ collides with Hayek (1945) on tacit knowledge and with worker self-management. Hayek's point also applies to DPEP: coarse categories, an annual cycle, and published technical recipes (REQ-DATA-03) still lose local know-how, and Algorithm 3c's federation MILP is a limited central solve over posted menus, not over hidden $\mathcal{T}_j$. Treating 3c as a Phase-1 oracle and a ratified aid, not an authority, is the intended boundary; it remains a tension, not a resolution.

4. Laibman (2014), Kornai (1980) and Weitzman (1974, 1980) on equilibrium, soft budgets, and the ratchet effect:

   - The Walrasian resemblance and dynamic entry (Laibman 2014; Kuch 2024): Laibman's abstract argues that the Albert–Hahnel model is designed in an "abstractly speculative" way, that its fear of authority and hierarchy has roots in classical anarchism, and that the result bears an "uncanny resemblance to Walrasian competitive market equilibrium." Hahnel replied in the following issue. This monograph does not attribute to Laibman any argument beyond his abstract. One practical consequence of the Walrasian reading is that an iterative procedure over a fixed list of producers says nothing about how new producers enter. DPEP's proposal: an accredited worker group refused accreditation by an Industry Federation (Kuch's gatekeeping concern) can appeal to a Cross-Industry Innovation Jury, selected by sortition and co-ratified by the Consumer Federation, backed by a dedicated R&D capital slice. This is untested (P7).
   - Hardening Soft Budget Constraints (Kornai 1980, 1986): If councils with $SB_j/SC_j < 1$ could obtain unlimited requests_appeal waivers, budget discipline would collapse (Kornai 1980, 1986; Weitzman 1974, 1980). DPEP limits appeal waivers to one planning cycle (requiring a physical retooling audit) and funds them from a capped Industry Transition Pool, so granting a waiver directly reduces peer councils' discretionary industry reserve.
   - Ratchet effect (Weitzman 1980)—open: Basing endowment charges on physical assets rather than last year's output removes one Soviet-style ratchet. It does not remove the fear that revealed capacity will feed next year's benchmarks, public blueprints (REQ-DATA-03), or SB/SC comparisons. Weitzman (1980) is cited at bibliographic level. Status: open.

5. Effort rating, surveillance, ecology, and limits (O'Neill 2024a): O'Neill makes three criticisms. First, distributing income by a metric of effort would involve pervasive surveillance and potential humiliation; this bears directly on the effort default discussed in Section 9.1.4. Second, a single price metric of opportunity cost and cost–benefit analysis cannot address value pluralism and incommensurability; he defends rational choice without commensurability and in-kind analysis in the tradition of Neurath and Kapp. Third, environmental limits must be taken more seriously than Hahnel does: substitution is necessary but not sufficient, and he proposes consumption and production "corridors" above sufficiency for all but below excess. DPEP's intended responses: (a) strict one-sided biophysical ceilings $(r^*, E_z^{\max})$, which are in-kind constraints rather than prices, set by long-run environmental planning, with an intergenerational trusteeship mandate that living cohorts cannot relax $E_z^{\max}$ without a supermajority plus a sortitioned youth/future-persons panel; (b) a per-capita exposure floor so that no neighborhood's modeled dose exceeds a published health threshold even if the regional CAP accepts the aggre-

gate E_z; (c) a plume sub-zone veto for the most-impacted $CC_h \subset CAP_z$; (d) earmarking $\geq 50\%$ of $p_z^{\text{pol}} E_z$ to remediation and preventative health. Rules (c) and (d) create hold-out and earmark-gaming problems (a sub-zone can veto to extract transfers; 50% earmarks invite recategorizing ordinary spending), and DPEP has no counterpart yet to an upper consumption corridor. Status: working rules, not solved mechanism design.

6. Governance of solver defaults and deliberative fatigue (Devine 1988; Albert 2003, chap. 18): Even without AI, local QP solver defaults could become de facto decisions if citizens rubber-stamp them. DPEP bans silent auto-submission of $> 5\%$ effort or category shifts, requires active choice among ≥ 3 options at milestone rounds, and measures rubber-stamping, comprehension, and the distribution of influence in Phase 1H (Table 9.2).

7. Black-market arbitrage and wage labor (Albert 2003, chap. 20): Prio3 is intended to keep personal bundles private and to reject reports that violate the encoded budget predicate if that circuit is specified and sound. An unauthorized firm still cannot draw intermediate inputs through the planning tree. Off-protocol barter remains a social question. No "mathematical prevention" of wage labor is claimed.

Three-Phase Experimental Program, Kill Gates, and Benchmarks

> The easiest "experiments" to implement are a variety of computer simulations in which various features of the system and assumptions about actors' behaviors can be altered to see how outcomes change… Conceivably one could run a "shadow" participatory economy within an existing economy.
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, "Feasibility," pp. 108, 129[1]

9.1 Why DPEP Uses Gated Phases Instead of a Single Monolithic Build

Attempting to build non-convex solvers, QUIC, aggregation, WASM, PQC, Prio3, and mobile UIs in one monolithic prototype would conflate mechanism validation with distributed-systems engineering. Each phase therefore ends in a kill gate: a test with a pre-registered pass condition, a fail condition, and a stated consequence. A gate that cannot fail is not a gate.

Table 9.1: Staged experimental program (budgets as in Table 1.2)

Phase & Stack	Core experiments	Duration & staff
Phase 1: Economic coordination (Python, NumPy, HiGHS, Clarabel)	Pre-registered instance set (below); Algorithms 3a–3d against a central LP/MILP oracle; scaling from 50 to 1,000 councils.	4–6 months; 1.0 FTE OR/economics
Phase 1H: Low-tech human probe (disclosed trusted server + web UI)	30–60 volunteers in 3 worker and 3 consumer councils run three milestone rounds of a simplified plan. Ethics review and consent first.	3–4 months; 1.0 FTE UX/dev; in parallel with Phase 1
Phase 2: Distributed coordination (Quint/TLA$^+$ model, then Rust dpep-node, CBOR, QUIC, Wasmtime)	Model-checked specification of states S2–S4, S7, and S8 first (Section 9.1.2); then a WAN testbed at $\geq 10^4$ simulated participants running both Layer-1 designs; fault injection (censorship, equivocation, 20% partition, stalled log operator); Prio3 verification and signature checks in the round.	6–8 months; 2.5 FTE
Phase 3: PQC, Prio3 & pilot (FIPS 203–205, Prio3 on DAP, native Android/iOS via UniFFI)	Device benchmarks; external audit; 10–30 cooperative field pilot, starting with the council-level trust model (Section 7.4.2) and the pilot fulfillment design (Section 7.5).	9–12 months; 4.5 FTE + audit

[1] Albert and Hahnel 1991a, chap. 6, 108, 129.

9.1.1 Kill Gates as Tests

Table 9.2 states each gate. All numeric thresholds are proposed, for author ratification, and each carries its reason. Consequences use four outcomes: proceed; narrow scope (continue only for the problem class or setting that passed); redesign and rerun (once, on a freshly pre-registered set); stop (publish the negative result and fund no later phase on that basis).

Table 9.2: Kill gates: pass and fail conditions, consequences, and reasons (thresholds proposed)

Gate	Pass (all required)	Fail (any)	Consequence and reason
1 Economic	For the best candidate: (a) $\geq 90\%$ of convex and Leontief instances cleared (Definition 4.2) within 25 rounds; (b) $\geq 70\%$ of staggered-lumpy instances cleared; (c) on Scarf and identical-lumpy instances, either cleared or correctly reported feasible, not cleared or infeasible with diagnostics in 100% of runs; (d) 100% of infeasible instances reported infeasible; (e) $\geq 70\%$ of endogenous-income instances cleared by 3d; (f) median welfare gap to the oracle $\leq 5\%$, 90th percentile $\leq 15\%$.	Fewer than 50% of convex and Leontief instances cleared; or any infeasible instance reported feasible; or median welfare gap $> 15\%$; or endogenous-income clearance $< 40\%$.	Proceed if all pass. Narrow scope if (a), (d), and (f) pass but (b) or (e) does not: Phase 2 then covers only the passing classes, with lumpy decisions left to federation MILP and ratification. Redesign and rerun once if between pass and fail. Stop if the fail condition holds after the rerun. Reasons: convex instances have known convergent methods, so failure there signals a design or implementation fault; 70% for lumpy instances reflects the one-good test, in which 57% of rounds clear; 25 rounds is about twice the 11.85-round pequod-cljs baseline; reporting an infeasible plan as feasible is never acceptable.
1H Human probe	(a) $\geq 80\%$ of participants complete all three milestones; (b) median score $\geq 70\%$ on a pre-registered comprehension quiz (prices, budgets, what an override does); (c) $\leq 25\%$ of solver drafts accepted without opening the change view; (d) of coded override reasons, $\geq 50\%$ are substantive (need, preference, local knowledge) rather than confusion; (e) the most active 20% of participants originate $\leq 60\%$ of adopted amendments.	Completion $< 60\%$; or median comprehension $< 50\%$; or $> 50\%$ of drafts accepted unopened; or the most active 20% originate $> 80\%$ of adopted amendments.	Proceed if all pass. Redesign and rerun (interface, facilitation, schedule) once if between. Stop human-facing claims and keep simulator research only if the fail condition holds after the rerun. Reasons: override rate alone rewards bad defaults, so it is paired with comprehension, reasons, and the spread of influence; (e) is a direct test for an emerging technical or articulate elite. Limits: 30–60 self-selected volunteers cannot show generalization to a population, long-run fatigue, or power dynamics at scale; with $n = 40$ a 95% interval on a proportion near 0.5 is about ± 15 points.

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Gate	Pass (all required)	Fail (any)	Consequence and reason
2 Distributed	Entry: the formal model of Section 9.1.2 passes model checking. (a) 100% of injected omissions and tamperings detected within $T_{\text{challenge}}$ over $\geq$ 1,000 fault-injection trials, for each Layer-1 design tested; (b) automated round time (excluding human deliberation), including Prio3 verification and signature checks, p95 $\leq$ 10 min at the tested scale ($\geq 10^4$ participants), and $\leq$ 60 min when extrapolated to 10^6 from measured per-item costs with $\leq$ 64 cores per aggregator; (c) total bytes and CPU $\leq 10\times$ an Architecture A baseline on the same workload; (d) for the witnessed log, a stalled operator is replaced and the round completed within one extra round window.	Any undetected tampering; or p95 $>$ 30 min at the tested scale; or overhead $> 50\times$ the A baseline.	Proceed if all pass. Narrow scope (smaller committees, coarse reports only, regional rather than national deployment) or redesign and rerun if between. Stop the distributed build and pilot on Architecture A if the fail condition holds. Reasons: detection is the point of Two-Root aggregation, so one miss fails it; 10 min fits the 1–10 min target of Table 5.4; a tenfold overhead is a reasonable price for removing a trusted operator, fiftyfold is not. If both Layer-1 designs pass, the simpler one to operate is preferred (Section 5.3.2).
3 Crypto, mobile, pilot	(a) External audit with no unresolved critical or high findings before field use; (b) on a 3 GB-RAM Android device: $\leq$ 150 kB in every round under the tiered signature policy (all-hybrid: $\leq$ 150 kB per ordinary round, $\leq$ 450 kB with a full quorum check; Section 7.7), peak RAM $\leq$ 150 MB, coarse report shard $\leq$ 1 s; (c) pilot: $\geq$ 70% of member councils still participating at the end, no unresolved privacy incident, and members' rated legitimacy of the plan not below their rating of the pre-pilot process; (d) fulfillment: daily rollups reconcile with depot stock within 1% of value, and no person's offline overspend exceeds one credit cap per day; (e) client apps are reproducibly built and available outside the app stores.	Unresolved critical audit finding; or any device metric more than 2$\times$ its cap; or pilot retention $<$ 40%.	Proceed to wider pilots if all pass. Narrow scope to the council-level trust model (no household cryptography) if (b) fails but (c) passes. Stop field use of custom circuits on an unresolved critical finding. Reasons: the bandwidth caps follow from the coarse 55 kB report plus the light-client options; 150 MB leaves room on entry-level phones; retention and legitimacy test whether people want to keep using the process, which no simulation can show.

Pre-registration. Before each gate, the fast-moving rows of the component maturity table (Table 6.3) are re-checked and the result is recorded with the gate's pre-registration. Before any candidate solver is run on them, the instance generators, random seeds, instance counts, metrics, thresholds, and analysis scripts are committed to a pre-registration directory in the project repository,[2] tagged with

[2] research/preregistration/.

a signed Git tag, and the tag's commit hash is deposited with an independent registry (for example the Open Science Framework). Results are reported against that tag. Deviations are logged with reasons in the same directory, and any re-run uses a freshly registered set. Phase 1H registers its quiz, coding scheme for override reasons, and influence measure in the same way, together with its ethics approval.

Pre-registered Phase 1 instance set. At least 20 instances in each family, at 50, 200, and 1,000 councils: (1) convex baselines (Cobb–Douglas and CES with fixed budgets); (2) Leontief input–output economies; (3) Scarf-style cycles; (4) identical lumpy plants; (5) staggered lumpy plants; (6) deliberately infeasible plans (caps below minimum requirements); (7) endogenous income through the Society Account; (8) binding ecological caps; (9) strategic misreporting, in which a fixed share of councils understate capacity by a fixed fraction, measuring the resulting price distortion and the misreporters' gain (reported as a diagnostic; no pass threshold is set until a baseline exists).

What a 1,000-council result can and cannot show. Phase 1 can show whether rounds-to-clear grow with the number of councils between 50 and 1,000 (a published log–log slope) and whether failure modes depend on scale. A flat or falling slope makes extrapolation plausible; a rising slope is evidence against national-scale feasibility. A pass at 1,000 councils does not show convergence at 10^6 participants: heterogeneity, the number of distinct goods, strategic behavior, and communication costs all change with scale, and Phase 2 measures the last of these separately.

9.1.2 Formal Specification Before Rust: a Phase 2 Entry Criterion

The state machine of Figure 4.1 has submission deadlines, grace windows, carry-forward, milestone and automated rounds, a two-bit participation bitmap, a challenge window, forced inclusion, aggregator replacement, and scoped replanning. Timeouts and liveness are where protocols of this kind usually fail, and they are hard to test by running code. Before any Phase 2 Rust work, states S2–S4, S7, and S8 are therefore written as a formal model in Quint or TLA$^+$ (Quint is used for the Malachite consensus specifications, and CometBFT keeps TLA$^+$ and Quint specifications)[3] and model-checked. At a minimum the model must establish:

1. no honest council holding a valid receipt is excluded from a finalized checkpoint;
2. every node computes the same $P^{(t+1)}$ for a finalized round;

[3] Quint, https://github.com/quint-co/quint (v0.33.1, October 28, 2026); Malachite's Quint specifications are in specs/consensus/quint of https://github.com/circlefin/malachite; CometBFT's in spec/ of https://github.com/cometbft/cometbft (all checked November 7, 2026).

3. a single partitioned council cannot halt a round;
4. a replaced aggregator cannot cause a report to be aggregated twice;
5. the challenge window always closes;
6. under the witnessed-log design, a stalled operator is replaced and the round completes;
7. in S7, with offline collection of rationed goods limited to the home depot, no person's offline overspend exceeds one credit cap per day before reconciliation.

The model costs about 1.0 FTE for 2–3 months ($20k–$45k at the assumed rate), included in the Phase 2 budget of Table 1.2. It becomes the reference against which the Rust implementation is tested.

9.1.3 Ethics, Consent and Data Protection

Phases 1H and 3 involve human participants and personal data. Before recruitment each requires: independent ethics review (a university research ethics committee if an academic partner is involved, otherwise an independent ethics panel whose membership is published); informed, written consent with a plain-language description of what is recorded, who can see it, and the right to withdraw and have data deleted; data minimization and a retention schedule; a data-protection impact assessment where GDPR or UK GDPR applies (Article 35); and fair compensation for participants' time. Phase 3 adds a data-processing agreement with each pilot cooperative and a named contact for complaints. These add 4–10 weeks of elapsed time before each phase and the non-labor costs shown in Table 1.2.

9.1.4 The Effort-Rating Default (proposed)

Equation (4.7) has two parameters with different meanings. $\bar{\epsilon}_{\max}$ caps a council's mean effort multiplier and so prevents council-wide grade inflation; the individual spread $\epsilon_{\max}$, with $|\epsilon_u - 1| \leq \epsilon_{\max}$, decides whether two people working the same hours in the same council can be paid differently for intensity. Maxim 3 asks for that difference. O'Neill (2024a) argues that rating it means pervasive surveillance and potential humiliation, and cooperatives often prefer flat hourly pay. This is a political choice, not a technical one. Proposed default (for author ratification): $\bar{\epsilon}_{\max} = 1.0$ in every council; individual spread $\epsilon_{\max} = 0$ (flat pay per accredited hour, with onerousness handled through job-complex balancing and hours) unless a council votes to allow peer-rated spread, capped at $\epsilon_{\max} = 0.15$ and reviewed after one plan year. Phase 1 simulates both settings; P6 tests them with cooperatives. Under this default DPEP honors Maxim 3 through hours and job balancing but not, by default, through intensity, and the monograph says so.

9.2 What a Working Demonstration Would Show

The phases and gates above are easier to judge against a concrete picture. This section describes the demonstration the program aims to be able to show, scene by scene, and states for each scene what it tests, when it could be shown, and its present status (Table 9.3). It describes a target, not software that exists.

Scene 1: councils plan. A visitor opens an application representing a small economy of 20 workplaces and 20 consumer councils. Worker councils propose what they can produce and the inputs they need; consumer councils propose what they want, including collective goods. Every participant sees, for each category, where proposals exceed what is available. Councils revise round by round, and the visitor watches the imbalance shrink. The iteration facilitation board computes and publishes new indicative prices; it cannot draft, alter, or replace any council's proposal (Sections 2.4 and 2.6). The demonstration economy is one in which rounds are known to converge. Whether they converge with indivisible plants and with incomes that respond to prices is what Kill Gate 1 tests (Section 4.4.3); whether real participants revise drafts rather than accept them is what Gate 1H tests.

Scene 2: a council goes offline. One council is disconnected. Its members keep working on their proposal on their own devices. If the council reconnects before the round closes, its fresh proposal counts; if it misses the deadline, its previous proposal is carried forward, and the participation bitmap records it as carried forward, not excluded (Section 5.3.1). At no point does the council hand its decision to anyone else to complete.

Scene 3: an operator cheats. A server operator alters a council's signed proposal, or leaves it out of the round. The alteration fails signature verification. The omission is exposed when the council presents its signed submission receipt within the challenge window, and inclusion is then forced and the operator publicly faulted (Figure 5.3). The design detects this manipulation; it does not prevent it, and it does not detect a council that misreports its own output, which rests on audit and institutions (Figure 7.5).

Scene 4: private household requests. Households submit their consumption requests split into shares held by two independent aggregators, run by different operators. The neighborhood total is correct, each report passes its validity checks, and neither aggregator sees any household's request (Section 7.4). Privacy holds if at least one aggregator is honest; correct totals need both. Proving privately that each request stays within the household's budget is not yet specified (Section 7.4.4).

What would count as failure. Each scene corresponds to pre-registered pass and fail conditions in Table 9.2. A demonstration that passes them would show that

the coordination mechanisms work at this scale and under these conditions. It would not show that they work at the scale of a society; Section 9.3 estimates what changes as scale grows.

Table 9.3: The demonstration scene by scene: claim, mechanism, phase, gate, and present status

Scene	Claim shown	Mechanism	Phase	Kill gate	Status
1. Councils plan	Coordination without a market or central office	Price rule and solvers (Section 4.4); states S2–S4 (Figure 4.1)	1, 1H	1 (clearing); 1H (drafts revised, comprehension)	Demonstrated for smooth Cobb–Douglas economies (Table 1.1); open for indivisible plants and price-sensitive incomes
2. A council goes offline	Councils keep control through partitions	Local council state; participation bitmap and carry-forward (Sections 5.3.1 and 5.4)	2	2 (fault injection, round time)	Specified; not built
3. An operator cheats	Alteration and omission detected, not prevented	Signatures, Two-Root inclusion, receipts, challenge window (Section 5.3.1)	2	2(a) (every injected omission and tampering detected)	Specified; not built
4. Private household requests	Aggregate privacy with two aggregators	Prio3 on DAP (Section 7.4)	3	3	Library measured on a desktop (Table 7.6); private budget proof open (Section 7.4.4)

9.3 Multi-Scale Analytical Projections (10^2 to 10^6 Participants)

Methodological note: Only the pequod-cljs CSV convergence counts, the Python stress tests, and the Prio3 benchmark have been executed.[4] Table 9.4 uses a 10-round annual plan with three fresh household-report rounds; Kill Gate 1's 25-round bound is a halt criterion, not the table's round count. Archive rows use the coarse single-key proposal sizes of Table 5.2, with Ed25519 council signatures under the tiered policy and, in brackets, hybrid signatures. Checkpoint rows assume an $n = 128$ committee; small federations could use smaller committees.

[4] Scripts and benchmark code are in research/analysis/ in the project repository.

Table 9.4: Analytical scaling projections across five population scales (10-round plan; SI units)

Metric (analytical estimate)	100	1,000	10,000	100,000	1,000,000
Total councils & federations (M)	24	51	430	4,120	40,600
Priced items ($G = \|\Omega\|$)	30	80	500	2,000	10,000
Council proposals / round	20	40	400	4,000	40,000
Raw leaf archive / round, Ed25519	51 kB	103 kB	1.0 MB	10 MB	103 MB
hybrid signatures	117 kB	235 kB	2.3 MB	23 MB	235 MB
Raw leaf archive / year, Ed25519	0.5 MB	1.0 MB	10 MB	103 MB	1.0 GB
hybrid signatures	1.2 MB	2.3 MB	23 MB	235 MB	2.3 GB
Mobile traffic / round (Role-LP), target	$\leq$ 150 kB every round (tiered); $\leq$ 450 kB with a hybrid full check				
Layer-1 checkpoints / year (425 blocks), Ed25519	3 MB	4 MB	11 MB	37 MB	177 MB
hybrid signatures	124 MB	125 MB	132 MB	158 MB	298 MB
Post-quantum anchors / year (tiered policy)	21 MB (3-of-5 SLH-DSA, BFT) or 11 MB (ML-DSA-44 cosignatures, log), at every scale				
Household-report CPU / aggregator / fresh round (desktop, coarse)	$<$ 0.1 s	0.25 s	2.5 s	25 s	4 core-min
Mobile peak RAM target (Role-LP)				$\leq$ 150 MB (unmeasured)	
Automated round time	Not estimated; measured in Phase 2 against Kill Gate 2				
Server hosting (excl. operations and audits)	\$0–\$5/mo	\$10–\$25/mo	\$80–\$150/mo	\$400–\$800/mo	\$3,000–\$6,000/mo

Open Theoretical, Institutional, Economic, and Technical Questions

> A simulation incorporates more detail than a differential economic model, but still misses the richness of living interaction. As a result, the full characteristics of participatory economics and the diverse solutions to institutional problems associated with it will ultimately only reveal themselves in actual operations.
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, p. 129[1]

We conclude our study by ranking the 17 most important unresolved questions across theoretical economics, institutional sociology, and distributed systems engineering, specifying how each will be investigated through either our simulation program ([SIM]) or human-in-the-loop cooperative pilots ([HUMAN]):

Table 10.1: Ranked research agenda of unresolved questions (priorities 1–17)

Rank	Unresolved Question	Domain	Experimental Investigation Protocol
P1	Convergence under non-convexities, setup costs & Leontief complementarities	[ECON / OR] ([SIM])	Replace unconstrained Cobb-Douglas in a new simulator (not by patching pequod-cljs FOCs); benchmark Algorithms 3a–3c on pre-registered instances.
P2	Stability of coupled endogenous income	[ECON] ([SIM])	Couple Society Account updates into active rounds with Algorithm 3d (damped budget fixed point; damping $\alpha \in (0, 1]$ is a relaxation, not a second numéraire, which remains $w_{\text{base}} \equiv 1$). Report cycling, multiple fixed points, and the effect of the one-round budget lag.
P3	Deliberative Complexity & Participation Fatigue in Multi-Round Planning	[INST / HCI] ([HUMAN])	Test the "3-Milestone Hybrid Schedule" (human rounds at $t = 1, 4, 8$ + automated intermediate solver rounds); measure NASA-TLX load.
P4	Biophysical Incommensurability vs. PDRM Pollution Pricing (O'Neill 2024a)	[ECOLOGY] ([SIM/GOV])	Verify hard barrier convergence when aggregate demand hits absolute biophysical caps $(r^*, E_z^{\max})$ above local CAP PDRM pricing.

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[1] Albert and Hahnel 1991a, chap. 6, 129.

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Rank	Unresolved Question	Domain	Experimental Investigation Protocol
P5	Coarse-to-Granular SKU Disaggregation & Slack Buffer Sizing	[ECON / OR] ([SIM])	Simulate 200 coarse reporting categories (the proposed privacy baseline, Section 7.4.2) mapped to 3,000 planning categories and 20,000 SKUs under stochastic preference drift, against slack $\eta \in [1\%, 10\%]$; measure the unmet-demand share it causes.
P6	Workplace Effort-Rating Dynamics: Solidarity vs. Surveillance or Parity	[INST] ([HUMAN])	Pilot with worker cooperatives comparing scaled effort multipliers vs. flat hourly parity ($\epsilon_u \equiv 1.0$) under BJC rotation.
P7	Preventing Industry Federation Cartelization & Gatekeeping (Kuch 2024)	[INST / ECON] ([HUMAN])	Test "Dual-Path Startup Accreditation" allowing rejected innovators to appeal to a sortition Innovation Jury + Consumer Federation.
P8	Calibrating Balanced Job Complexes Across Disparate Industries	[INST / OR] ([SIM/HUM])	Model cross-workplace BJC rotation as a bipartite matching problem in EFB accounting for travel & skill constraints.
P9	Inter-Regional Investment Apportionment Across Unequal Endowments	[ECON] ([SIM])	Compare per-capita vs. Porto Alegre need-weighted regional investment matrices over 10-year horizons.
P10	Reproductive & In-Home Caring Labor Integration (DEP Ch. 10)	[FEMINIST ECON] ([HUMAN])	Convene Women's and Caregivers' Caucuses to define care-allowance rules and household BJC norms.
P11	Byzantine Resilience Under Prolonged Regional Network Partitions	[DIST-SYS] ([SIM])	Benchmark Regional Proxy Extrapolation accuracy when 10% of councils are partitioned for multiple rounds.
P12	Standardization of Compact Post-Quantum Threshold Signatures	[CRYPTO] ([SW-ARCH])	Councils use k-of-n multi-signatures (64 B per signer with Ed25519 under the proposed tiered policy, 3,373 B hybrid) until a standard post-quantum threshold signature exists; CryptoSuiteId agility allows the switch. NIST opened a call for multi-party threshold schemes in January 2026 (NIST IR 8214C), a reference process rather than a standards competition. Proposal sizes depend on k (Table 5.2).
P13	Privacy anonymity sets in small hamlets ($m < 30$)	[PRIVACY] ([SW-ARCH])	Apply $m_{\min} = 30$ per coarse reporting category and aggregate rare categories at Ward level with $m_{\text{rare}} = 250$; measure differencing leakage between milestone releases under the carry-forward rule of Section 7.4.3.

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Rank	Unresolved Question	Domain	Experimental Investigation Protocol
P14	Endpoint Security vs. Open Modifiability on User Smartphones	[SECURITY] ([SW-ARCH])	Use local hardware keystore unlock + social anomaly verification without mandatory proprietary OS attestation lock-in; split hybrid custody until Android holds post-quantum keys in hardware (Section 7.6); this needs native clients, since a web app cannot reach the keystore (Section 5.5).
P15	Bootstrapping a "Parallel Economy" Inside Surrounding Markets	[ADOPTION] ([HUMAN])	Implement a Dual-Price Fiat Reserve Bridge in Sandström's Society Account for federated cooperatives (PEPE Ch. 6 p. 129).
P16	Private and offline intra-year fulfillment (state S7)	[PRIVACY / DIST-SYS] ([SW-ARCH/HUM])	Specify and test a point-of-use design that hides what a person collects and bounds offline double spending (Section 7.5); until then the household privacy claim covers annual requests only.
P17	Layer-1 design: BFT checkpoints or a witnessed log	[DIST-SYS] ([SIM])	Run both on the Phase 2 testbed, including a stalled log operator and its replacement (Section 5.3.2); prefer the simpler design if both pass Kill Gate 2.

CHAPTER 11

Practical Adoption Strategy, Cooperative Pilots, and Theory of Transition

> Conceivably one could run a "shadow" participatory economy within an existing economy. This could occur in a subset of the whole economy, whether geographic or by industrial sector… not only [to] serve as a laboratory to test participatory planning, but also as a school, both educating the public and developing the means to implement participatory planning without undue disruption.
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, "Experiment 2: Developing a Parallel Economy," pp. 129–130[1]

11.1 Distinguishing Research Simulation from Practical Cooperative Coordination

To avoid equating small-scale cooperative software with a full post-capitalist economy, DPEP separates two distinct applications:

1. The Scientific Research Platform (Phases 1–2): A computational laboratory for testing whether multi-tier iterative planning algorithms converge under realistic industrial non-convexities, input-output networks, and strategic incentives.
2. The Practical Cooperative Coordination Suite (Phase 3): Modular local-first tools allowing existing worker cooperatives, consumer buying clubs, community land trusts (CLTs), and participatory budgeting municipalities to coordinate procurement, shared equipment, and inter-cooperative production within a surrounding capitalist market economy.

11.2 Theory of Transition: Market Boundaries, Voluntary Exit, and Legal Exposure

Any voluntary federation of cooperatives experimenting with participatory planning inside a surrounding market economy must navigate four structural realities of a pilot, plus two that remain open even for a universal economy:

1. Market-priced inputs, external debts, and taxes (the dual-price boundary): Participating cooperatives cannot pay external steel suppliers, commercial

[1] Albert and Hahnel 1991a, chap. 6, 129–130.

Table 11.1: Six early practical applications and their institutional requirements

Application	DPEP Modules Used	Viable Inside Market Economy?	Institutional / Ownership Changes Needed for Full Parecon Semantics
1. Cooperative Joint Procurement	Layer 2/3 demand aggregation across food co-ops & worker co-ops.	Yes (Immediately): Legal B2B purchasing cooperative.	Requires pooled working-capital reserves to buffer seasonal supplier price swings.
2. Shared Equipment & Tool Trusts	Stand-Alone Capital Asset ledger charging user-right fees (P_k).	Yes: Via a Cooperative Equipment Trust or CLT.	Member co-ops transfer title or long-term lease rights of heavy machinery/kitchens to the shared trust.
3. Community Mutual-Aid Distribution	Layer 3/4 Maxim-4 need allocation & non-transferable credits.	Yes: Non-commercial mutual-aid association.	Bounded by the volume of goods produced by volunteer or grant-funded community labor.
4. Inter-Cooperative Production Planning	Quarterly ADMM/tâtonnement clearing across 10–50 federated co-ops.	Partially (Dual-Price Regime): Needs Fiat Reserve Bridge.	External raw materials, rent, debt service, and taxes must still be settled in fiat currency (Eq. 11.1).
5. Municipal Participatory Budgeting	Layer 4 "What-If" explorer, Need Matrix & Prio3 voting.	Yes: Municipal public capital budgeting.	Vulnerable to municipal coalition reversals and fiscal austerity (as in Porto Alegre's 2017 suspension).
6. Regional Co-op Supply Networks	Multi-year investment (σ_{inv}) & EFB worker reallocation.	Partially (e.g., Mondragón / Cecosesola).	Full Parecon requires abolishing enterprise profit retention and equalizing BJC empowerment across plants.

landlords, bank mortgages, or national tax authorities in internal effort credits. Every pilot federation must enforce an explicit External Fiat Clearing Constraint alongside its internal Society Account:

$$\sum_{j \in \mathcal{J}} P_{\text{ext}}^{\text{fiat}} \cdot \bar{a}_j^{\text{export}} + D_{\text{fiat}}^{\text{dues}} \geq \sum_{j \in \mathcal{J}} P_{\text{ext}}^{\text{fiat}} \cdot \underline{a}_j^{\text{import}} + F^{\text{taxes}} + F^{\text{debt}} \tag{11.1}$$

where external imports $\underline{a}_j^{\text{import}}$ carry an internal shadow price $\lambda_{\text{fiat}} P_{\text{ext}}^{\text{fiat}}$ reflecting the scarcity of external fiat revenue.

2. Strict Voluntarism, Right of Exit, and Non-Participants (Learning from Aragon 1936): Informed by the conflict between Aragonese collectives and "individualist" smallholders (Casanova 2005), membership in any DPEP pilot or transitional federation must be strictly voluntary with a constitutionally codified Right of Exit: exiting individuals or cooperatives settle their net savings credits and refundable capital shares according to pre-agreed cooperative bylaws without confiscation, and non-participating artisans/households trade with the federation across transparent external interfaces.

3. Graduated sanctions without coercive labor discipline: Following Ostrom (1990), failure to meet council commitments triggers: (a) automatic SB_j / SC_j accounting adjustment, (b) UFB peer mediation, and (c) for persistent bad-faith breach, revocation of federation accreditation and access to shared federation inputs/credit—never forced labor. Losing access to federation inputs is mild in a voluntary pilot with an external market to exit into, and severe in a universal economy where exit is hard. The same rule therefore has two political meanings; a full-economy constitution needs a due-process and subsistence-access clause that a pilot can omit.

4. Jurisdiction-Specific Legal Exposure (Informational Analysis, Not Legal Advice):

   - Competition / Antitrust Law (US Sherman Act §1; EU TFEU Art. 101; UK Competition Act 1998): Independent corporations that coordinate prices and output quotas through an IFB risk prosecution as a horizontal price-fixing cartel unless structured under statutory cooperative/agricultural exemptions (e.g., US Capper-Volstead Act, EU producer-organization rules) or incorporated as operating divisions of a single secondary cooperative / common-ownership consortium.
   - Taxation of Multilateral Barter & Clearing (US IRS Form 1099-B; EU-/UK VAT): Multilateral credit clearing is generally taxable at fair market value; pilot deployments must generate statutory fiat-equivalent tax reports where required by domestic law.
   - Data Protection & Right to Erasure (EU/UK GDPR Art. 17): DPEP is designed to avoid immutable personal ledger data by keeping personal profiles local, deleting Prio3 input shares when the challenge window closes

(Section 7.4.3), and storing qualitative Lists behind mutable council pointers with tombstone erasure. Pilots additionally need the ethics and data-protection steps of Section 9.1.3.

Open even for a universal economy. Foreign trade (how a DPEP polity prices and rations extra-federation goods, manages a foreign-exchange constraint, and avoids reproducing center–periphery extraction) and defense (who commands organized coercion, how it is paid, and how it is made compatible with sortition and dual-federation ratification) are not specified for either a pilot or a full economy. They remain open political questions, not protocol features.

11.3 Bridged Adoption: Interoperability with Existing Systems

DPEP does not ask anyone to leave existing systems before using it. It runs on ordinary phones and computers over the ordinary internet (Chapter 5), and each application in Table 11.1 can be adopted alone. What a co-operative or federation already uses (bookkeeping software, bank accounts, invoicing, tax filings, decision tools) stays in place while DPEP is tried alongside it. The fiat clearing constraint (Eq. 11.1) is the economic bridge between the federation and the market economy around it. Table 11.2 lists the technical connections it needs. None of them is specified yet; they are requirements for the practical suite of Phase 3.

Running in parallel, and rolling back. A group trying DPEP should be able to run it in shadow mode: proposals, accounts, and planning rounds are computed alongside the existing system, compared with it, and bind no one until a council ratifies a switch. If the trial fails, the group exports its data and stops. The right of exit in Section 11.2 therefore has to include data: a council that leaves takes its full history with it in open formats. Low switching costs and safe exit decide adoption, as the barter networks of Argentina (Section 2.9) and the failed regional currencies discussed in Section 11.4 show from opposite directions.

Design Principle 11.1 (Bridged Adoption). Every DPEP module offered to existing organizations must (a) run in shadow mode alongside the systems it may replace, binding no one until a council ratifies the change; (b) export all of a council's data in open, documented formats at any time; and (c) be removable without loss of that data. Identity checks required by banks or states are carried out at the federation's legal boundary and are never linked to council credentials.

Where the bridge ends. The bridges above let DPEP coexist with the market economy; they do not make the coexistence permanent. Full participatory-economy semantics still require the ownership and remuneration changes listed

[2] Valueflows, https://www.valueflo.ws/; Cyclos, https://www.cyclos.org/; Open Food Network, https://openfoodnetwork.org/; Loomio, https://www.loomio.com/ (all checked November 2026).

Table 11.2: Technical bridges between DPEP and existing systems (requirements; none yet specified)

Bridge	Existing system	What DPEP must provide
Bookkeeping	A co-operative's existing accounting software	Import and export of journal entries in open formats (and the OECD Standard Audit File for Tax where a tax authority requires it), with a published mapping from SB_j / SC_j to the existing chart of accounts.
Fiat payments	Bank accounts and bank payment files (ISO 20022)	A fiat account held by the federation's legal entity; automatic reconciliation of its flows against Eq. 11.1.
Invoicing and tax	Electronic invoicing (EN 16931, Peppol where mandated) and statutory returns	Fiat-equivalent valuation of internal transactions for the reports that domestic law requires (Section 11.2).
Identity at the boundary	Banks' and states' identity checks	Checks performed once, by the legal entity that faces the bank or the state. Council credentials and household reports never pass across the boundary, and nothing links them to legal identity (Section 7.3).
Co-operative and solidarity tools	Open Food Network, Loomio, Decidim, mutual-credit software such as Cyclos	Adapters for orders, decisions, and balances. Valueflows, an open vocabulary for economic flows between organizations, is a candidate shared format.[2]
Complementary currencies	Mutual-credit networks and regional vouchers	Treated either as an external unit across the bridge or as a transitional internal unit (Section 11.4).
Open protocol standards	DAP aggregation servers; C2SP transparency-log witnesses and monitors	With two aggregators, household aggregation runs on DAP and existing servers; a witnessed-log Layer 1 uses C2SP checkpoint and cosignature formats, so independent witnesses and monitors can follow DPEP's log (Sections 7.4 and 5.3.2).

in the last column of Table 11.1. Those changes are made step by step, by the participants' own decisions, not as a precondition for using the software.

11.4 Complementary Currencies as Allies and Bridges

Regional and complementary currencies are the existing practice closest to DPEP's transition layer. Kennedy, Lietaer, and Rogers (2012) describe three building blocks: circulating vouchers (Chiemgauer, BerkShares, Brixton Pound), usually pegged to or redeemable in national currency and sometimes subject to demurrage, a holding fee that encourages spending over hoarding; exchange rings or mutual-credit systems (WIR, business barter networks, time banks), in which a trade creates a debit on one account and an equal credit on another, so that balances sum to zero; and micro-credit banks lending at low or no interest (Banco Palmas). Their aims are to keep money circulating within a region, provide interest-free liquidity, stabilize members' trade in downturns, and govern money democratically, all inside the market economy.[3]

[3] Kennedy, Lietaer, and Rogers 2012, chaps. 2, 4, 6, and 7 (read in full).

Where they align with DPEP. Mutual credit is a small-scale form of the double-entry accounting that DPEP takes from Sandström: zero-sum balances, credit limits, and published accounts correspond to the SB_j / SC_j ledgers and the Society Account (Chapter 4). Joint procurement, mutual aid, and inter-co-operative clearing (Table 11.1) are what exchange rings already do. And demurrage aims at the same end as Parecon's consumption credits, which are earned by effort, are not transferable, and earn no interest, so that money circulates and does not become capital.

Where they differ. A complementary currency changes the money, not the allocation mechanism. Market prices and competition between enterprises remain; an enterprise paid in a regional currency still maximizes its own revenue, and external costs are still left outside its accounts. Nor does a new currency change who manages production (Section 2.4): a firm can pay in Chiemgauer and keep its hierarchy. Governance is often in the hands of a founding group, a non-profit, or a co-operative bank rather than assemblies of users. Complementary currencies pass Brinton's Test (Section 2.13.2) only to the extent that their users govern them.

Lessons for DPEP. The same practice records three warnings.

1. Rapid growth without verifiable issuance fails. Kennedy et al.'s account of Argentina, drawing on Peter North, describes idealism, growth too rapid for crisis conditions, corruption, and eventual collapse, with networks spread by franchised start-up kits. This is the failure that DPEP's signed records and public budget checks address (Section 2.9).
2. Critical mass, burnout, and distance. Stuttgart's Rössle currency closed at the end of 2011 after four years. Its organizers named a burned-out core group, too few active users, and members spread so far apart that transaction costs rose.[4] DPEP's answers are to count governance work inside paid hours (Section 2.5) and to start pilots among organizations that already trade closely with one another.
3. Running costs, legal status, and tax decide survival. Professional services paid in a complementary currency are taxable (informal trades usually are not), and two of the examples in People Money (Banco Palmas and Talente Tauschkreis Vorarlberg) let members pay local taxes in the local currency,[5] which points to useful partnerships with municipalities.

An adoption path. A federation can use an existing currency network as a bridge in three stages.

[4] Kennedy, Lietaer, and Rogers 2012, chap. 6, "Learning from a currency that failed."
[5] Kennedy, Lietaer, and Rogers 2012, chap. 4, "The exchange ring."

1. Use the currency as it is. Co-operatives already in a mutual-credit network keep using it, and DPEP's procurement and mutual-aid modules record their trades in that unit.
2. Treat it as an external partner. Trade inside the federation is cleared in DPEP's own accounts, and the currency network becomes a partner across the bridge, as national currency is in Eq. 11.1.
3. Hand coordination to planning. The planning modules take over what a currency cannot do, and non-transferable consumption credits replace circulating money inside the federation. The regional currency remains a means of trade with non-participants.

DPEP can also give these networks something they lack: credits that cannot be forged and issuance that anyone can verify, privacy for household consumption (where exchange rings typically publish every member's balance), offline operation, and federation across regions.

Cryptocurrencies, tokens and supply control. Readers familiar with cryptocurrencies may ask why DPEP issues no token, and why its credits have no rules for minting and burning to control inflation. In DPEP, supply control comes from the plan, not from monetary policy. Household budgets B_u are computed inside the planning process rather than issued by a monetary authority; at a cleared plan they sum to the value of planned consumption (Chapter 4). The public budget check $p \cdot \sum y \leq \sum B_u$ (Section 7.6) keeps total claims within planned output. There is no floating supply to expand or contract. Consumption credits are also deliberately non-transferable: they cannot be traded, lent at interest, or accumulated as capital. A transferable token would bring back what participatory economics removes: speculation, accumulation, and influence in proportion to holdings, which is why the book rejects token-weighted governance (Section 2.1). Public token ledgers would also expose who pays whom, where DPEP keeps household consumption private. Some blockchain-based community currencies are closer to the practice above than to speculative cryptocurrency. Sarafu, run by the Kenyan non-profit Grassroots Economics, moved its community vouchers onto blockchain tokens in 2018, and its 2020–2021 transaction records, covering some 40,000 users during the COVID-19 pandemic and a unit roughly equal to one Kenyan shilling, have been published for research.[6] Circles, on Gnosis Chain, lets each verified person mint one unit per hour of a personal currency that loses 7% of its value per year, and makes these currencies exchangeable along a web of trust between people.[7] Both show that blockchains can carry mutual credit and demurrage. Both keep tokens transferable and leave coordination to markets, so they belong at the same bridge as regional currencies, in the first or second stage of the path above. Stablecoins

[6] Mattsson, Criscione, and Ruddick 2022; Mattsson, Criscione, and Takes 2023. Both cited at bibliographic level.

[7] Circles documentation, "Why Build on Circles?" https://docs.aboutcircles.com/overview/why-build-on-circles (checked November 2026).

could at most serve as a payment channel at a pilot's fiat edge, with counterparty and regulatory risk, never as an internal unit.

What DPEP can take from these experiments is the tooling, not the token. Signed records, issuance rules that anyone can verify, and wallets that work on cheap phones serve non-transferable credits as well as they serve transferable ones. Whether a federation trades with such networks is a decision for its councils at the bridge. Inside the federation, what each household can claim is settled by the plan the councils ratify, not by a token supply schedule.

Six-Discipline Critical Examination and Remaining Weaknesses

Table 12.1 lists the strongest remaining criticisms, grouped by discipline. For each it gives a severity, the evidence that would settle it, the phase that would produce that evidence, and whether it could stop the project. Severity has three levels: Critical (if confirmed, the program as conceived should stop or be fundamentally redesigned), Major (forces narrowing of scope or a significant redesign), and Moderate (a known kind of fix exists, but it is not yet specified or tested). Nothing in the table is marked resolved.

Table 12.1: Strongest remaining criticisms: severity, settling evidence, and phase

Criticism	Severity	Evidence that would settle it	Phase	Could stop?
Participatory economics and democratic theory				
Participation at scale: ordinary people will not sustain multi-round planning; turnout and attention will collapse after the novelty fades.	Critical	Completion, comprehension, and fatigue (NASA-TLX) in Phase 1H; retention over a full plan year in pilots.	1H, 3	Yes
A technical elite forms around the software: facilitators, solver experts, and articulate members set defaults and dominate amendments, re-creating a coordinator class.	Critical	Share of adopted amendments originated by the most active 20%; who sets solver defaults; rotation of facilitation roles (Kill Gate 1H(e)).	1H, 3	Yes
Federations become central planning authorities in disguise (Kuch 2024).	Major	Count and scope of federation decisions in pilots; outcomes of Innovation Jury appeals (P7).	3	No (narrows)
Flat effort pay by default ($\epsilon_{\max} = 0$) honors Maxim 3 only through hours and job balancing; rating intensity invites surveillance (O'Neill 2024a).	Moderate	Cooperative pilots comparing flat and peer-rated pay (P6).	3	No
Cryptographic invisibility removes the 1991 channel of collective criticism of consumption.	Moderate	Members' stated preferences in pilots that offer both modes.	3	No
Operations research				
No candidate solver is shown to clear Leontief, lumpy, and endogenous-income economies; 3b's theorem covers only fixed-budget convex cases, and 3d has no convergence result.	Critical	Kill Gate 1 on the pre-registered set, against an oracle.	1	Yes

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Criticism	Severity	Evidence that would settle it	Phase	Could stop?
One-good smoothing results do not transfer to interdependent markets.	Major	Lumpy Leontief families in the instance set.	1	No (narrows)
Rounds-to-clear may grow with the number of councils.	Major	Log–log scaling slope from 50 to 1,000 councils.	1	Yes
Distributed systems				
Round cost at 10^6 (0.10–1.45 GB archives and 10^6 report verifications per aggregator; signature checks measured at 2–10 s per desktop core) may not fit a usable round time.	Major	Kill Gate 2 measurements at $\geq 10^4$ with extrapolation from measured per-item costs.	2	Yes (for D; A remains)
Under a witnessed log, one operator per round can stall planning; recovery is an institutional replacement procedure.	Major	Kill Gate 2(d): stalled-operator trials.	2	No (BFT remains)
Phones and app stores: store removal and push-notification metadata expose participants to the platform owners.	Moderate	Distribution outside the stores, reproducible client builds, content-free notifications (Kill Gate 3(e)).	3	No
Annual committee capture lasts a year; a 30% corrupt pool gives a 21% chance of an unsafe draw.	Major	Rotation frequency; operation of the interim threshold beacon (Section 7.2.1); a post-quantum beacon (open). Not settled by an experiment.	2	No
Cryptography				
Aggregate correctness needs all aggregators; the public budget check limits only gross overspend.	Major	Pilot use of non-nested aggregators; audit.	3	No
No construction for budget credentials and one-report-per-person; the budget circuit is unspecified and price-dependent.	Critical (for household crypto only)	A specified, audited construction: an interim classical credential with threshold issuance covers one report per person (Section 7.2.1); budget binding and the circuit remain open. Until then the council-level trust model.	3	Yes (household mode only)
Carry-forward cannot re-aggregate reports; milestone-only revision freezes household demand between milestones.	Moderate	Planning-accuracy cost in simulation (P5) and pilots.	1, 3	No
Consumption at the point of use (S7) is not private: depots and home councils see what each person collects, so the household privacy claim covers annual requests only.	Major	A specified point-of-use design with offline double-spend bounds, tested in a pilot (P16; Kill Gate 3(d)).	3	No (narrows the claim)
Under the tiered signature policy, a covert CRQC before a migration trigger could forge current round objects; migrating many devices is operationally hard.	Moderate	Rehearsed suite migration; per-round k-of-n post-quantum anchoring within 24 h.	2, 3	No

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Criticism	Severity	Evidence that would settle it	Phase	Could stop?
Cooperative practice				
30–60 self-selected volunteers say little about a general population.	Major	Larger, more representative pilots after Phase 1H.	3	No
Sanctions that cut federation inputs are mild in a pilot and severe in a universal economy.	Moderate	Constitutional due-process and subsistence clause (open).	–	No
Skeptical economist				
Ratchet effects and incentive limits (Weitzman, Hurwicz) remain; public blueprints do not remove them.	Major	Strategic-misreporting family in Phase 1 (diagnostic); pilots.	1, 3	No
The labor floor $p^{\text{lab}} \geq w_{\text{base}}$ blocks surplus clearing.	Moderate	Phase 1 instances with labor surplus.	1	No
Foreign trade and defense are unspecified.	Major	Not settled by any experiment here; a political design question.	–	No (scope limit)

APPENDIX A

Scientific Claims, Evidence, and Validation Status Register

Each claim has eight separate fields: identifier, claim, classification, status (first line) and evidence, assumptions, limitations and required validation (second line). Classifications: Established (peer-reviewed theorem or standard), Source-supported (checked in primary literature, audited code or a measurement), Analytical (derived from models or arithmetic in this monograph), Conjectural (hypothesis or judgment), Experimental target (a goal to be tested). Statuses: Verified, Partially verified, Unverified, Refuted.

Table A.1: Master register of scientific and engineering claims

Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-ECON-01	Economic convergence (reference): the continuous-time FMPE gradient process converges.	Class: Established	Status: Verified (as a cited theorem; not a DPEP result)
Evidence: Arrow–Hurwicz–Uzawa (1958); Heal (1973), Tier 3.	Assumptions: Strict convexity; gross substitutability.	Limitations: Fails on Scarf-type and non-convex economies.	Validation: Reference only.
CLM-ECON-02	dep1ex61–100.csv: Year 1 reaches 5% in 11.85 rounds and 3% in 19.20; Year 2 5% in 6.58.	Class: Source-supported	Status: Verified
Evidence: Audit script[1] on pequod-cljs @ 8bd46b5c.	Assumptions: Fixed \$5,000 incomes; Cobb–Douglas FOCs.	Limitations: Other subsets differ (01–50: 11.18); no wall-clock log.	Validation: Reproduce in the Phase 1 simulator.
CLM-ECON-03	Eq. (4.15) plus trust regions does not clear Scarf's cycle (endowment incomes) or a single lumpy plant; with fixed budgets it clears Scarf's consumers.	Class: Analytical	Status: Verified
Evidence: Simulation script[2]: min imbalance 10.5% and 50.4%; fixed-budget control $< 0.02\%$.	Assumptions: The stress-test economies.	Limitations: Not a general impossibility proof. The Scarf failure comes from endowment income, which Parecon does not use.	Validation: Instance families 3–4 in Phase 1.

[1] research/analysis/verify_r2.py in the project repository.
[2] research/analysis/verify_economics.py in the project repository.

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Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-ECON-04	With the toothache increment, $\sum b + c/k > 1$ for every one of 750,000 worker-council records in the 70 generators.	Class: Source-supported	Status: Partially verified
Evidence: Dataset audit script[3]; LFS objects SHA-256 checked.	Assumptions: 20% of councils selected at run time.	Limitations: Perverse supply predicted; whether committed CSVs include toothache runs is unknown (exponents not recorded); Clojure not run; DEP Ch. 9 not read at page level.	Validation: Run pequod-cljs with toothaches.
CLM-ECON-05	One good, $K = 20$ staggered plants: min imbalance 0.40% over 500 rounds; last-100-round max 8.4%.	Class: Analytical	Status: Partially verified
Evidence: Simulation script[4]; sweep median 6.0%, worst 9.5%.	Assumptions: $a \in [60, 80]$; thresholds in $[8.5, 11.5]$.	Limitations: One good; not Leontief.	Validation: Lumpy Leontief family in Phase 1.
CLM-ECON-06	$\sum_u B_u = (1 - \sigma_{\text{inv}}) Y^{\text{GDP}}$, and $\sigma_{\text{inv}} Y^{\text{GDP}} = p \cdot D^{\text{inv}}$ at a cleared plan.	Class: Analytical	Status: Partially verified
Evidence: Algebra in Section 4.2 and Table 4.2.	Assumptions: Y^{allow} is cash transfers only.	Limitations: No numerical test on a full Society Account ledger.	Validation: Phase 1 ledger test.
CLM-ECON-07	Algorithm 3b (Deng et al. Algorithm 4 with slack) converges on fixed-budget convex problems.	Class: Established	Status: Partially verified
Evidence: Deng et al. (2017) Thm. 2.3–2.4 (read in full); test B in the analysis script.[5]	Assumptions: Convex $\mathcal{T}_c$; fixed budgets; P_i condition (2.11).	Limitations: Not endogenous income, non-convexity, or dual renormalization.	Validation: Phase 1 convex families.
CLM-ECON-08	Some candidate (3a–3d) clears full DPEP instances (Leontief, lumpy, endogenous income, strategic).	Class: Experimental target	Status: Unverified
Evidence: None yet.	Assumptions: Pre-registered instance set.	Limitations: Toy tests only; 3d has no convergence result.	Validation: Kill Gate 1.

[3] research/analysis/verify_pequod_datasets.py in the project repository.
[4] research/analysis/verify_r2.py in the project repository.
[5] research/analysis/verify_r3.py in the project repository.

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Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-ECON-09	One-sided feasibility ($D \leq S$) is an adequate success criterion.	Class: Analytical	Status: Refuted
Evidence: Single plant: feasible at round 3 with 42% unused output (reference test script)[6].	Assumptions: –	Limitations: –	Validation: Replaced by Definition 4.2.
CLM-TECH-01	Leaf transport plus BFT takes 9–27 s per round at $N = 10^6$.	Class: Analytical	Status: Refuted
Evidence: Assumed 5.2 kB proposals; reworked sizes give 0.24–1.46 GB per round (Table 5.2).	Assumptions: –	Limitations: Omitted report verification and signature checks.	Validation: Replaced by Kill Gate 2 thresholds.
CLM-TECH-02	Computational scalability: Prio3 verification costs ≈ 0.25 ms (coarse) to 3.5 ms (dense) per report per aggregator.	Class: Source-supported	Status: Partially verified
Evidence: libprio 0.18.1 benchmark (Table 7.6).	Assumptions: One desktop core; VDAF draft 18 code.	Limitations: No phone or Raspberry Pi measurement; network excluded.	Validation: Device benchmarks, Kill Gate 3.
CLM-TECH-03	Prio3SumVec report sizes follow the VDAF size formula.	Class: Source-supported	Status: Verified
Evidence: The size model[7] matches libprio byte-for-byte on 9 configurations.	Assumptions: libprio chunk-length rule.	Limitations: Custom budget circuit sizes unknown.	Validation: Recompute when the circuit is specified.
CLM-TECH-04	Mobile feasibility: ≤ 150 kB in every round with a full checkpoint check under the tiered signature policy (≤ 450 kB with a hybrid full check); RAM ≤ 150 MB.	Class: Experimental target	Status: Unverified
Evidence: Byte budget analytical (Section 7.7); 21–101 kB per round computed.	Assumptions: Coarse 55 kB report; Ed25519 or aggregated BLS checkpoint signatures; native client.	Limitations: No device test.	Validation: Kill Gate 3.
CLM-TECH-05	Operating cost: hosting 100 servers $\approx$ \$3k–\$6k/month.	Class: Analytical	Status: Unverified
Evidence: \$30–\$60 per server-month (assumed).	Assumptions: Commodity servers.	Limitations: Excludes audits, facilitation, research, and energy.	Validation: Phase 2 and 3 cost accounting.

[6] research/analysis/reference_tests_r3.py in the project repository.
[7] research/analysis/parameters.py in the project repository.

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Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-TECH-06	Architecture D outranks A, B, and C at baseline weights.	Class: Conjectural	Status: Partially verified
Evidence: Expert scores; arithmetic and perturbation test in the analysis script.[8]	Assumptions: Table 6.1 scores.	Limitations: A overtakes D when simplicity, throughput, and cost carry $\geq 63\%$ of the weight.	Validation: Phases 2–3 replace judgements with measurements.
CLM-TECH-07	Component maturity: every function except threshold signatures, anonymous credentials, the beacon, capability-certificate formats, and post-quantum HPKE has a standardized, deployed primitive.	Class: Source-supported	Status: Partially verified
Evidence: Table 6.3; maturity dossier[9] (web checks, November 2026).	Assumptions: Ratings are judgements on the M0–M5 scale.	Limitations: Primitive maturity is not library or combination maturity; nine combinations are original.	Validation: Re-check before each gate; Kill Gate 3 audit.
CLM-SEC-01	Privacy: Prio3 hides an individual report if at least one aggregator is honest.	Class: Source-supported	Status: Partially verified
Evidence: VDAF-22 §9 (full text).	Assumptions: Non-colluding aggregators; $m_{\min} = 30$.	Limitations: Budget circuit unspecified; differencing between releases.	Validation: Phase 3 audit.
CLM-SEC-02	Censorship resistance: Two-Root aggregation with receipts detects omitted or tampered proposals.	Class: Analytical	Status: Unverified
Evidence: Design argument (Section 5.3.1).	Assumptions: ≥ 1 honest relay; clients keep receipts.	Limitations: Does not bind aggregator shares.	Validation: Kill Gate 2(a).
CLM-SEC-03	Post-quantum security: the FIPS 203/204/205 primitives are standardized.	Class: Established	Status: Verified
Evidence: NIST FIPS 203–205 (2024).	Assumptions: Module-LWE/SIS and hash assumptions.	Limitations: No whole-system PQ claim; BLAKE3 128-bit; VDAF no PQ claim.	Validation: Phase 3 review.
CLM-SEC-04	Cryptographic integrity: aggregate correctness requires every aggregator to be honest.	Class: Source-supported	Status: Verified
Evidence: VDAF-22 §9 quotation.	Assumptions: Standard Prio3.	Limitations: The public budget check limits only gross overspend.	Validation: Non-nested aggregators in pilots.

[8] research/analysis/verify_r3.py in the project repository.
[9] reports/STACK_MATURITY.md in the project repository.

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Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-SEC-05	Sybil resistance (partial): INV-4B bounds credited hours given a true roster.	Class: Analytical	Status: Partially verified
Evidence: Eq. (4.7).	Assumptions: Roster and SB/SC honest.	Limitations: Falsified rosters and carousel fraud need physical audit.	Validation: Pilot audits.
CLM-SEC-06	Sybil resistance (household): one report per person without a population register.	Class: Conjectural	Status: Unverified
Evidence: No post-quantum construction (Section 7.4.4); interim classical credential adopted (Section 7.2.1).	Assumptions: An anonymous credential with threshold issuance and nullifiers.	Limitations: Budget binding open; interim scheme not post-quantum.	Validation: Research; council-level trust meanwhile.
CLM-SEC-07	An $n = 128$ annual draw with $p = 20\%$ corrupt gives a 0.022% unsafe committee.	Class: Analytical	Status: Verified
Evidence: Exact binomial computation.[10]	Assumptions: i.i.d. draws; failure at $k \geq 43$.	Limitations: $p = 30\%$ gives 21%; capture lasts a year.	Validation: Beacon design.
CLM-SEC-08	Prio3 reports cannot be carried forward into later rounds' aggregates.	Class: Source-supported	Status: Verified
Evidence: VDAF-22 §7.2.3, §9.4.	Assumptions: Standard Prio3.	Limitations: Aggregate-level carry-forward is still possible.	Validation: –
CLM-SEC-09	An off-the-shelf BFT engine accepts post-quantum validator keys.	Class: Source-supported	Status: Verified
Evidence: Cosmos SDK documentation (CometBFT 0.40, Cosmos SDK 0.55).	Assumptions: Releases as documented.	Limitations: Pure ML-DSA-65, not hybrid; ≈ 331 kB of signatures per block at 100 validators; BLS keys only behind a build flag, without vote aggregation (CometBFT v0.40.0, July 27, 2026).	Validation: Kill Gate 2.
CLM-SEC-10	An interim classical credential keeps past showings unlinkable after a CRQC.	Class: Conjectural	Status: Unverified
Evidence: None for a chosen scheme; RFC 9474 states no assumption for blindness.	Assumptions: Statistically zero-knowledge presentation and statistical blinding.	Limitations: If false, a CRQC reveals who submitted, not what.	Validation: Scheme selection before Phase 3.

[10] research/analysis/verify_r2.py in the project repository.

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Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-SEC-11	Canary batches detect indiscriminate tampering with aggregate shares.	Class: Analytical	Status: Unverified
Evidence: Argument in Section 7.4.	Assumptions: Per-batch aggregate shares; batches indistinguishable to aggregators.	Limitations: Misses tampering aimed at batches known to be real.	Validation: Fault injection in Phase 3.
CLM-TECH-08	Signature verification at 10^6: 40k–200k council signatures per round verify in 2–10 s on one desktop core even when all are hybrid.	Class: Source-supported	Status: Partially verified
Evidence: bench_signatures (ed25519-dalek 3.0.0, libcrux-ml-dsa 0.0.11, blst 0.3.17; Apple M4 Pro).	Assumptions: 32-byte digests signed; batches of 64 for Ed25519.	Limitations: Desktop only; no community-server or phone measurement.	Validation: Kill Gate 2.
CLM-TECH-09	A witnessed transparency log provides history, split-view resistance, and attribution for Layer 1 without a consensus engine.	Class: Analytical	Status: Unverified
Evidence: C2SP tlog-witness; Certificate Transparency precedent (Section 5.3.2).	Assumptions: Fewer than a quorum of witnesses collude with the operator; witnesses recompute totals.	Limitations: Liveness rests on one operator per round; replacement is institutional.	Validation: Kill Gate 2(d).
CLM-SEC-12	Under the tiered signature policy, per-round k-of-n post-quantum anchoring bounds a future CRQC's power to rewrite round history.	Class: Analytical	Status: Unverified
Evidence: Argument in Section 7.2.1.	Assumptions: Anchoring within 24 h of round close by 3 of 5 federations (BFT) or 11 of 16 witnesses (log); migration before a CRQC exists.	Limitations: Does not protect live round objects from a covert CRQC before the trigger; anchor-key custody unspecified.	Validation: Phase 3 review; migration rehearsal.
CLM-SEC-13	Household consumption is private at the point of use (intra-year fulfillment).	Class: Experimental target	Status: Unverified
Evidence: None; the pilot design makes pickups visible to depots and home councils (Section 7.5).	Assumptions: —	Limitations: Household privacy covers annual requests only.	Validation: P16; Kill Gate 3(d).
CLM-POL-01	Democratic scalability: ordinary participants complete, understand, and genuinely shape multi-round planning.	Class: Experimental target	Status: Unverified
Evidence: None yet.	Assumptions: Phase 1H design.	Limitations: $n = 30$–60 self-selected volunteers.	Validation: Kill Gate 1H.

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Claim ID / Evidence	Claim / Assumptions	Classification / Limitations	Status / Validation
CLM-POL-02	No technical elite dominates proposals and defaults.	Class: Experimental target	Status: Unverified
Evidence: None yet.	Assumptions: Influence measure of Kill Gate 1H(e).	Limitations: Small, short probe.	Validation: Kill Gate 1H and pilots.
CLM-POL-03	Neighborhood councils of $\sim$ 50–200 are workable.	Class: Conjectural	Status: Unverified
Evidence: Speaking-time arithmetic.	Assumptions: One shared meeting hour.	Limitations: Not a finding of Mansbridge or Dahl & Tufte.	Validation: Pilots.

APPENDIX B

Complete List of Abbreviations

Because this monograph integrates participatory economics, labor and cooperative history, operations research, distributed systems, and post-quantum cryptography, this list defines every abbreviation used in this book. The core mathematical notation is summarized at the start of Chapter 4.

A. Participatory Economics, Accounting, and Bibliographic Abbreviations

Abbreviation	Full Term	Definition & Role in Monograph
BJC	Balanced Job Complex	Bundle of workplace tasks balanced for empowerment and desirability within and across workplaces (PEPE Ch. 2).
CAP	Community of Affected Parties	Ecological/geographic collective of residents impacted by a specific pollutant in an airshed or watershed (DEP Ch. 7).
CC (CC_h)	Consumers' Council	Democratic neighborhood consumer council or nested ward/city/regional consumer federation.
CFB	Consumption Facilitation Board	Informational board assisting consumer councils and federations in evaluating collective consumption projects (LF Ch. 6).
CLT	Community Land Trust	Member-controlled trust holding land or long-lived assets outside market sale (Chapter 11).
CLUEA	Consejo Levantino Unificado de Exportación Agrícola	Unified Levantine Council for Agricultural Export organized by the CNT and UGT in Valencia (1936–1939).
CNT	Confederación Nacional del Trabajo	Anarcho-syndicalist National Confederation of Labor in the 1936–1939 Spanish Revolution.
CPR	Common-Pool Resource	Shared natural or human-made resource system governed by collective-choice rules (Ostrom 1990).
DAANES	Democratic Autonomous Administration of North and East Syria	Rojava's confederal administration, discussed as a historical precedent (Chapter 2).
DEP	Democratic Economic Planning	Primary reference volume by Robin Hahnel et al. (Routledge, 2021); cited as Hahnel 2021.
DPEP	Distributed Participatory Economics Protocol	The open-source decentralized coordination protocol specified in this monograph.
EFB	Employment Facilitation Board	Facilitation board assisting workers transferring between councils and balancing cross-workplace BJCs (LF Ch. 6).
EJPE	Erasmus Journal for Philosophy and Economics	Academic journal hosting the Winter 2024 symposium on Hahnel's Democratic Economic Planning.

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Abbreviation	Full Term	Definition & Role in Monograph
FMPE	Formal Model of a Participatory Economy	Mathematical formalization of participatory planning analyzed in PEPE Chapter 5 (pp. 73–106).
FTE	Full-Time Equivalent	Unit of engineering or research staffing effort across roadmap scenarios.
GA	General Assembly	Direct plenary deliberative assembly of a workplace, village collective, or horizontal movement.
GDP	Gross Domestic Product	Aggregate social value of net final output (Y^{GDP}) cleared in the Society Account.
GDPR	General Data Protection Regulation	EU data-protection law governing pilot data handling, retention, and erasure.
GHG	Greenhouse Gas	Atmospheric emissions (CO_2-equivalent) subject to hard biophysical ceilings ($E_z^{\max}$).
HFB	Household Facilitation Board	Facilitation board assisting individuals and families with housing and neighborhood membership (LF Ch. 6).
IF (IF_k)	Industry Federation	Federation of worker councils within industry k managing sector capacity, R&D, and SB/SC appeals.
IFB	Iteration Facilitation Board	Informational board/engine that aggregates supply and demand and updates indicative prices via public rules.
LF	Looking Forward	Primary reference volume by Michael Albert and Robin Hahnel (South End Press, 1991); cited as Albert and Hahnel 1991b.
Parecon / PE	Participatory Economics	Post-capitalist economic model developed by Michael Albert and Robin Hahnel.
PB	Participatory Budgeting / Planning Bureau	Municipal democratic budgeting (Porto Alegre, 1989); also denotes the abstract Planning Bureau in FMPE Ch. 5.
PCL	Post-Capitalist Labs	Open-source research and engineering collective publishing this monograph (https://github.com/Post-Capitalist-Labs).
PDRM	Pollution Demand-Revealing Mechanism	Iterative procedure pricing local pollutant damages via CAP emission tolerances and credits (DEP Ch. 7).
PEP	Participatory Economy Project	Research collective maintaining participatoryeconomy.org and the 2023 User Story Mapping.
PEPE	The Political Economy of Participatory Economics	Primary technical reference volume by Michael Albert and Robin Hahnel (Princeton University Press, 1991); cited as Albert and Hahnel 1991a.
PFB	Production Facilitation Board	Board assisting worker councils with production externalities, scale economies, and technology choice (LF Ch. 6).
PIPA	Practically Identical Production Activities	Equivalence classes $\{PIPA\}_i$ partitioning workplace tasks by qualitative empowerment effect (PEPE Ch. 2, p. 26).

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Abbreviation	Full Term	Definition & Role in Monograph
PLAC	Parecon: Life After Capitalism	Comprehensive institutional and daily-life synthesis of Participatory Economics by Michael Albert (Verso Books, 2003); cited as Albert 2003.
SB / SC	Social Benefit / Social Cost	Ratio $SB_j(P)/SC_j(P)$ measuring the social value of a worker council's outputs relative to the opportunity cost of its inputs and endowments (≥ 1 constraint).
SDF	Syrian Democratic Forces	Military coalition of North and East Syria; party to the 2026 integration agreement (Chapter 2).
SKU	Stock Keeping Unit	Granular product variant (size, color, style) resolved at intra-year fulfillment from coarse planning categories.
TFEU	Treaty on the Functioning of the European Union	EU treaty whose Article 101 governs coordination between independent enterprises (Chapter 11).
UFB	Updating Facilitation Board	Facilitation board mediating intra-year plan adjustments when demand shifts or supply shocks exceed slack (LF Ch. 6).
UGT	Unión General de Trabajadores	Socialist General Union of Workers that co-organized industrial and agrarian collectives in Spain (1936–1939).
WC (WC_j)	Workers' Council	Democratic workplace collective where each worker has one vote and holds a Balanced Job Complex.

B. Operations Research, AI, Distributed Systems, and Cryptography Abbreviations

Abbreviation	Full Term	Definition & Role in Monograph
ADMM	Alternating Direction Method of Multipliers	Augmented-Lagrangian primal-dual method; Algorithm 3b is the proximal Jacobian variant of Deng et al. (2017), a research candidate.
AI	Artificial Intelligence	Machine-learned models; restricted in DPEP to advisory and accessibility roles (Chapter 8).
AP / CP	Availability / Consistency under Partition	CAP theorem properties: local-first drafting is AP (available offline), while Epoch BFT price finality is CP.
ARIMA / ETS	AutoRegressive Integrated Moving Average / Error-Trend-Seasonality	Classical statistical time-series forecasting models used for baseline demand and SKU disaggregation.
BBS	Boneh–Boyen–Shacham signatures	Pairing-based signatures used for anonymous credentials; not post-quantum.
BFT	Byzantine Fault Tolerance	Distributed consensus property guaranteeing safety even when up to $f < n/3$ validator nodes act maliciously.
BHT	Brassard–Høyer–Tapp algorithm	Quantum collision-finding algorithm; relevant to hash security margins.

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Abbreviation	Full Term	Definition & Role in Monograph
BLAKE3	BLAKE3 Cryptographic Hash	Hash function with 256-bit output and a 128-bit security target; used for the Two-Root aggregation trees and content addressing.
BLS	Boneh–Lynn–Shacham signatures	Pairing-based aggregatable signatures over BLS12-381; used in threshold randomness beacons and proposed for aggregated round checkpoints; not post-quantum.
C2SP	Community Cryptography Specification Project	Publishes the tlog-checkpoint, tlog-cosignature and tlog-witness specifications for witnessed transparency logs (Section 5.3.2).
BSP	Bulk-Synchronous Parallel	Execution model alternating parallel local computation rounds with synchronized global aggregation checkpoints.
CapCert	Capability Certificate	Chained, signed certificates encoding council membership and revocable delegation. Format to be chosen (UCAN 1.0 and Biscuit are candidates); signed under the active CryptoSuiteId (Section 7.2.1).
CBOR (CDE)	Concise Binary Object Representation	RFC 8949 Canonical Deterministic Encoding used for all signed DPEP messages with fixed 10^{-6} micro-unit scale.
CFRG	Crypto Forum Research Group	IRTF group that develops cryptographic specifications such as VDAF.
CID	Content Identifier	Self-certifying cryptographic hash address referencing a redactable qualitative List or plan blob.
CPU / RAM	Central Processing Unit / Random-Access Memory	Processor time and working memory, used in device and server cost estimates.
CRDT	Conflict-Free Replicated Data Type	Data structure (Loro / Automerge) allowing offline concurrent edits in council workspaces that merge deterministically.
CRQC	Cryptographically Relevant Quantum Computer	Future fault-tolerant quantum computer capable of breaking classical elliptic-curve and RSA cryptography via Shor's algorithm.
CSV	Comma-Separated Values	Plain-text table format of the pequod-cljs run logs.
DA	Data Availability	Verification that all leaf proposals of a round root are downloadable by auditing nodes (0.24–1.46 GB raw per round at $N = 10^6$; Table 5.2).
DAO	Decentralized Autonomous Organization	Blockchain governance construct critiqued in Chapter 2 for conflating token voting with council democracy.
DAP	Distributed Aggregation Protocol	IETF protocol for running VDAFs between clients and aggregators.
DP	Differential Privacy	Calibrated noise added to released totals to bound what can be learned about one person.
DTN	Delay-Tolerant Networking	Store-and-forward networking (including LoRa mesh and USB data-mules) for intermittently connected regions.

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Abbreviation	Full Term	Definition & Role in Monograph
EEIO / LCA	Environmentally Extended Input-Output / Lifecycle Assessment	Matrix accounting methods estimating cradle-to-gate resource and emission footprints across supply chains.
FIPS	Federal Information Processing Standards	NIST post-quantum cryptographic standards finalized in August 2024: FIPS 203, FIPS 204, and FIPS 205.
FLP	Fully Linear Proof	VDAF-22 proof on secret shares; binds report validity, not aggregator honesty.
FN-DSA	FFT over NTRU-Lattice Digital Signature Algorithm	Falcon-based signature standard (FIPS 206, draft) with smaller signatures than ML-DSA.
FOC	First-Order Conditions	Optimality conditions from setting derivatives of the objective to zero.
FRI / STARK	Fast Reed-Solomon IOP / Scalable Transparent ARgument of Knowledge	Discarded household-budget design; not part of Architecture D.
GC	Garbage Collection	Automatic memory reclamation in managed language runtimes.
HCI	Human–Computer Interaction	Study of interface usability, used in the Phase 1H probe.
HPKE	Hybrid Public Key Encryption	IETF scheme (RFC 9180) used by DAP to encrypt report shares to aggregators.
HQC	Hamming Quasi-Cyclic	Code-based key-encapsulation mechanism selected by NIST in 2025 as a backup to ML-KEM.
HRU	Household Reporting Unit	Per-category integer unit for household Prio3 reports, converted exactly to protocol micro-units (Section 7.4.1).
IDL	Interface Definition Language	Language-neutral description of message schemas (Listings 4.1–4.2).
IEEE-754	IEEE Floating-Point Standard	Hardware floating-point representation replaced in DPEP core rules by deterministic 64-bit fixed-point integers.
IETF / IRTF	Internet Engineering / Research Task Force	Bodies that publish Internet standards (RFCs) and research drafts.
IIS	Irreducible Infeasible Subsystem	Minimal set of constraints that makes an optimization problem infeasible.
KEM	Key-Encapsulation Mechanism	Public-key method for establishing a shared secret key (e.g., ML-KEM).
KKT	Karush-Kuhn-Tucker Conditions	First-order necessary and sufficient optimality conditions for constrained nonlinear/conic optimization.
LAN / WAN	Local / Wide Area Network	Council-premises network versus links between councils and federations.
LFS	Large File Storage	Git extension storing large data files outside the repository history.
LLM	Large Language Model	Generative text model; restricted in DPEP to translation, accessibility, and summaries.
LP / QP / MILP	Linear / Quadratic / Mixed-Integer Linear Programming	Mathematical optimization paradigms solved locally via Clarabel.rs and HiGHS.
LWE / SIS	Learning With Errors / Short Integer Solution	Lattice problems whose module variants underlie ML-KEM and ML-DSA.
M0–M5	Maturity levels	Component maturity scale from unspecified (M0) to ubiquitous (M5) (Section 6.3).

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Abbreviation	Full Term	Definition & Role in Monograph
MAPE	Mean Absolute Percentage Error	Accuracy metric used to audit and backtest statistical demand forecasts.
MCP	Mixed Complementarity Problem	Joint formulation of optimality and market-clearing conditions solved by PATH (Algorithm 3d).
ML-DSA-65	Module-Lattice Digital Signature Algorithm	NIST FIPS 204 (Dilithium3) post-quantum digital signature scheme combined in hybrid mode with Ed25519.
ML-KEM-768	Module-Lattice Key-Encapsulation Mechanism	NIST FIPS 203 (Kyber-768) post-quantum key-establishment scheme combined in hybrid mode with X25519.
MLS	Messaging Layer Security	IETF RFC 9420 protocol providing end-to-end encrypted group messaging inside councils.
MPC	Secure Multiparty Computation	Candidate hardening of aggregator correctness (Table 7.5); not the Phase-1 construction.
NASA-TLX	NASA Task Load Index	Standardized workload questionnaire proposed for measuring participation fatigue in Phase 1H and Phase 3.
NIST	National Institute of Standards and Technology	US standards body that published FIPS 203–205 (post-quantum cryptography).
NMT / TTS	Neural Machine Translation / Text-to-Speech	Local accessibility models translating qualitative Lists across languages and assisting visually impaired users.
OR	Operations Research	Discipline of mathematical optimization, linear programming, and supply-chain clearing.
P2P	Peer-to-Peer	Decentralized networking architecture without central client-server gatekeepers.
PoC	Proof of Concept	A proposed, unbuilt reference implementation (Phases 1–3, Chapter 9); no implementation exists.
PQ	Post-Quantum	Prefix for schemes designed to resist quantum attacks (see PQC).
PQC	Post-Quantum Cryptography	Cryptographic algorithms resistant to both classical and quantum (Shor/Grover) attacks.
PRG	Pseudorandom Generator	Deterministic expansion of a short random seed into a long pseudorandom string.
Prio3 / VDAF	Verifiable Distributed Aggregation Function	IETF CFRG draft-irtf-cfrg-vdaf-22 (Informational). Privacy needs one honest aggregator; correct totals need all of them.
PWA	Progressive Web Application	Web client used for read-only exploration, accessibility, and the Phase 1H probe; not the primary participant client, because it cannot reach hardware keystores (Section 5.5).
Quint / TLA$^+$	Specification languages	Executable specification languages based on the temporal logic of actions, used to model-check the protocol state machine before Phase 2 Rust work (Section 9.1.2).
QUIC	Quick UDP Internet Connections	Modern encrypted, multiplexed transport protocol (quinn / iroh) used for Layer 2 federation sync.

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Abbreviation	Full Term	Definition & Role in Monograph
RFC	Request for Comments	IETF document series that includes Internet standards.
Role-LP / CC / SR / PCW / BFT / GW	DPEP Node Roles 1–6	Lightweight Participant (LP), Council Coordinator (CC), Storage Replica (SR), Planning Compute Worker (PCW), Epoch Validator (BFT), and Delay-Tolerant Gateway (GW).
SDK	Software Development Kit	Libraries and tools for building on a platform (e.g., Cosmos SDK).
SHA-2 / SHA-3	Secure Hash Algorithm 2 / 3	NIST hash-function families (e.g., SHA-256, SHA3-256; FIPS 180-4, 202).
SLH-DSA	Stateless Hash-Based Digital Signature Algorithm	NIST FIPS 205 (SPHINCS+) signature standard relying solely on hash functions, used for long-term root anchors.
SQL / REST	Structured Query Language / Representational State Transfer	Relational database queries and web APIs of the conventional Architecture A.
TLS	Transport Layer Security	Standard protocol for encrypted network channels; used inside QUIC.
TPS	Transactions Per Second	Throughput metric shown in Chapter 5 to be irrelevant to epoch-synchronized participatory planning.
TTL	Time-To-Live	Hard expiration timer on delegate capability certificates and emergency replanning playbooks.
Two-Root	Two-Root BLAKE3 aggregation	Proposal Merkle root plus category-totals root (Section 5.3.1). Replaces internal-vector Sparse Merkle-Sum Trees.
UCAN	User Controlled Authorization Network	Capability-token format; a candidate format for CapCerts.
UI / UX	User Interface / User Experience	Screens and interaction design of participant clients.
UniFFI	Universal Foreign Function Interface	Mozilla binding generator exposing the Rust dpep-core library to native Kotlin (Android) and Swift (iOS).
VRF / VDF	Verifiable Random / Delay Function	Primitives for unbiasable public randomness; candidate post-quantum beacon building blocks.
WASM	WebAssembly	Deterministic, sandboxed binary instruction format (Wasmtime) executing IFB price-update rules.
X25519 / Ed25519	Curve25519 key exchange / signatures	Classical elliptic-curve halves of DPEP's hybrid schemes.
XOF	Extendable-Output Function	Hash function with output of any requested length (e.g., TurboSHAKE128 in VDAF).

Glossary

This glossary defines the terms a reader from one discipline may meet from another. The first section explains the core ideas in three vocabularies: for advocates and cooperators, who care about who decides and who is protected; for economists, who care about prices, information, convergence, and incentives; and for technologists, who care about threat models, guarantees, and what runs where. The remaining sections define terms by area, each with the section of the book where it is treated in full. Status labels follow the book: demonstrated, specified, or open. Abbreviations are listed in Appendix B.

The Core Ideas in Three Vocabularies

Idea	Advocates	Economists	Technologists
DPEP itself	Software that helps councils plan the economy together, without a market and without a central planning office.	An implementation of Albert and Hahnel's participatory planning procedure: iterative indicative-price planning, with councils as the only proposers.	A local-first protocol: councils hold their own state, federation servers aggregate, deterministic WASM computes prices, and a checkpoint layer fixes each round's history.
A planning round	Everyone proposes, sees where demand exceeds what's available, and revises.	One iteration of a tâtonnement-like procedure: proposals, then excess demand, then a price update.	One pass through states S2–S4: signed proposals, Two-Root aggregation, the price update, a checkpoint.
The facilitation board	It does the arithmetic and publishes it. It can't write or change anyone's plan.	A price-adjustment rule applied publicly, with no discretion over allocations.	A deterministic WASM function that anyone can recompute; a proposal is valid only if a council signed it.
Two-Root aggregation	You can check that your council's proposal was counted and that the totals add up.	A verifiable record of every proposal and every category total, so the excess-demand signal can be audited.	Two BLAKE3 Merkle trees (proposals and category totals), a 512-byte inclusion path for phones, and full recomputation for auditors.
Offline councils	Losing the internet doesn't hand your decision to anyone else.	Participation is robust to missed rounds: the last proposal stands in.	Local-first state with CRDT merging; the participation bitmap marks each council fresh, carried forward or excluded.
Cheating detection	If someone changes or drops your proposal, your receipt proves it and forces it back in.	An audit mechanism against omission by intermediaries (it does not cover misreporting by a council).	Signed proposals, signed receipts, the challenge window and forced inclusion: detection, not prevention.

Idea	Advocates	Economists	Technologists
Household privacy	What your household asks for stays private; only neighborhood totals are seen.	Demand is aggregated privately; individual bundles are never revealed.	Prio3 secret-sharing over two aggregators: privacy needs one honest aggregator, correct totals need both.
Post-quantum security	Records stay trustworthy even if powerful quantum computers arrive.	(Rarely needed.) The integrity of the plan record over decades.	Hybrid ML-KEM for confidentiality; a tiered signature policy; k-of-n post-quantum anchoring of each round's roots.
Kill gates	Each stage has a test it can fail, and failing means stopping or changing course.	Pre-registered falsification criteria with defined consequences.	Go/no-go thresholds measured on testbeds before building further.

Self-Management and Institutions (Chapters 2–3)

- Workers' council (WC, WC_j). A democratic workplace collective: one worker, one vote. Each worker holds a balanced job complex. Economists: the supply-side planning unit. (§4.1)
- Consumers' council (CC, CC_h). A neighborhood council of consumers, nested into ward, city and regional federations. Economists: the demand-side planning unit. (§4.1)
- Federation (IF, consumer federations). A body of councils joined by industry or geography. It is an instrument of the councils, not a level above them. (§4.1, §2.6)
- Balanced job complex (BJC). A bundle of tasks balanced for empowerment and desirability, so no one specializes in deciding while others only execute. (§3.2)
- Coordinator class. Albert and Hahnel's term for managers and experts who monopolize empowering work, as distinct from owners. Brinton's history of Russia 1917–21 is the book's main case. (§2.1, §2.4)
- Brinton's Test (Design Principle 2.1). A feature is justified only if it increases people's confidence, autonomy, participation and self-activity, not their dependence on experts. (§2.13.2)
- Plan factory (Castoriadis). A technical body that computes alternatives and their consequences but never decides. DPEP's facilitation board is designed on this rule. (§2.6)
- Revocable delegation / recall. A delegate can be withdrawn by the council that sent them, at any time. In DPEP, a delegate certificate is revoked by one council vote. (§2.3, §7.7)
- Sortition. Selection by lot, as in Athens. DPEP uses it for audit, juries and facilitation oversight; it supports councils and does not replace them. (§2.2,

§7.7)

- Euthynai. The Athenian end-of-term audit of officials. The book's name for physical audits of things cryptography can't check, such as output, emissions and rosters. (§2.2, Fig. 7.5)
- Community of Affected Parties (CAP). Residents affected by a specific pollutant in an airshed or watershed. They set emission tolerances. (§4.3)
- Productive commons. Productive assets belong to society. Councils are charged for using inherited capital, which avoids the "collective egoism" of Barcelona in 1936. (§2.5)
- Bridged adoption (Design Principle 11.1). DPEP runs alongside existing systems, such as bookkeeping, payments and identity, through specified bridges, with a right of exit. (§11.3)

Economics and Planning (Chapter 4)

- Indicative prices ($P^{(t)}$). Prices published each round as estimates of social opportunity cost. They guide revisions; they are not market prices. (§4.2)
- Priced items (Ω, $G = |\Omega|$). The master catalog of everything priced: private, public and intermediate goods, resources, labor, capital and pollutants. G is the number of items. (Definition 4.1)
- Social benefit / social cost (SB/SC). A council's outputs valued against the opportunity cost of its inputs; proposals must reach SB/SC ≥ 1. (§4.3)
- Excess demand / imbalance (ρ_i). The normalized gap between demand and supply in a category. Planning aims to bring it under a tolerance τ. (§4.4)
- Feasible, cleared, socially acceptable (Definition 4.2). A plan can be feasible while still wasteful. Cleared means the imbalances are within tolerance. Socially acceptable means people endorse it. Kill Gate 1 requires a cleared plan. (§4.4)
- Tâtonnement. Iterative price adjustment toward balance. Its convergence is guaranteed only in special cases, which is the central open economic question. (§4.4.3)
- Scarf test. A classic economy in which simple price adjustment cycles forever. The book's diagnosis is that the instability comes from incomes that move with prices. (§4.4.3)
- Non-convexity / lumpy (indivisible) plants. A plant you either build whole or not at all, which breaks the smooth mathematics behind price convergence. (§4.4)
- Endogenous income. Incomes that depend on the plan itself, such as effort-rated pay. (§4.4)

- Algorithms 3a–3d. Research-candidate solvers (damped tâtonnement, proximal ADMM, federation MILP, damped budget fixed point). None has a general convergence guarantee. (§4.4.4)
- Effort rating (ϵ). A multiplier for effort and sacrifice. The proposed default is flat hourly pay unless a council votes otherwise. (§9.1.4)
- Slack buffer (SL). A planned surplus so that small intra-year shocks need no replanning. (§4.2)
- Society Account (Sandström). Double-entry accounting for a democratic economy. DPEP uses it for its accounting identities. (§4.2)
- Pollution Demand-Revealing Mechanism (PDRM). Hahnel's procedure for pricing local pollution through CAPs. (§4.3)

The Protocol: How DPEP Runs (Chapters 4–5, 7)

- State machine (S0–S9). The annual cycle: planning rounds, ratification, the operating year, intra-year fulfillment (S7), and scoped replanning (S8). (Fig. 4.1)
- Milestone rounds / automated rounds. Households submit fresh private reports only at milestone rounds (proposed t = 1, 4 and 8). The rounds between them carry aggregates forward. (§7.4.3, Fig. 7.4)
- Participation bitmap. Two bits per council each round: fresh proposal, carried forward, or excluded. It makes censorship distinguishable from a network partition. (§5.3.1)
- Carry-forward. A council that misses a round's deadline keeps its previous proposal, recorded as carried, not dropped. (§5.3.1)
- Signed receipt / challenge window / forced inclusion. A council receives a signed receipt for each submission. If its proposal is omitted, it presents the receipt within the challenge window (proposed: 14 days), and inclusion is forced. (§5.3.1)
- Checkpoint. A small per-round record of the roots and quorum signatures that fixes the round's history. (§5.3, §7.2)
- Intra-year fulfillment (S7). Collecting goods at depots through the year. The pilot design uses council allocation for staples and receipts held by the home council for rationed goods. (§7.5)
- Credit cap / home depot. An offline-overspend bound: rationed goods are checked online, or collected offline only at your home depot, so overspending is limited to one cap per person per day. (§7.5)
- Household Reporting Unit (HRU). The integer unit for private household reports, converted exactly to protocol units. (§7.4.1)

- Coarse report. The privacy baseline: about 200 categories, about 55 kB per report. (§7.4.2)
- Node roles (Role-LP, CC, SR, PCW, BFT, GW). Phone, council coordinator, storage replica, planning compute worker, epoch validator, delay-tolerant gateway. (Table 5.5)

Distributed Systems (Chapters 5–6)

- Local-first. Your device holds your data and works offline; servers sync and aggregate but don't own the data. (§5.3)
- CRDT. A data structure that merges offline edits made by council members without conflicts. (§5.3)
- Architecture A / B / C / D (and D-log). Federated SQL/REST; pure peer-to-peer; a per-transaction blockchain; DPEP's four-layer hybrid (D-log uses a witnessed log instead of BFT). D scores highest at baseline weights, but A wins when simplicity and cost dominate. (§6.1)
- Four layers.
  - Layer 4: human deliberation and the local solver.
  - Layer 3: local-first council and household state.
  - Layer 2: federated aggregation and the data plane.
  - Layer 1: checkpoints and the deterministic price engine. (Fig. 5.1)
- BFT (Byzantine fault tolerance). Consensus that stays correct while fewer than a third of validators misbehave. CometBFT is the reference engine. (§5.3.2)
- Witnessed log (C2SP). The alternative Layer 1: one operator appends each round, and independent witnesses cosign it. It is simpler than BFT but gives up liveness. (§5.3.2)
- Deterministic WASM. Price rules compiled so that every node computes exactly the same result. (§5.3)
- Canonical CBOR. One exact byte encoding for every signed message, so signatures can be checked by anyone. (§4.5)
- Why not a blockchain. Planning is a bulk, round-based computation, not millions of transactions per second. Transactions per second is the wrong metric. (§5.2)

Cryptography and Privacy (Chapter 7)

- Prio3 / VDAF / DAP. An IETF standard track for computing sums over secret-shared reports. In DPEP, households split each report between two aggregators. (§7.4)

- Aggregator (K = 2). A server that sees only random-looking shares. Two are recommended for pilots; a third is optional. (§7.4)
- Privacy vs. correctness asymmetry. Privacy needs at least one honest aggregator; correct totals need all of them. (Fig. 7.3)
- Budget credential / one report per person. An unlinkable per-epoch credential that limits each person to one report. Binding each household's budget inside the private report is open research. (§7.4.4)
- Post-quantum cryptography (PQC) / CRQC. Algorithms resistant to quantum computers; a CRQC is a quantum computer able to break today's public-key cryptography. (§7.2)
- Hybrid (X25519 + ML-KEM; Ed25519 + ML-DSA). Classical and post-quantum algorithms combined, so an attacker must break both. (§7.2)
- Tiered signature policy. Per-round objects are signed classically (small and fast). Long-lived objects are signed hybrid. Each round's roots get a post-quantum anchor from several parties. (§7.2.1)
- Post-quantum anchor (k of n). Under BFT, three of five federations sign each round's roots with SLH-DSA; under the log design, witnesses add ML-DSA-44 cosignatures. (Table 7.3)
- Crypto-agility / CryptoSuiteId. Every signed object names its algorithm suite, so suites can be replaced through a rehearsed four-step migration. (Fig. 7.1)
- Trust map. What cryptography guarantees, what is only detected, and what rests on institutions and audit. (Fig. 7.5)
- Sortition beacon. Public randomness for drawing lots that no one can bias. The design is open. (§7.7)

The Research Program and Evidence (Chapters 1, 6, 9–12)

- Three tiers of evidence. Demonstrated (code or data audited), analytical estimate, and open research target. Use these labels when explaining status. (Table 1.1)
- Phases 1, 1H, 2, 3.
  - Phase 1: economic simulator.
  - Phase 1H: human probe with 30–60 volunteers.
  - Phase 2: distributed testbed, with a formal model first.
  - Phase 3: cryptography, mobile and cooperative pilot. (Table 9.1)
- Kill gate. A pre-registered pass/fail test at the end of a phase, with a stated consequence: proceed, narrow, redesign and rerun, or stop. (§9.1.1, Table 9.2)

- Pre-registration. Instances, thresholds and analysis scripts are committed before the run, so results can't be tuned after the fact. (§9.1.1)
- Formal model (Quint / TLA+). A model-checked specification of states S2–S4, S7 and S8, written before any Phase 2 Rust code. (§9.1.2)
- Maturity levels (M0–M5). A scale for components, from unspecified (M0) to ubiquitous (M5). (Table 6.3)
- The demonstration. Four scenes: councils plan; a council goes offline; an operator cheats; households report privately. Each scene is tied to a gate and a status. (§1.1.1, §9.2, Table 9.3)
- Shadow mode. A group runs DPEP alongside its existing tools, binding no one until a council ratifies a switch. (§11.3)
- Claims register. Every scientific and engineering claim, with its evidence, assumptions, limitations and required validation. (Appendix A)

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